

Q1. What is the primary function of a Load Balancer in the context of AWS cloud infrastructure?

- A.** It stores backups of EC2 instances across Availability Zones
- B.** It controls security groups to restrict network access
- C.** It distributes incoming traffic across multiple targets in different Availability Zones
- D.** It provisions new EC2 instances when traffic increases

Correct Answer:

- C.** It distributes incoming traffic across multiple targets in different Availability Zones

Explanation:

Load balancers in AWS (e.g., Elastic Load Balancing - ELB) are designed to automatically distribute incoming application or network traffic across multiple targets, such as EC2 instances, containers (ECS), Lambda functions, or even IP addresses. They improve fault tolerance and ensure high availability by routing traffic to healthy targets in different Availability Zones.

<https://docs.aws.amazon.com/elasticloadbalancing/latest/userguide/what-is-load-balancing.html>

Incorrect Options Explained:

A. It stores backups of EC2 instances across Availability Zones

This is the role of services like Amazon S3 or AWS Backup, not load balancers.

B. It controls security groups to restrict network access

Security groups and network ACLs handle access control, not traffic distribution.

D. It provisions new EC2 instances when traffic increases

This is handled by Auto Scaling, not by the load balancer itself.

Q2. In the context of an AWS Elastic Load Balancer, what is the role of a listener, and why is defining a default rule important?

- A.** A listener encrypts data in transit and requires a default rule to apply rate limiting to the target group
- B.** A listener monitors backend instance health and routes traffic using availability metrics
- C.** A listener checks for incoming client connections and uses the default rule when no other conditions are met
- D.** A listener logs incoming requests and automatically assigns priorities to all target groups

Correct Answer:

C. A listener checks for incoming client connections and uses the default rule when no other conditions are met

Explanation:

In AWS, a listener is a process that checks for incoming connection requests based on the protocol and port you define (e.g., HTTP on port 80 or HTTPS on port 443).

Each listener must have at least one default rule, which defines how to route requests if none of the custom listener rules match. This rule specifies a target group, a condition (such as a path pattern), and a priority to determine the order of rule evaluation.

<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-listeners.html>

Incorrect Options Explained:

A. *A listener encrypts data in transit and requires a default rule to apply rate limiting to the target group*

Listeners can use HTTPS for encrypted traffic, but they do not apply rate limits that are handled by other AWS services (e.g., AWS WAF or API Gateway).

B. *A listener monitors backend instance health and routes traffic using availability metrics*
Health checks are performed by the load balancer itself, not the listener.

D. *A listener logs incoming requests and automatically assigns priorities to all target groups*
Listeners do not assign priorities automatically; you manually define rule priorities. Logging is done through Access Logs, a separate feature.

Q3. Which of the following best describes the primary benefit of using AWS Direct Connect over a standard internet connection?

A. It provides automatic encryption and data loss prevention between VPCs

B. It allows private, dedicated network connectivity between your on-premises network and AWS, bypassing the public internet

C. It automatically replicates EC2 instances across Availability Zones without user configuration

D. It enables serverless applications to be hosted entirely outside the AWS Cloud

Correct Answer:

B. It allows private, dedicated network connectivity between your on-premises network and AWS, bypassing the public internet

<https://docs.aws.amazon.com/directconnect/latest/UserGuide/Welcome.html>

Incorrect Options Explained:

- **A.** *It provides automatic encryption and data loss prevention between VPCs*
Direct Connect does not encrypt traffic by default; you must implement encryption separately (e.g., IPsec VPN over Direct Connect).
- **C.** *It automatically replicates EC2 instances across Availability Zones without user configuration*
EC2 replication across AZs must be configured by the user using services like Auto Scaling or Load Balancers.
- **D.** *It enables serverless applications to be hosted entirely outside the AWS Cloud*
Serverless apps (e.g., using AWS Lambda) still run within AWS; Direct Connect doesn't enable external hosting.

Q4. You are designing a highly available network architecture using AWS Direct Connect. Which configuration ensures automatic failover while maintaining active traffic flow through both connections?

- A.** Active/Passive with Border Gateway Protocol (BGP)
- B.** Virtual Private Gateway with Route 53 failover
- C.** Active/Active (BGP multipath) configuration
- D.** Public Virtual Interface over a single connection

Correct Answer:

C. Active/Active (BGP multipath) configuration

Explanation:

The Active/Active (BGP multipath) setup in AWS Direct Connect enables both connections to be active simultaneously, facilitating load balancing of traffic and automatically rerouting it if one path fails. It's the default configuration for high availability using Border Gateway Protocol (BGP).

<https://docs.aws.amazon.com/directconnect/latest/UserGuide/resiliency.html>

Incorrect Options Explained:

- *A. Active/Passive with BGP*
Only one connection is active; the other is used only during failover, not ideal for active load sharing.
- *B. Virtual Private Gateway with Route 53 failover*
Route 53 handles DNS-based failover, not Direct Connect link-level redundancy.
- *D. Public Virtual Interface over a single connection*
This setup lacks redundancy; a single link means no failover capability.

Q6. What is the CloudFormation Designer primarily used for?

- A.** Monitoring EC2 instance performance in real time
- B.** Creating, viewing, and modifying CloudFormation templates using a graphical interface
- C.** Managing IAM policies across multiple AWS accounts
- D.** Performing backup and recovery of S3 buckets

Correct Answer:

- B.** Creating, viewing, and modifying CloudFormation templates using a graphical interface

Explanation:

CloudFormation Designer is a visual tool provided by AWS that lets users design infrastructure templates using a drag-and-drop interface. It also includes built-in JSON/YAML editors for direct editing of the templates.

<https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/working-with-templates-cfn-designer.html>

Incorrect Options Explained:

- *A. Monitoring EC2 instance performance in real time*
That is done using Amazon CloudWatch, not CloudFormation Designer.

- *C. Managing IAM policies across multiple AWS accounts*
IAM policies are managed through IAM, and cross-account access involves AWS Organizations or STS, not CloudFormation Designer.
- *D. Performing backup and recovery of S3 buckets*
AWS Backup or lifecycle policies manage S3 backups—not CloudFormation Designer.

Q7.

Which of the following statements is **true** regarding **Amazon CloudWatch Metrics**?

- A.** CloudWatch metrics can be deleted manually to reduce storage costs
- B.** Memory and disk usage metrics for EC2 instances are available by default
- C.** Metric timestamps can be backdated up to one month if needed
- D.** CloudWatch metrics automatically expire after 15 months if no new data is published

Correct Answer:

- D.** CloudWatch metrics automatically expire after 15 months if no new data is published

Explanation:

CloudWatch metrics expire after 15 months if no new data points are published. This retention policy helps manage storage automatically. As new data arrives, older data beyond 15 months is dropped. You cannot manually delete a metric.

https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/cloudwatch_concepts.html

Incorrect Options Explained:

- *A. CloudWatch metrics can be deleted manually to reduce storage costs*
Metrics cannot be manually deleted; they expire based on age and data updates.
- *B. Memory and disk usage metrics for EC2 instances are available by default*
These require the CloudWatch Agent to be installed on the EC2 instance.
- *C. Metric timestamps can be backdated up to one month if needed*
CloudWatch only accepts timestamps up to 2 weeks in the past.

Q8.

Which of the following is a key benefit of using Amazon S3 Glacier Deep Archive for data storage?

- A. It allows instant data retrieval with millisecond latency
- B. It is designed for frequently accessed data across multiple regions
- C. It provides the lowest-cost storage option for long-term data retention with retrieval times up to 48 hours
- D. It automatically replicates data to on-premises tape libraries for backup

Correct Answer:

C. It provides the lowest-cost storage option for long-term data retention with retrieval times up to 48 hours

Explanation:

Amazon S3 Glacier Deep Archive is ideal for rarely accessed data, such as backups and archives. It offers very low storage costs, and while retrieval is not instant, bulk retrieval can return large amounts of data within up to 48 hours, and standard restores are available within 12 hours or less. Data is stored across at least 3 Availability Zones with 11 nines (99.999999999%) durability.

<https://aws.amazon.com/s3/storage-classes/glacier/>

Incorrect Options Explained:

- *A. It allows instant data retrieval with millisecond latency*
That's true for S3 Standard or S3 Intelligent-Tiering, not Glacier Deep Archive.
- *B. It is designed for frequently accessed data across multiple regions*
Glacier Deep Archive is meant for infrequent access, not frequent or real-time access.
- *D. It automatically replicates data to on-premises tape libraries for backup*
AWS does not replicate to on-premises systems; it stores data within the AWS Cloud infrastructure.

Q9.

Which of the following statements best describes how Amazon S3 Object Lock protects objects?

- A. It encrypts objects during upload and automatically deletes them after 30 days
- B. It prevents objects from being deleted or overwritten for a specified time or until a legal hold is removed
- C. It moves objects to Glacier storage for long-term protection
- D. It compresses objects to save space and locks the entire S3 bucket permanently

Correct Answer:

B. It prevents objects from being deleted or overwritten for a specified time or until a legal hold is removed

Explanation:

Amazon S3 Object Lock enables you to lock individual object versions to prevent deletion or modification. You can enforce this via a retention period (fixed time) or a legal hold (remains until explicitly removed). It is useful for regulatory compliance and data immutability. Object Lock only works with versioned buckets.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html>

Incorrect Options Explained:

- *A. It encrypts objects during upload and automatically deletes them after 30 days*
Encryption and lifecycle policies are separate features; Object Lock prevents deletion, not triggers it.
- *C. It moves objects to Glacier storage for long-term protection*
Object Lock is not a storage class; it works with objects regardless of their storage class.
- *D. It compresses objects to save space and locks the entire S3 bucket permanently*
Locking is done per object version, not for the entire bucket; and it does not compress objects.

Q10.

Which of the following is the recommended method for interacting with Amazon S3, considering current API support?

- A. Using the SOAP API over HTTP for full feature compatibility
- B. Using the REST API with virtual hosting and HTTP requests
- C. Using FTP for faster object transfers
- D. Using only the AWS CLI as the exclusive method for S3 access

Correct Answer:

- B.** Using the REST API with virtual hosting and HTTP requests

Explanation:

Amazon S3 supports multiple access methods, but SOAP support over HTTP is deprecated, and new features are not added to the SOAP API. The REST API is the recommended method for interacting with S3 and can be used with virtual hosting via the HTTP Host header. Alternatively, AWS SDKs (built on REST) offer a more user-friendly interface.

<https://docs.aws.amazon.com/AmazonS3/latest/API/Welcome.html>

Incorrect Options Explained:

- *A. Using the SOAP API over HTTP for full feature compatibility*
SOAP over HTTP is deprecated, and new S3 features are not supported on SOAP.
- *C. Using FTP for faster object transfers*
Amazon S3 does not support FTP; transfers are done via HTTP-based APIs or tools like AWS CLI.
- *D. Using only the AWS CLI as the exclusive method for S3 access*
While the AWS CLI is useful, it is not the only method—REST API and SDKs are also commonly used.

Q11.

How can you use CloudWatch Log Streams to better understand the execution of your AWS Lambda function?

- A. By enabling encryption at rest for Lambda artifacts
- B. By writing custom logging statements in your Lambda code to track execution flow
- C. By enabling automatic backups of function logs every 24 hours
- D. By configuring S3 triggers to create CloudWatch dashboards

Correct Answer:

- B. By writing custom logging statements in your Lambda code to track execution flow

Explanation:

AWS Lambda automatically sends invocation logs to CloudWatch Logs. Developers can add custom logging statements (e.g., `console.log()` in Node.js or `print()` in Python) within the Lambda code. These logs appear in CloudWatch Log Streams, helping you trace the execution flow, debug errors, and measure performance.

<https://docs.aws.amazon.com/lambda/latest/dg/monitoring-cloudwatchlogs.html>

Incorrect Options Explained:

- A. *By enabling encryption at rest for Lambda artifacts*
Encryption at rest secures data but doesn't help analyze execution flow.
- C. *By enabling automatic backups of function logs every 24 hours*
CloudWatch Logs retains data, but there is no built-in automatic backup for logs in this way.
- D. *By configuring S3 triggers to create CloudWatch dashboards*
S3 triggers invoke Lambda functions, but they don't generate dashboards directly.

Q12.

In an Amazon Aurora DB cluster, what is the primary role of the **Aurora Replica**?

- A. It performs both read and write operations on a separate storage volume
- B. It mirrors the primary instance to provide instant backups only
- C. It supports read-only operations and can be promoted during failover
- D. It replicates data from the primary instance using snapshot-based syncing

Correct Answer:

C. It supports read-only operations and can be promoted during failover

Explanation:

Aurora Replicas are read-only instances that connect to the same shared storage as the primary. They help offload read traffic and are also candidates for automated failover if the primary instance becomes unavailable. You can configure failover priority for each replica.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replica.html>

Incorrect Options:

- A. Aurora Replicas do not perform write operations and do not use separate storage volumes.
- B. They do not act only as backup mirrors; they handle real-time read workloads.
- D. Aurora uses shared storage, not snapshot-based replication for read replicas.

Q13.

What happens in an Aurora DB cluster when the primary DB instance fails?

- A. The database becomes read-only until a new cluster is manually provisioned
- B. An Aurora Replica is automatically promoted to the new primary DB instance
- C. Data is lost unless a recent snapshot is restored
- D. A secondary storage volume is mounted as a temporary write target

Correct Answer:

B. An Aurora Replica is automatically promoted to the new primary DB instance

Explanation:

Aurora offers automatic failover. If the primary instance fails, Aurora promotes a replica (based on priority settings) to act as the new primary, maintaining high availability without manual intervention.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Managing.html#Aurora.Managing.FaultTolerance>

Incorrect Options:

- A. Aurora ensures high availability by automatic failover—it doesn't go read-only indefinitely.
- C. Aurora provides durable shared storage, so no data is lost on instance of failure.
- D. Aurora does not use secondary volumes; it relies on shared cluster storage.

Q14.

Which components are required to establish a VPN connection between your on-premises data center and an Amazon VPC?

- A. A NAT gateway and a Route 53 hosted zone
- B. An internet gateway and an EC2 instance with a public IP
- C. A virtual private gateway in AWS and a customer gateway in your data center
- D. A load balancer and a CloudFront distribution

Correct Answer:

C. A virtual private gateway in AWS and a customer gateway in your data centre

Explanation:

To set up a site-to-site VPN with AWS, you need two components:

- A Virtual Private Gateway (VGW) on the AWS side, attached to your VPC
- A Customer Gateway (CGW) on your end, typically a physical or software-based VPN device in your data centre

These work together to establish a secure IPsec VPN tunnel.

https://docs.aws.amazon.com/vpn/latest/s2svpn/VPC_VPN.html

Incorrect Options Explained:

- **A.** NAT Gateway and Route 53 are unrelated to VPN connections.
- **B.** An internet gateway allows public internet access but doesn't establish a VPN tunnel.
- **D.** Load balancers and CloudFront are used for content delivery and traffic distribution, not for VPN connectivity.

Q15.

Which of the following correctly describes a **VPN-only subnet** in an Amazon VPC?

- A. It connects directly to the internet using an internet gateway
- B. It does not have any route to external networks
- C. It uses a virtual private gateway to connect to a remote network over a VPN
- D. It is used to host public-facing applications like web servers

Correct Answer:

- C. It uses a virtual private gateway to connect to a remote network over a VPN

Explanation:

A VPN-only subnet is designed to route traffic through a virtual private gateway (VGW) instead of an internet gateway. This setup is commonly used for private, secure connectivity between your AWS VPC and your on-premises data center over a VPN connection.

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Subnets.html

Incorrect Options Explained:

- A. That describes a public subnet, which is connected to the internet via an internet gateway.
- B. VPN-only subnets do have routes, but through a VGW, not the internet.
- D. Public subnets are used for hosting public-facing apps, not VPN-only subnets.

Q16.

You are managing a VPC peering connection that is currently in the pending-acceptance state. As the owner of the requester VPC, what action must you take if you need to add a new CIDR block to your VPC?

- A. Add the CIDR block immediately since changes are allowed at any stage
- B. Accept the peering request first, then add the CIDR block
- C. Delete the current peering request, add the CIDR block, and then create a new peering request
- D. Ask the acceptor VPC owner to add a matching CIDR block first

Correct Answer:

C. Delete the current peering request, add the CIDR block, and then create a new peering request

Explanation:

When a VPC peering connection is in the pending-acceptance state, the requester VPC owner is not allowed to add new CIDR blocks. To proceed, they must delete the current peering request, modify the CIDR block, and re-initiate a new peering request.

<https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html>

Incorrect Options Explained:

- A. CIDR block modifications are not allowed in the pending-acceptance state for the requester.
- B. The requester cannot accept the request—they can only initiate or delete it.
- D. The acceptor adding a matching CIDR block would likely cause an overlap conflict, not solve it.

Q17.

What is the primary purpose of associating a route table with a subnet in an Amazon VPC?

- A.** To define the encryption rules for data in transit
- B.** To control which EC2 instance types can be launched
- C.** To specify the routes for outbound traffic from the subnet
- D.** To set user access permissions for the subnet

Correct Answer:

C. To specify the routes for outbound traffic from the subnet

Explanation:

Every subnet in a VPC must be associated with a route table, which defines how outbound traffic is directed, for example, whether it goes to an internet gateway, a NAT gateway, or stays within the VPC. If a custom route table is not explicitly associated, the main route table is used by default.

Incorrect Options Explained:

- **A.** Route tables don't handle **encryption**—that's done by security groups, TLS, or VPN.
- **B.** EC2 instance types are controlled by **region and quotas**, not route tables.
- **D.** User access is managed via **IAM policies**, not route tables.

Q18.

Which of the following statements is true about Security Groups in Amazon VPC?

- A.** A newly created security group allows all inbound and outbound traffic by default
- B.** You can associate only one security group with an EC2 instance
- C.** Security groups control traffic at the instance level and are associated with network interfaces
- D.** If no security group is specified, the instance will be blocked from all network communication

Correct Answer:

C. Security groups control traffic at the instance level and are associated with network interfaces

Explanation:

Security groups in AWS act as virtual firewalls that control inbound and outbound traffic at the instance level. They are actually applied to the network interfaces attached to the instance. You can associate up to 5 security groups with an instance.

Incorrect Options Explained:

- **A.** A newly created security group has no inbound rules, and only one outbound rule allowing all outbound traffic.
- **B.** You can associate up to 5 security groups with an instance, not just one.
- **D.** If no security group is specified, the instance is associated with the default security group, which allows inbound traffic from other instances in the same group and outbound traffic to all destinations.

Q19.

Which of the following is a valid use case for Amazon VPC Flow Logs?

- A. Encrypting data in transit between EC2 instances
- B. Diagnosing overly restrictive security group rules by analysing network traffic
- C. Creating custom IAM roles for EC2 traffic analysis
- D. Automatically patching network vulnerabilities in your VPC

Correct Answer:

- B. Diagnosing overly restrictive security group rules by analysing network traffic

Explanation:

VPC Flow Logs capture information about IP traffic going to and from network interfaces in your VPC. This data helps you monitor traffic, verify if traffic is being blocked, and determine whether security group rules or NACLs are too restrictive.

<https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs.html>

Incorrect Options Explained:

- A. Flow Logs only monitor traffic; they do not encrypt it.
- C. IAM roles are unrelated to Flow Logs; they control permissions, not traffic analysis.
- D. Flow Logs do not fix or patch vulnerabilities—they only report on traffic patterns.

Q20.

Which of the following statements about network interfaces (ENIs) in Amazon EC2 is correct?

- A. A primary network interface can be detached and moved to another instance
- B. Each network interface must have exactly one IPv6 address
- C. A network interface can include multiple private IPv4 addresses and one or more security groups
- D. Public IPv4 addresses are assigned to all secondary network interfaces by default

Correct Answer:

- C. A network interface can include multiple private IPv4 addresses and one or more security groups

Explanation:

An Elastic Network Interface (ENI) can include:

- A primary private IPv4 address
- One or more secondary private IPv4 addresses
- One or more security groups
- Optionally, a public IPv4 address (for eth0)
- IPv6 addresses, MAC address, EIP, etc.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-eni.html>

Incorrect Options Explained:

- A. The primary ENI (eth0) cannot be detached from an instance.
- B. IPv6 addresses are optional, and you can have more than one—or none.
- D. Public IPv4 addresses are only auto-assigned to the primary interface (eth0), not to secondary ENIs by default.

Q21.

Which of the following statements is true about route tables in an Amazon VPC?

- A. A subnet can be associated with multiple route tables simultaneously
- B. You can delete the main route table if it's no longer in use
- C. Route tables use the most specific matching rule to determine the traffic path
- D. Route tables automatically update to include new gateways or connections

Correct Answer:

C. Route tables use the most specific matching rule to determine the traffic path

Explanation:

Route tables apply the longest prefix match rule, meaning they select the most specific route that matches the destination IP address. This ensures accurate and optimized routing decisions.

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Route_Tables.html

Incorrect Options Explained:

- A. A subnet can be associated with only one route table at a time.
- B. The main route table cannot be deleted, though it can be replaced with a custom one.
- D. You must manually update route tables to add routes for new gateways or connections.

Q22.

Which of the following is required to allow an EC2 instance in a public subnet to access the internet via IPv4?

- A.** Assign an IAM role to the instance
- B.** Enable CloudTrail logs on the subnet
- C.** Attach an Internet Gateway to the VPC and update the route table
- D.** Enable auto-scaling for the VPC

Correct Answer:

- C.** Attach an Internet Gateway to the VPC and update the route table

Explanation:

To enable internet access for an EC2 instance via IPv4, you must:

- Attach an Internet Gateway (IGW) to your VPC
- Ensure the route table for the subnet has a route to the IGW
- Assign a public IP or Elastic IP to the instance
- Configure security groups and NACLs to allow relevant traffic

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Internet_Gateway.html

Incorrect Options Explained:

- **A.** IAM roles control permissions to AWS services, not internet connectivity.
- **B.** CloudTrail logs are for auditing, not for managing internet access.
- **D.** Auto Scaling helps manage instance count, but does not provide internet access.

Q23.

What is the primary function of an Egress-Only Internet Gateway in an Amazon VPC?

- A.** To allow both inbound and outbound IPv6 traffic between the Internet and your VPC
- B.** To provide outbound-only IPv6 access from your VPC to the internet, blocking inbound connections
- C.** To route IPv4 traffic through a NAT instance for private subnets
- D.** To enable DNS resolution for instances in the VPC

Correct Answer:

B. To provide outbound-only IPv6 access from your VPC to the internet, blocking inbound connections

Explanation:

An Egress-Only Internet Gateway (EOIGW) enables outbound-only IPv6 traffic from instances in your VPC to the internet. It prevents the internet from initiating IPv6 connections to your instances, thereby enhancing security for IPv6-enabled environments.

<https://docs.aws.amazon.com/vpc/latest/userguide/egress-only-internet-gateway.html>

Incorrect Options Explained:

- A. Egress Only Internet Gateways do not allow inbound traffic—only outbound is permitted.
- C. That describes a NAT gateway/instance, which is for IPv4, not IPv6.
- D. DNS resolution is managed by AmazonProvidedDNS or Route 53, not an Egress Only Internet Gateway.

Q24.

Which of the following statements accurately describes the behavior and configuration requirements of a NAT Gateway in AWS?

- A.** It allows inbound connections from the internet to instances in a private subnet
- B.** It must be deployed in a private subnet and requires a public IP to function
- C.** It enables instances in private subnets to initiate internet-bound traffic using a public subnet's Elastic IP
- D.** It supports VPC peering and VPN traffic routing by default

Correct Answer:

C. It enables instances in private subnets to initiate internet-bound traffic using a public subnet's Elastic IP

Explanation:

A NAT Gateway is placed in a public subnet, where it uses an Elastic IP to allow outbound-only internet access for instances in private subnets. It blocks inbound connections, supports TCP, UDP, and ICMP, and is commonly used to access AWS services or download software updates.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html>

Incorrect Options Explained:

- A. NAT Gateways do not allow inbound connections from the Internet.
- B. A NAT Gateway must be deployed in a public subnet, not a private one.
- D. NAT Gateways cannot route traffic over VPC endpoints, VPNs, Direct Connect, or peering connections.

Q25.

Which of the following is true about Elastic IP addresses (EIPs) in AWS?

- A. They are automatically released when the associated instance is stopped
- B. You are charged extra only when the EIP is actively used by a running instance
- C. EIPs are static public IPv4 addresses that remain allocated to your account until you release them
- D. You can associate an Elastic IP with multiple instances at the same time

Correct Answer:

C. EIPs are static public IPv4 addresses that remain allocated to your account until you release them

Explanation:

Elastic IPs (EIPs) are static public IPv4 addresses provided by AWS. They remain associated with your account until you explicitly release them, regardless of whether they're in use or not. AWS charges you when an EIP is not associated with a running instance or is attached to a stopped instance.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/elastic-ip-addresses-eip.html>

Incorrect Options Explained:

- A. EIPs are not automatically released when the instance stops—you must release them manually.
- B. Charges apply when the EIP is not associated with a running instance—not when actively used.
- D. An EIP can be associated with only one instance or network interface at a time.

Q26.

In AWS Systems Manager Automation, what is a step?

- A.** A backup of the Automation document stored in S3
- B.** A user permission policy applied to Automation executions
- C.** A single initiated action performed during an Automation execution, usually per target
- D.** A log file is automatically generated after the Automation completion

Correct Answer:

- C.** A single initiated action performed during an Automation execution, usually per target

Explanation:

A step in Systems Manager Automation represents a discrete action executed as part of an Automation document (runbook). You can choose to run all steps simultaneously or execute them step-by-step, and each step typically targets a specific managed instance or resource.

<https://docs.aws.amazon.com/systems-manager/latest/userguide/systems-manager-automation.html>

Incorrect Options Explained:

- A. A step is not a backup; it's an execution unit in the automation workflow.
- B. IAM policies handle permissions, not step definitions.
- D. Steps generate logs, but are not logs themselves.

Q27.

What is the primary purpose of using **Resource Groups** in AWS Systems Manager?

- A.** To group IAM users based on roles and policies
- B.** To manage and monitor a collection of resources in the same AWS Region that match defined criteria
- C.** To create backup copies of resources across multiple AWS Regions
- D.** To enforce security rules at the subnet level in a VPC

Correct Answer:

B. To manage and monitor a collection of resources in the same AWS Region that match defined criteria

Explanation:

Resource Groups allow you to organize AWS resources (like EC2, S3, Lambda, etc.) that share common tags or attributes within the same region. You can then apply Systems Manager tools like Automation, Compliance, and Monitoring on the group as a unit.

<https://docs.aws.amazon.com/ARG/latest/userguide/welcome.html>

Incorrect Options Explained:

- A. IAM groups manage users, not AWS infrastructure resources.
- C. Resource Groups don't manage backups or cross-region replication.
- D. Subnet-level security is handled using Security Groups and Network ACLs, not Resource Groups.

Q28.

Which of the following statements is **true** about AWS access keys used for programmatic access?

- A.** An access key can be used to log in to the AWS Management Console
- B.** If the secret access key is lost, it can be retrieved from the IAM dashboard
- C.** Access keys consist of an access key ID and a secret access key used for API requests
- D.** Access keys can be shared with team members to simplify access management

Correct Answer:

C. Access keys consist of an access key ID and a secret access key used for API requests

Explanation:

Access keys are used for programmatic access to AWS services via the CLI, SDKs, or API. Each access key includes two parts:

- Access Key ID (visible)
- Secret Access Key (only shown once at creation)

If the secret key is lost, it cannot be retrieved—you must regenerate a new pair.

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_access-keys.html

Incorrect Options Explained:

- A. Console logins use username and password, not access keys.
- B. You cannot retrieve a lost secret key—only regenerate it.
- D. Access keys are user-specific credentials and must not be shared.

Q29.

Which of the following is a **key feature** of Amazon Athena?

- A.** It requires provisioning EC2 instances to run queries
- B.** It is serverless and includes a built-in SQL query editor
- C.** It only supports querying data stored in DynamoDB
- D.** It is used for managing IAM roles and permissions

Correct Answer:

B. It is serverless and includes a built-in SQL query editor

Explanation:

Amazon Athena is a serverless interactive query service that lets you analyze data in Amazon S3 using standard SQL. It includes a web-based query editor, so you can run queries directly from the AWS Management Console without setting up infrastructure.

<https://docs.aws.amazon.com/athena/latest/ug/what-is.html>

Incorrect Options Explained:

- A. Athena is serverless—you do not need to provision or manage servers.
- C. Athena supports S3-based datasets, not just DynamoDB.
- D. IAM role and permission management is handled via AWS IAM, not Athena.

Q30.

What is the primary purpose of **AWS Budgets**?

- A.** To manage VPC routing and subnet allocations
- B.** To create IAM roles and assign user permissions
- C.** To monitor and control AWS usage and costs by setting spending limits
- D.** To schedule automatic backups of EC2 instances

Correct Answer:

- C.** To monitor and control AWS usage and costs by setting spending limits

Explanation:

AWS Budgets allows you to track your AWS usage, cost, and reserved instance utilization by setting custom budget thresholds. You can receive notifications when your spending approaches or exceeds the defined limits.

<https://docs.aws.amazon.com/cost-management/latest/userguide/budgets-managing-costs.html>

Incorrect Options Explained:

- A. VPC routing is handled in the VPC console, not through Budgets.
- B. IAM is used to manage roles and permissions, not costs.
- D. EC2 backups are managed via AWS Backup or custom scripts, not AWS Budgets.

Q31.

What is the primary purpose of AWS Billing Conductor?

- A. To encrypt billing data using customer-managed keys
- B. To create IAM roles for managing budget thresholds
- C. To group accounts for a consolidated view of cost and usage
- D. To automatically delete unused resources to save costs

Correct Answer:

- C. To group accounts for a consolidated view of cost and usage

Explanation:

AWS Billing Conductor allows you to assign linked accounts to billing groups, providing a customized and aggregated view of cost and usage across your workloads. This is especially helpful for large organizations managing multiple accounts.

<https://docs.aws.amazon.com/billingconductor/latest/userguide/what-is-billing-conductor.html>

Incorrect Options:

- A. Encryption of billing data is handled by AWS services, but that's not Billing Conductor's core function.
- B. IAM roles are managed through IAM, not via Billing Conductor.
- D. Billing Conductor **does not delete resources**—it tracks and groups cost data.

Q32.

Which of the following is **true** when using AWS Billing Conductor?

- A. Each linked account can belong to multiple billing groups simultaneously
- B. Billing Conductor provides an aggregated view of monthly cost data for assigned account groups
- C. Billing Conductor automatically applies service quotas based on billing group usage
- D. You must enable Billing Conductor for each AWS Region separately

Correct Answer:

- B. Billing Conductor provides an aggregated view of monthly cost data for assigned account groups

Explanation:

Billing Conductor gives aggregated, customizable billing views for accounts assigned to a billing group. This helps organizations track costs more effectively by internal departments or projects.

<https://docs.aws.amazon.com/billingconductor/latest/userguide/what-is-billing-conductor.html>

Incorrect Options:

- A. An account can belong to only one billing group at a time.
- C. Billing Conductor does not manage quotas; it's focused on billing and cost tracking.
- D. Billing Conductor is a global service, not regional.

Q33.

When configuring a cost and usage report in AWS, what does selecting a time granularity of "Daily" mean?

- A.** The report will display total usage only for weekends
- B.** The report will break down cost and usage data by each individual day
- C.** The report will update every minute with real-time usage
- D.** The report will only include daily charges from EC2 instances

Correct Answer:

- B.** The report will break down cost and usage data by each individual day

Explanation:

Time granularity determines how the data in your cost and usage report is aggregated. Choosing "Daily" means you will receive cost and usage data broken down by day, helping you analyze daily trends and spending patterns.

<https://docs.aws.amazon.com/cur/latest/userguide/data-columns.html#time-granularity>

Incorrect Options Explained:

- A. The report includes all days, not just weekends.
- C. Real-time reporting is not supported; CUR updates are typically delayed.

- D. The report includes all AWS services, not just EC2, unless specifically filtered.

Q34.

Why does AWS ensure that Availability Zones within a Region are located in the same country?

- A.** To reduce storage costs by limiting international traffic
- B.** To improve instance launch speed across continents
- C.** To comply with data sovereignty laws requiring sensitive data to remain within a specific country
- D.** To ensure that all AWS services are automatically replicated worldwide

Correct Answer:

C. To comply with data sovereignty laws requiring sensitive data to remain within a specific country

Explanation:

Data sovereignty refers to the legal requirement that certain data must be stored within the boundaries of a particular country. AWS designs its Availability Zones within a Region to be geographically separated (for fault tolerance) but still within the same country to help meet compliance requirements related to data privacy and storage.

<https://aws.amazon.com/about-aws/global-infrastructure/>

Incorrect Options Explained:

- A. Cost savings are a factor, but data sovereignty is the main reason for same-country AZs.
- B. Launch speed is improved by proximity, but compliance is the key factor here.
- D. AWS services are not automatically replicated globally; replication must be configured.

Q35.

What is the main benefit of using a Content Delivery Network (CDN) such as Amazon CloudFront?

- A.** It encrypts all traffic within a private VPC by default
- B.** It stores user credentials in regional edge caches

- C. It delivers content faster by caching it at edge locations closer to users
- D. It provisions compute resources automatically across multiple regions

Correct Answer:

- C. It delivers content faster by catching it at edge locations closer to users

Explanation:

A CDN, like Amazon CloudFront, uses a network of edge locations and regional edge caches to serve content faster to users by caching it closer to their geographic location. This reduces latency and improves the overall performance of web applications.

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Introduction.html>

Incorrect Options Explained:

- A. VPC-level encryption is unrelated to CDN functionality.
- B. CDNs do not store credentials—only static and dynamic content.
- D. CDNs do not provision compute resources—that's handled by services like EC2 or Lambda.

Q36.

What is the purpose of having multiple Availability Zones within an AWS Region?

- A. To ensure higher data compression for storage resources
- B. To offer high availability, fault tolerance, and scalability through physically separated data centers
- C. To allow users to deploy resources across different AWS accounts
- D. To enable cross-account billing and automated backups

Correct Answer:

- B. To offer high availability, fault tolerance, and scalability through physically separated data centers

Explanation:

Each AWS Region consists of multiple Availability Zones, which are physically separated data centers with redundant power, networking, and connectivity. These zones are interconnected with low-latency, high-throughput networks, making it possible to build highly available and fault-tolerant applications.

https://aws.amazon.com/about-aws/global-infrastructure/regions_az/

Incorrect Options Explained:

- A. Data compression is unrelated to the purpose of Availability Zones.
- C. AZs are used for infrastructure resilience, not account separation.
- D. Cross-account billing and backups are managed by other AWS services, not through AZs.

Q37.

Which AWS infrastructure component is used by CloudFront to cache content closer to end-users for low-latency delivery?

- A. AWS Local Region
- B. AWS Local Zone
- C. Edge Location
- D. Availability Zone

Correct Answer:

- C. Edge Location

Explanation:

Edge locations are part of AWS's global network of Points of Presence (PoPs) and are used by Amazon CloudFront to cache and deliver content to users with low latency. These locations are geographically distributed and help serve content faster from the nearest possible point.

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Introduction.html>

Incorrect Options Explained:

- A. A Local Region is a full AWS datacenter but not specifically used for content caching.
- B. A Local Zone brings services closer to cities, but it's not used for content delivery in CloudFront.
- D. Availability Zones are part of standard AWS Regions, not PoPs for caching.

Q38.

What is one benefit of Amazon Aurora's cloning feature?

- A. It automatically converts the database engine to PostgreSQL during cloning
- B. It allows quick and space-efficient duplication of a database, including sharing across AWS accounts
- C. It disables backups on the cloned database to improve performance
- D. It requires manual export and import of all database objects to perform cloning

Correct Answer:

B. It allows quick and space-efficient duplication of a database, including sharing across AWS accounts

Explanation:

Amazon Aurora supports fast and efficient cloning of a database cluster. Clones use copy-on-write technology, which minimizes storage usage and speeds up the process. Aurora also allows sharing cloned databases with other AWS accounts, making testing and development easier.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/aurora-cloning.html>

Incorrect Options Explained:

- A. Aurora does not auto-convert engines during cloning—you must choose the same engine.
- C. Backups are not disabled during cloning; normal backup operations continue.
- D. Aurora cloning does not require manual export/import; it's an automated process.

Q39.

What is a key characteristic of an Amazon Aurora cluster volume?

- A. It stores data only in the primary Availability Zone to reduce replication overhead
- B. It is a virtual storage volume that spans multiple Availability Zones with automatic data replication
- C. It requires manual configuration to replicate data between regions
- D. It can only support a single database engine per Region

Correct Answer:

B. It is a virtual storage volume that spans multiple Availability Zones with automatic data replication

Explanation:

An Aurora cluster volume is a virtual storage layer designed for high availability and durability. It automatically replicates data across multiple Availability Zones in the same Region to ensure fault tolerance and fast recovery.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/aurora-storage.html>

Incorrect Options Explained:

- A. Aurora replicates data across multiple AZs, not just one.
- C. Cross-AZ replication is automatic—you do not configure it manually.
- D. Aurora supports multiple engines (MySQL & PostgreSQL) within the same region.

Q40.

How does **Amazon Aurora** handle a failure in one of the disk volume segments that make up a cluster volume?

- A.** It automatically fails over to a new Region and deletes the failed volume
- B.** It stops the database cluster until the failed segment is manually replaced
- C.** It detects the failure, repairs the segment automatically, and uses data from other volumes to restore consistency
- D.** It sends a snapshot of the failed volume to S3 for archival and shuts down the cluster

Correct Answer:

C. It detects the failure, repairs the segment automatically, and uses data from other volumes to restore consistency

Explanation:

Amazon Aurora is designed for high availability and fault tolerance. When a disk segment in the cluster volume fails, Aurora automatically repairs it using redundant data from other copies in the cluster volume. This ensures the repaired segment contains current and consistent data, without requiring manual intervention.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/aurora-storage.html>

Incorrect Options Explained:

- A. Aurora does not need to migrate the database just for a disk failure—storage is already redundant across AZs.
- B. The engine does not restart or roll back due to storage segment failures.
- D. Aurora does not require manual snapshots or halt writes—repairs happen automatically and seamlessly.

Q41.

What is one of the main benefits of creating Aurora Replicas across multiple Availability Zones?

- A.** It allows automatic encryption of data at rest
- B.** It enables synchronous replication to reduce latency and improve backup reliability
- C.** It prevents the need for database snapshots during backups
- D.** It disables read traffic to ensure high write performance

Explanation:

When Aurora Replicas are created in different Availability Zones, Amazon RDS synchronously replicates data from the primary DB instance. This provides high availability, data redundancy, and helps minimize I/O freezes and latency spikes during backups, making the system more resilient.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/aurora-architecture.html#aurora-storage>

Incorrect Options Explained:

- A. Aurora does support encryption, but it's not related to the benefit of cross-AZ replication.
- C. Snapshots are still used for backups; replication helps improve performance during backups but doesn't replace snapshots.
- D. Read traffic is not disabled; Aurora Replicas are designed to offload read traffic from the primary instance.

Q42.

How does Amazon Aurora ensure high availability and data durability across Availability Zones?

- A.** By replicating data to Amazon S3 in every Region
- B.** By maintaining 6 copies of data across 3 Availability Zones and automatically recovering in a healthy zone if needed
- C.** By creating daily snapshots and distributing them across Regions
- D.** By running only a single database instance to reduce complexity

Correct Answer:

- B.** By maintaining 6 copies of data across 3 Availability Zones and automatically recovering in a healthy zone if needed

Explanation:

Amazon Aurora automatically maintains six copies of your data, with two copies in each of three different Availability Zones. If there's a failure in one zone, Aurora can automatically recover using healthy copies from another zone without data loss, providing high availability and fault tolerance.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/aurora-storage.html>

Incorrect Options Explained:

- A. Aurora does not replicate to S3 for database durability—S3 is a separate storage service.
- C. Snapshots are useful for backups but do not provide real-time fault tolerance.

- D. Aurora relies on replication and redundancy, not single-instance deployments, for high availability.

Q43.

Which of the following is true about instance store volumes in Amazon EC2?

- A.** They retain data even after the instance is stopped
- B.** They can be detached and reattached to other instances like EBS volumes
- C.** They are temporary storage volumes that are deleted when the instance is stopped or terminated
- D.** They support snapshot backups and are automatically encrypted

Correct Answer:

C. They are temporary storage volumes that are deleted when the instance is stopped or terminated

Explanation:

Instance store volumes provide temporary block-level storage for an EC2 instance. The data is lost when the instance is stopped, terminated, or fails, making it suitable only for ephemeral data. In contrast, EBS-backed instances can be stopped and restarted without data loss.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/InstanceStorage.html>

Incorrect Options Explained:

- **A.** Instance store data is not persistent—it is deleted when the instance is stopped or terminated.
- **B.** Instance store volumes cannot be detached or reattached; they are tied to the host hardware.
- **D.** Instance store volumes do not support snapshots and are not automatically encrypted.

Q44.

Which of the following statements best describes Instance Store-backed EC2 instances?

- A.** They allow you to stop and restart the instance without losing data
- B.** They automatically back up all data to Amazon S3 on shutdown

- C. They lose all data on the instance store volumes if the instance is terminated or fails
- D. They support EBS volumes for persistent storage by default

Correct Answer:

- C. They lose all data on the instance store volumes if the instance is terminated or fails

Explanation:

Instance Store-backed EC2 instances use ephemeral storage directly attached to the physical host. This storage is not persistent, so any data stored is lost upon termination, failure, or hardware issues. Therefore, it's recommended to back up critical data regularly to persistent storage, like Amazon S3 or EBS.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/InstanceStorage.html>

Incorrect Options Explained:

- A. Instance store-backed instances cannot be stopped—only started or terminated.
- B. AWS does not automatically back up instance store data—you must do it manually.
- D. EBS volumes are not attached by default to instance store-backed instances; they're a separate storage type.

Q45.

Which AWS service is fully managed and is used to automate the creation, management, and deployment of Amazon Machine Images (AMIs)?

- A. AWS CodeDeploy
- B. Amazon EC2 Auto Scaling
- C. EC2 Image Builder
- D. AWS Systems Manager Session Manager

Correct Answer:

- C. EC2 Image Builder

Explanation:

EC2 Image Builder is a fully managed AWS service that simplifies the process of automating the creation, patching, testing, and distribution of Amazon Machine Images (AMIs). It helps ensure that your images are secure, up-to-date, and compliant with organizational standards.

<https://docs.aws.amazon.com/imagebuilder/latest/userguide/what-is-image-builder.html>

Incorrect Options Explained:

- A. AWS CodeDeploy is used for automated application deployments, not AMI creation.
- B. EC2 Auto Scaling launches instances from existing AMIs but does not manage or create them.
- D. Session Manager provides secure shell access to EC2 instances, not image creation capabilities.

Q46.

What is a key benefit of using Reserved Instances in Amazon EC2?

- A. They automatically scale based on incoming traffic without user configuration
- B. They allow you to pause billing when the instance is idle
- C. They offer a lower hourly rate in exchange for committing to a one- or three-year term
- D. They support instant switching between Regions without reconfiguration

Correct Answer:

- C. They offer a lower hourly rate in exchange for committing to a one- or three-year term

Explanation:

Reserved Instances (RIs) allow you to make a low, upfront payment and commit to using an instance for a 1-year or 3-year term, in return for a significantly reduced hourly rate compared to On-Demand pricing. AWS offers Standard and Convertible RI classes, with Convertible RIs allowing instance type changes

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/reserved-instances-types.html>

Incorrect Options Explained:

- A. Auto scaling is handled by Auto Scaling Groups, not Reserved Instances.

- B. Billing for RIs is continuous over the term; there is no pause.
- D. RIs are tied to a specific Region, and switching requires modification or exchange in case of Convertible RIs.

Q47.

Which of the following best describes data durability in services like Amazon S3?

- A.** Data is stored in a single Availability Zone for faster retrieval
- B.** Data is replicated across multiple Availability Zones to achieve 99.999999999% durability
- C.** Durability ensures that data can only be accessed by the root user
- D.** Durability guarantees that deleted data can be recovered for 30 days

Correct Answer:

B. Data is replicated across multiple Availability Zones to achieve 99.999999999% durability

Explanation:

Durability measures how reliably your data is stored over time. Amazon S3 achieves 11 nines (99.999999999%) durability by automatically replicating data across multiple Availability Zones, ensuring it is not lost even in the case of component or AZ failures.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/DataDurability.html>

Incorrect Options Explained:

- **A.** Data is stored in a single Availability Zone for faster retrieval
This is incorrect because durability relies on multi-AZ replication. Storing data in a single AZ increases the risk of loss in case of failure.
- **C.** Durability ensures that data can only be accessed by the root user
This is false. Access control is managed by IAM policies and bucket permissions, not by durability.
- **D.** Durability guarantees that deleted data can be recovered for 30 days
This is incorrect. Once data is deleted (unless versioning or a lifecycle policy like S3 Object Lock is used), durability does not apply. Durability is about protecting data while it exists, not restoring deleted items.

Q48.

Which of the following best describes the regional nature of Amazon ECS (Elastic Container Service)?

- A.** ECS resources are automatically replicated across all AWS Regions
- B.** ECS is a regional service, meaning clusters and tasks are specific to the AWS Region in which they are created
- C.** ECS tasks run globally and are not tied to any specific Region
- D.** ECS uses Availability Zones instead of Regions to manage container placement

Correct Answer:

B. ECS is a regional service, meaning clusters and tasks are specific to the AWS Region in which they are created.

Explanation:

Amazon ECS is a regional service. This means that clusters, services, and tasks you create in ECS are scoped to the Region in which you provision them. Resources are not automatically replicated across Regions, and you must configure deployments separately for each Region if needed

<https://docs.aws.amazon.com/AmazonECS/latest/developerguide/what-is-ecs.html>

Incorrect Options Explained:

- **A.** ECS does not replicate resources across Regions automatically; this must be done manually.
- **C.** ECS tasks are tied to the Region in which the ECS cluster resides.
- **D.** ECS does consider Availability Zones for task placement within a Region, but the service is defined at the regional level, not zonal.

Q49.

Which of the following statements best describes the role of a container image in a containerized application?

- A. It is a temporary file system that stores application logs during runtime
- B. It is a read-only template that includes everything needed to run an application
- C. It is a database snapshot used for container storage
- D. It stores environment variables for runtime configuration only

Correct Answer:

- B. It is a read-only template that includes everything needed to run an application

Explanation:

A container image is a read-only template used to create containers. It includes everything required to run the application: the application code, runtime, system tools, system libraries, and dependencies. Containers launched from the image use this consistent environment for execution.

<https://docs.aws.amazon.com/whitepapers/latest/overview-containers/overview-containers.html>

Incorrect Options Explained:

- A. The file system during runtime is writable, but the image itself is read-only and doesn't store logs.
- C. Container images are not database snapshots; they define the environment for the app.
- D. While containers may use environment variables, images contain the full application stack, not just configs.

Q50.

What is a key feature of Amazon Simple Notification Service (SNS)?

- A. It stores large volumes of structured data for analytics
- B. It uses the publish-subscribe model and pushes notifications to subscribers
- C. It provides a database engine for mobile app backends
- D. It requires clients to poll regularly for new messages

Correct Answer:

- B. It uses the publish-subscribe model and pushes notifications to subscribers

Explanation:

Amazon SNS is a fully managed messaging service that supports the publish-subscribe

(pub-sub) model. Publishers send messages to a topic, and subscribers (such as email, SMS, Lambda, or SQS endpoints) receive the messages via a push mechanism, eliminating the need for polling.

<https://docs.aws.amazon.com/sns/latest/dg/welcome.html>

Incorrect Options Explained:

- A. SNS is not designed for data storage or analytics—it's a messaging service.
- C. SNS is not a database; it's used for event-driven notifications.
- D. SNS uses push notifications, not polling.

Q51.

Which AWS service allows you to send push notifications such as email and SMS messages to individuals or groups?

- A.** Amazon S3
- B.** Amazon EC2
- C.** Amazon SNS
- D.** Amazon RDS

Correct Answer:

- C.** Amazon SNS

Explanation:

Amazon Simple Notification Service (SNS) is a fully managed messaging service that supports push-based delivery of messages via email, SMS (text messaging), mobile push, and more. It allows messages to be sent to individuals or entire groups through topics.

<https://docs.aws.amazon.com/sns/latest/dg/sns-email-notifications.html>

Incorrect Options Explained:

- A. Amazon S3 is a storage service, not a messaging system.
- B. Amazon EC2 provides compute resources, not messaging.
- D. Amazon RDS is used for relational databases, not for sending notifications.

Q52.

Which AWS service allows you to configure automatic scaling using scaling plans that include both dynamic and predictive scaling?

- A. AWS Auto Scaling
- B. AWS CloudTrail
- C. Amazon EC2
- D. AWS Config

Correct Answer:

A. AWS Auto Scaling

Explanation:

AWS Auto Scaling enables you to automatically scale your AWS resources (such as EC2 instances, ECS services, DynamoDB tables, etc.) using scaling plans. These plans can include both dynamic scaling (based on real-time demand) and predictive scaling (based on historical trends and machine learning).

<https://docs.aws.amazon.com/autoscaling/plans/userguide/what-is-aws-auto-scaling.html>

Incorrect Options Explained:

- B. AWS CloudTrail is used for logging API activity, not scaling resources.
- C. Amazon EC2 provides compute instances, but does not handle scaling plans directly.
- D. AWS Config tracks resource configuration history, not scaling.

Q53.

What is the main purpose of an Auto Scaling Group in Amazon EC2?

- A. To assign unique IAM roles to each instance in your VPC
- B. To group EC2 instances with different configurations for auditing
- C. To manage a group of EC2 instances with similar characteristics for automated scaling and health monitoring
- D. To migrate EC2 instances between Regions for disaster recovery

Correct Answer:

C. To manage a group of EC2 instances with similar characteristics for automated scaling and health monitoring

Explanation:

An Auto Scaling Group (ASG) allows you to define and manage a collection of EC2 instances that share the same configuration. It helps automatically scale the number of instances up or down based on demand and ensures that healthy instances are maintained according to the desired count.

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/AutoScalingGroup.html>

Incorrect Options Explained:

- A. IAM roles are not managed at the group level in an ASG.
- B. ASGs manage instances with similar, not different, configurations.
- D. ASGs are region-specific and do not handle cross-region migrations.

Q54.

What is the purpose of a launch template in an Auto Scaling Group?

- A.** To configure security group rules for the entire VPC
- B.** To automate the patching of EC2 instances across Regions
- C.** To define configuration details such as AMI ID, instance type, and security groups for EC2 instances
- D.** To replicate EC2 instances into S3 buckets for backup

Correct Answer:

C. To define configuration details such as AMI ID, instance type, and security groups for EC2 instances

Explanation:

A launch template is used to define the EC2 instance configuration—including the Amazon Machine Image (AMI), instance type, key pair, block device mappings, and security groups.

It allows consistent, repeatable deployment of instances, especially in Auto Scaling Groups.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-launch-templates.html>

Incorrect Options Explained:

- A. Security group rules are managed separately, not through launch templates.
- B. Launch templates don't automate patching; AWS Systems Manager is used for that.
- D. EC2 instances aren't replicated into S3 via launch templates; Amazon Machine Images (AMIs) are used for backup purposes.

Q55.

What is a launch configuration used for in an Amazon EC2 Auto Scaling group?

- A.** To schedule instance backups across multiple regions
- B.** To monitor the performance of running EC2 instances
- C.** To define the configuration details for EC2 instances launched by the Auto Scaling group
- D.** To apply IAM policies to users accessing the EC2 Dashboard

Correct Answer:

- C.** To define the configuration details for EC2 instances launched by the Auto Scaling group

Explanation:

A launch configuration is a template that defines how EC2 instances should be configured when launched by an Auto Scaling group. It includes settings such as the AMI ID, instance type, key pair, security groups, and block device mapping. Although AWS now recommends launch templates, launch configurations are still supported.

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/LaunchConfiguration.html>

Incorrect Options Explained:

- A. Backup scheduling is not handled by launch configurations.
- B. Monitoring is done through CloudWatch, not launch configurations.
- D. IAM policy application is unrelated to instance configuration templates.

Q56.

What is the primary function of CloudTrail Lake in AWS?

- A.** It provides real-time monitoring for EC2 instance metrics
- B.** It allows aggregation, immutable storage, and querying of activity logs
- C.** It automatically deletes old logs to save storage space
- D.** It replicates IAM roles across AWS accounts

Correct Answer:

- B.** It allows aggregation, immutable storage, and querying of activity logs

Explanation:

CloudTrail Lake is a managed data lake feature in AWS CloudTrail that enables users to aggregate, store immutably, and query activity logs across AWS accounts and Regions. It supports querying through the AWS SDK, CLI, and console, making it easier to analyze user and service activity.

<https://docs.aws.amazon.com/awscloudtrail/latest/userguide/cloudtrail-lake.html>

Incorrect Options Explained:

- A. Real-time monitoring of instance metrics is handled by Amazon CloudWatch, not CloudTrail Lake.
- C. CloudTrail Lake retains logs immutably; they are not auto-deleted for cost savings.
- D. IAM role replication across accounts is not a feature of CloudTrail Lake.

Q57.

Which of the following is true about creating an organization trail in AWS CloudTrail?

- A.** It can be created by any account within the AWS Organization
- B.** It only logs events from the management account by default
- C.** It must be created in the management account and logs events across all member accounts
- D.** It is limited to a single Availability Zone

Correct Answer:

C. It must be created in the management account and log events across all member accounts

Explanation:

An organization trail in AWS CloudTrail allows you to log all events across all accounts in an AWS Organization. However, only the management account has the permission to create and manage organization trails. Once created, it captures events from every account in the organization.

<https://docs.aws.amazon.com/awscloudtrail/latest/userguide/creating-trail-organization.html>

Incorrect Options Explained:

- A. Only the management account can create an organization trail.
- B. It captures events from all accounts, not just the management account.
- D. Trails are global or regional, not tied to an Availability Zone.

Q58.

What happens automatically in Amazon Route 53 after you successfully register a domain name?

- A.** A private hosted zone is created and linked to your VPC
- B.** The domain is locked for editing for 30 days
- C.** A public hosted zone is created with the same name as the registered domain
- D.** An SSL certificate is issued for the domain

Correct Answer:

C. A public hosted zone is created with the same name as the registered domain

Explanation:

When you register a domain using Amazon Route 53, it automatically creates a public hosted zone with the same name as the domain. This hosted zone includes default DNS records (such as name servers and start of authority) to help you begin managing DNS immediately.

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/domain-register.html>

Incorrect Options Explained:

- A. Private hosted zones must be created manually and are used with Amazon VPCs, not automatically set up on domain registration.
- B. There is no 30-day lock for editing DNS settings after domain registration in Route 53.
- D. SSL certificates are managed separately (e.g., via AWS Certificate Manager), not automatically issued upon registration.

Q59.

What is the purpose of an alias record in Amazon Route 53?

- A.** To create temporary DNS redirects for testing
- B.** To route traffic to non-AWS domains using CNAME records
- C.** To route traffic directly to AWS resources like S3 buckets or CloudFront distributions
- D.** To encrypt DNS queries between hosted zones

Correct Answer:

- C.** To route traffic directly to AWS resources like S3 buckets or CloudFront distributions

Explanation:

Alias records in Route 53 are special types of DNS records that let you route traffic to AWS resources, such as Amazon S3 buckets, CloudFront distributions, Elastic Load Balancers, and API Gateway endpoints, without needing to use an IP address or a CNAME at the zone apex.

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-to-aws-resources.html>

Incorrect Options Explained:

- A. Alias records are not for temporary use or testing; they're used for production routing.
- B. Alias records are used for AWS-managed destinations, not external domains.

- D. DNS encryption is not handled by alias records but by services like DNS over HTTPS (DoH) or TLS at the application layer.

Q60.

Which statement best describes the Simple Routing Policy in Amazon Route 53?

- A.** It routes traffic based on user location to the nearest endpoint
- B.** It splits traffic randomly across multiple AWS Regions
- C.** It routes internet traffic to a single resource and allows multiple values in one record, such as multiple IPs
- D.** It automatically adjusts traffic routing based on resource health checks

Correct Answer:

C. It routes internet traffic to a single resource and allows multiple values in one record, such as multiple IPs

Explanation:

The Simple Routing Policy is the most basic routing type in Route 53. It routes traffic to a single resource, such as a web server or an S3 website endpoint. While you cannot create multiple records with the same name and type, you can provide multiple values (like multiple IPs) in the same record.

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy.html#routing-policy-simple>

Incorrect Options Explained:

- A. That describes Geolocation or Latency-based routing, not simple routing.
- B. Weighted routing policy is used to split traffic, not simple routing.
- D. Health checks can be associated with simple routing, but failover routing is specifically designed for health-based traffic redirection.

Q61.

Which of the following programming languages are **natively supported by AWS Lambda**?
(Choose all that apply)

- A.** Node.js
- B.** Java
- C.** C#

- D. PHP
- E. Go
- F. Python
- G. Ruby
- H. PowerShell

Correct Answers:

- A. Node.js
- B. Java
- C. C#
- E. Go
- F. Python
- G. Ruby
- H. PowerShell

Explanation:

AWS Lambda provides **native runtime support** for the following languages:

- **Node.js**
- **Java**
- **C#** (via .NET Core)
- **Go**
- **Python**
- **Ruby**
- **PowerShell** (via .NET Core for Windows scripting)

PHP is not natively supported, but can be run using custom runtimes via AWS Lambda's [Lambda Runtime API](#).

<https://docs.aws.amazon.com/lambda/latest/dg/lambda-runtimes.html>

Q62.

What best describes a function in AWS Lambda?

- A. A container image that manages VPC networking
- B. A monitoring tool for tracking AWS usage and billing

- C. A script or program that processes events and returns a response
- D. A physical server that runs user-defined software

Correct Answer:

- C. A script or program that processes events and returns a response

Explanation:

In AWS Lambda, a function is your application logic—a script or program written in a supported language (e.g., Python, Node.js, Java). When Lambda is invoked, it passes an event to the function, which executes the code and optionally returns a response.

<https://docs.aws.amazon.com/lambda/latest/dg/welcome.html>

Incorrect Options Explained:

- A. Lambda functions may use containers, but this refers to deployment packaging, not the function itself.
- B. AWS Billing and Cost Explorer tools monitor usage—not Lambda.
- D. Lambda is a serverless service, meaning users don't manage physical servers.

Q63.

Which of the following are true about AWS Lambda layers? (*Choose 2 answers*)

- A. Layers allow you to package and share libraries or dependencies across multiple Lambda functions
- B. Layers are used to store function execution logs in CloudWatch
- C. Layers help separate in-development code from stable dependencies
- D. Layers are required for running Lambda functions in VPCs
- E. Layers automatically replicate function code across multiple Regions

Correct Answers:

- A. Layers allow you to package and share libraries or dependencies across multiple Lambda functions
- C. Layers help separate in-development code from stable dependencies

Explanation:

- Lambda layers are a mechanism for sharing libraries, custom runtimes, and dependencies across multiple Lambda functions.
- They help manage code more efficiently by separating changing function code from stable or reusable components.

Incorrect Options Explained:

- B. Logs are stored in CloudWatch, but layers are not used for logging.
- D. Layers are not required for VPC access; VPC configuration is separate.
- E. Layers are Region-specific and do not automatically replicate across Regions.

Q64.

You are setting up a site-to-site VPN to securely connect your on-premises data center to your AWS VPC. Which component should you attach to your VPC to act as the VPN concentrator on the AWS side of the connection?

- A.** Internet Gateway
- B.** NAT Gateway
- C.** Virtual Private Gateway
- D.** VPC Peering Connection

Correct Answer:

- C.** Virtual Private Gateway

Explanation:

A Virtual Private Gateway (VGW) is the VPN endpoint on the AWS side of a site-to-site VPN. It enables secure communication between your on-premises data center and your VPC over an IPsec VPN tunnel.

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_VPN.html

Incorrect Options Explained:

- A. Internet Gateway allows public internet access, not VPN connectivity.
- B. NAT Gateway enables instances in a private subnet to access the internet, not for VPN use.
- D. VPC Peering is used for connecting two VPCs, not on-premises networks.

Q65.

What is the primary use of **AWS PrivateLink**?

- A.** To connect VPCs across Regions using public IPs
- B.** To privately connect your VPC to AWS services and partner services without using public internet
- C.** To manage IAM policies across multiple AWS accounts
- D.** To automatically assign public IP addresses to EC2 instances in private subnets

Correct Answer:

B. To privately connect your VPC to AWS services and partner services without using public internet

Explanation:

AWS PrivateLink provides private connectivity between your VPC and supported AWS services, services in other AWS accounts, or AWS Marketplace partner services without exposing traffic to the public internet. It uses interface VPC endpoints powered by Elastic Network Interfaces (ENIs).

<https://docs.aws.amazon.com/privatelink/latest/userguide/what-is-privatelink.html>

Incorrect Options Explained:

- *A. To connect VPCs across Regions using public IPs*
This describes VPC Peering or Transit Gateway with inter-region support, not PrivateLink. PrivateLink does not use public IPs or support cross-region VPC-to-VPC peering.
- *C. To manage IAM policies across multiple AWS accounts*
This relates to AWS Organizations or Resource Access Manager (RAM), not PrivateLink. IAM policy management is outside the scope of networking services.
- *D. To automatically assign public IP addresses to EC2 instances in private subnets*
This is a function of Auto-assign public IP settings in subnet configuration and has no relation to PrivateLink, which is used for private connectivity.

Q66.

You have a VPC with both public and private subnets. You want instances in the private subnet to access the internet for software updates, but you do not want them to be reachable from the internet. Which of the following should you use?

- A.** Internet Gateway
- B.** NAT Gateway in a public subnet
- C.** VPC Peering Connection
- D.** Egress-Only Internet Gateway in a public subnet

Correct Answer:

- B.** NAT Gateway in a public subnet

Explanation:

A NAT Gateway enables instances in a private subnet to initiate outbound traffic to the internet, such as downloading software updates, while preventing inbound connections from the internet. The NAT Gateway itself must reside in a public subnet with a route to the Internet Gateway.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html>

Incorrect Options Explained:

- *A. Internet Gateway*
Provides direct internet access to instances in a public subnet, but cannot be used alone by instances in a private subnet.
- *C. VPC Peering Connection*
Used to connect two VPCs privately, not to access the internet.
- *D. Egress-Only Internet Gateway in a public subnet*
Only supports IPv6 traffic and is not suitable for IPv4-based internet access typically used for software updates.

Q67.

Which of the following IP addresses in an AWS subnet CIDR block are reserved and cannot be assigned to instances?

(Choose all that apply)

- A. The first IP address in the CIDR block
- B. The second IP address in the CIDR block
- C. The third IP address in the CIDR block
- D. The last IP address in the CIDR block
- E. Any IP address not in the reserved range
- F. The fifth IP address in the CIDR block

Correct Answers:

- A.** The first IP address
- B.** The second IP address
- C.** The third IP address
- D.** The last IP address
- F.** The fifth IP address

Explanation:

In every AWS subnet, five IP addresses are reserved and cannot be assigned to EC2 instances:

1. First IP – Network address

2. Second IP – VPC router
3. Third IP – Reserved for AWS DNS
4. Fourth IP – Reserved for future use (by AWS)
5. Last IP – Network broadcast address (though AWS does not support broadcast)

So in a /24 subnet like 192.168.1.0/24, the IPs 192.168.1.0, .1, .2, .3, and .255 are reserved.

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Subnets.html#subnet-sizing

Incorrect Option Explained:

- E. *Any IP address not in the reserved range*
This is not correct to select, because it refers to assignable IPs. The question asked for reserved ones.

Q68.

You are managing two VPCs: VPC-A (requester) and VPC-B (accepter), which are in a pending-acceptance peering state. You realise that VPC-A needs to add a new CIDR block. What is the correct sequence of actions you should take to comply with AWS peering rules?

- A. Add the CIDR block directly to VPC-A while the peering connection is pending
- B. Ask the owner of VPC-B to accept the peering connection, then add the CIDR block to VPC-A
- C. Delete the peering request from VPC-A, add the CIDR block, and then create a new peering request
- D. Add the CIDR block to both VPCs at the same time without deleting the peering connection

Correct Answer:

- C. Delete the peering request from VPC-A, add the CIDR block, and then create a new peering request

Explanation:

In a pending-acceptance state, the requester VPC (VPC-A) is not allowed to add CIDR blocks. The only way to proceed is to delete the pending peering request, add the CIDR block, and then re-initiate the peering connection.

<https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html#add-cidr>

Incorrect Options Explained:

- A. You cannot add CIDR blocks to the requester VPC in a pending-acceptance state
- B. Even if VPC-B accepts the connection, you won't be able to add the CIDR if the request wasn't updated
- D. Simultaneous CIDR block additions are not allowed and may result in overlapping CIDR conflicts

Q69.

Which AWS service provides a resource inventory, configuration history, and change notifications to support security and governance?

- A. AWS CloudTrail
- B. AWS Config
- C. AWS Systems Manager
- D. AWS Inspector

Correct Answer:

B. AWS Config

Explanation:

AWS Config is a fully managed service that lets you track changes to AWS resources, maintain a resource inventory, record configuration history, and receive notifications about changes. It's widely used for auditing, compliance, and governance.

<https://docs.aws.amazon.com/config/latest/developerguide/WhatIsConfig.html>

Incorrect Options Explained:

- A. *AWS CloudTrail* tracks API activity, not detailed configuration history or resource inventory
- C. *AWS Systems Manager* helps manage infrastructure but does not track configuration changes
- D. *AWS Inspector* is used for automated security assessments, not configuration tracking

Q70.

Which of the following statements are true about Amazon Aurora? *(Choose 2 answers)*

- A.** It is a fully managed relational database engine compatible with MySQL and PostgreSQL
- B.** It is used only for storing NoSQL documents
- C.** It provides automatic failover, replication, and backup
- D.** It is a self-managed service that requires customer patching and maintenance
- E.** It supports only PostgreSQL

Correct Answers:

- A.** It is a fully managed relational database engine compatible with MySQL and PostgreSQL
- C.** It provides automatic failover, replication, and backup

Explanation:

- A. Amazon Aurora is designed to be compatible with MySQL and PostgreSQL, allowing applications using these engines to easily migrate.
- C. Aurora supports high availability through replication, automatic failover, and automated backups as part of its managed service.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Welcome.html>

Incorrect Options Explained:

- B. Aurora is not a NoSQL service; it's a relational database engine.
- D. Aurora is fully managed by AWS; customers do not handle patching or maintenance.
- E. Aurora supports both MySQL and PostgreSQL, not just PostgreSQL.

Q71.

Which AWS tool helps you evaluate and improve your cloud architecture by answering a set of best-practice questions?

- A.** AWS Trusted Advisor
- B.** AWS Well-Architected Tool
- C.** AWS Config
- D.** AWS Budgets

Correct Answer:

- B.** AWS Well-Architected Tool

Explanation:

The AWS Well-Architected Tool helps you review and improve your cloud workloads by evaluating them against AWS best practices across six pillars (e.g., security, reliability, performance efficiency). You answer evaluation questions to identify risks and areas for improvement.

<https://docs.aws.amazon.com/wellarchitected/latest/userguide/what-is-aws-wa.html>

Incorrect Options Explained:

- A. *AWS Trusted Advisor* provides checks for cost, security, and service limits, but doesn't offer architecture reviews.
- C. *AWS Config* tracks configuration changes, not architectural design reviews.
- D. *AWS Budgets* is used to monitor and control AWS spending, not architecture best practices.

Q72.

Which of the following is not one of the six pillars of the AWS Well-Architected Framework?

- A. Security
- B. Cost Optimization
- C. Automation
- D. Sustainability

Correct Answer:

C. Automation

Explanation:

The six pillars of the AWS Well-Architected Framework are:

1. Security
2. Reliability
3. Performance Efficiency
4. Operational Excellence
5. Cost Optimization
6. Sustainability

Automation is a technique used within several pillars but is not a pillar by itself.

<https://docs.aws.amazon.com/wellarchitected/latest/framework/welcome.html>

Incorrect Options Explained:

- A. Security is one of the core pillars focused on data protection and access control.
- B. Cost Optimization aims to run systems efficiently while reducing unnecessary expenses.
- D. Sustainability was added as the sixth pillar to help organizations reduce their environmental impact.

Q73.

What is the main objective of the Security Pillar in the AWS Well-Architected Framework?

- A. To optimise resource usage and reduce cloud costs
- B. To assess and mitigate risks while protecting data, systems, and assets
- C. To improve application performance across regions
- D. To deploy applications as fast as possible

Correct Answer:

- B. To assess and mitigate risks while protecting data, systems, and assets

Explanation:

The Security Pillar focuses on implementing measures that protect information, systems, and assets, using risk assessments and mitigation strategies. This helps deliver business value securely and reliably.

<https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html>

Incorrect Options Explained:

- A. Refers to the Cost Optimization pillar, not security.
- C. Linked to the Performance Efficiency pillar.
- D. Emphasised in the Operational Excellence pillar, not security.

Q74.

Which of the following are **true** about Amazon EBS snapshots? (*Choose 2 answers*)

- A. EBS snapshots are automatically deleted when the instance is terminated
- B. Snapshots are stored in Amazon S3
- C. You can create point-in-time snapshots of EBS volumes
- D. Snapshots cannot be copied to other AWS Regions

Correct Answers:

- B. Snapshots are stored in Amazon S3
- C. You can create point-in-time snapshots of EBS volumes

Explanation:

- B. EBS snapshots are persisted to Amazon S3, though not directly visible in your S3 bucket.
- C. Snapshots provide point-in-time backups of EBS volumes, which can be used to restore or create new volumes.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/EBSSnapshots.html>

Incorrect Options Explained:

- A. Snapshots are not deleted when the instance is terminated — they persist independently.
- D. Snapshots can be copied across AWS regions for backup, migration, or disaster recovery.

Q75.

Which of the following statements about Amazon EBS snapshots are correct? (*Choose 2 answers*)

- A. Snapshots are persisted to Amazon S3 storage
- B. Snapshots can be copied across AWS Regions
- C. Snapshots are deleted automatically when an instance stops
- D. Snapshots are stored in your S3 bucket and visible by default

Correct Answers:

- A. Snapshots are persisted to Amazon S3 storage
- B. Snapshots can be copied across AWS Regions

Explanation:

- A. Amazon EBS snapshots are persisted to Amazon S3, although not directly accessible like typical S3 objects.
- B. EBS snapshots can be copied across regions, supporting backup, disaster recovery, and migration use cases.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/EBSSnapshots.html>

Incorrect Options Explained:

- C. Snapshots are not automatically deleted when an instance stops or terminates — they remain until manually deleted.
- D. While stored in S3, snapshots are not visible in your S3 bucket and are managed through the EBS console or API, not the S3 interface.

Q76.

Which of the following performance metrics provided by Amazon CloudWatch can help you monitor and optimise Amazon EBS volumes? (*Choose 2 answers*)

- A. Bandwidth
- B. Route propagation
- C. Latency
- D. Availability Zone name

Correct Answers:

- A. Bandwidth
- C. Latency

Explanation:

- A. Bandwidth measures the amount of data transferred over time, helping assess if your volume throughput meets application needs.
- C. Latency reflects the time it takes to complete I/O operations, helping detect bottlenecks.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-monitoring.html>

Incorrect Options Explained:

- B. *Route propagation* relates to networking (e.g., route tables in VPC), not EBS volume performance.
- D. *Availability Zone name* is a location attribute, not a performance metric.

Q77.

Which of the following are true about AWS Backup? (*Choose 2 answers*)

- A. AWS Backup is an automated and centralized backup service for AWS resources
- B. AWS Backup supports backups for Amazon DynamoDB and Amazon RDS
- C. AWS Backup replaces the need for IAM policies when securing backups
- D. AWS Backup is used only for Amazon S3 buckets

Correct Answers:

- A. AWS Backup is an automated and centralized backup service for AWS resources
- B. AWS Backup supports backups for Amazon DynamoDB and Amazon RDS

Explanation:

- A. AWS Backup provides a centralized and automated way to manage backups across multiple AWS services.
- B. It integrates with services like Amazon EBS, Amazon RDS, Amazon EFS, Amazon DynamoDB, and AWS Storage Gateway to simplify backup management.

<https://docs.aws.amazon.com/aws-backup/latest/devguide/whatisbackup.html>

Incorrect Options Explained:

- C. AWS Backup does not eliminate the need for IAM policies; IAM is still required to manage permissions for backup operations.
- D. AWS Backup is not limited to S3 — in fact, it does not natively back up S3 objects; S3 has its own versioning and replication options.

Q78.

Which of the following are true about a multi-region trail in AWS CloudTrail? (*Choose 2 answers*)

- A.** It records events from all AWS regions
- B.** It stores logs only in the default region where the trail was created
- C.** It delivers event log files to an S3 bucket that you specify
- D.** It can only be enabled through the AWS CLI

Correct Answers:

- A.** It records events from all AWS regions
- C.** It delivers event log files to an S3 bucket that you specify

Explanation:

- A. A multi-region trail captures API events from all AWS regions, improving governance and compliance visibility across global resources.
- C. You can specify an S3 bucket to store the log files generated by the trail, regardless of the region in which the events occurred.

<https://docs.aws.amazon.com/awscloudtrail/latest/userguide/cloudtrail-concepts.html>

Incorrect Options Explained:

- B. Multi-region trails do not restrict logs to the default region — they collect from all regions.
- D. While the AWS CLI can be used to enable trails, the AWS Console also fully supports this (it's the default method for most users).

Q79.

In AWS CloudTrail, what is the term used to describe a record of activity performed by a user, role, or service?

- A.** Audit log
- B.** Event
- C.** Snapshot
- D.** Resource action

Correct Answer:

B. Event

Explanation:

In AWS CloudTrail, an event represents a record of an activity that occurred in your AWS account. This activity may be an API call made by a user, an IAM role, or an AWS service.

<https://docs.aws.amazon.com/awscloudtrail/latest/userguide/cloudtrail-event-reference.html>

Incorrect Options Explained:

- A. *Audit log* is a general term but not the specific object used in CloudTrail
- C. *Snapshot* refers to EBS or RDS backups, not activity logs
- D. *Resource action* is not a defined CloudTrail concept

Q80.

Which of the following are true about Amazon SNS mobile and messaging notifications?
(Choose 2 answers)

- A.** SNS supports sending push notifications to mobile devices like iOS and Android
- B.** SNS can only send messages to devices located in the United States
- C.** SNS can fan out messages to multiple protocols such as SMS and email
- D.** SNS only supports HTTP-based delivery and not email or SMS

Correct Answers:

- A.** SNS supports sending push notifications to mobile devices like iOS and Android
- C.** SNS can fan out messages to multiple protocols such as SMS and email

Explanation:

- A. Amazon SNS allows mobile push notifications to platforms like iOS, Android, Fire OS, Windows, and Baidu.
- C. SNS uses the fan-out pattern to deliver the same message to multiple endpoints such as SMS, email (SMTP), mobile push, and HTTP/S.

<https://docs.aws.amazon.com/sns/latest/dg/welcome.html>

Incorrect Options Explained:

- B. SNS supports SMS delivery to 200+ countries, not just the U.S.

- D. SNS supports multiple protocols including email and SMS, not only HTTP.

Q81.

Which of the following statements are true about Standard and Convertible Reserved Instances? *(Choose 2 answers)*

- A. Standard Reserved Instances offer the highest discount but allow limited modifications
- B. Convertible Reserved Instances can be exchanged for different attributes
- C. Convertible Reserved Instances can be sold on the Reserved Instance Marketplace
- D. Standard Reserved Instances cannot be sold during their term

Correct Answers:

- A. Standard Reserved Instances offer the highest discount but allow limited modifications
- B. Convertible Reserved Instances can be exchanged for different attributes

Explanation:

- A. Standard RIs provide higher discounts but allow limited changes and can be sold on the Reserved Instance Marketplace.
- B. Convertible RIs allow you to exchange them for other Convertible RIs with different attributes (e.g., instance type, OS).

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-reserved-instances.html>

Incorrect Options Explained:

- C. Convertible RIs cannot be sold on the Reserved Instance Marketplace.
- D. Standard RIs can be sold on the marketplace after a 30-day hold period.

Q82.

Which of the following are true about AWS security groups used as firewalls? *(Choose 2 answers)*

- A. Security groups allow you to specify allowed inbound and outbound traffic by protocol and port
- B. Security groups are stateless, requiring separate rules for inbound and outbound traffic

- C. Security groups act at the instance level, not the subnet level
- D. Security groups block all traffic by default, including outbound traffic

Correct Answers:

- A. Security groups allow you to specify allowed inbound and outbound traffic by protocol and port
- C. Security groups act at the instance level, not the subnet level

Explanation:

- A. You can configure security groups to allow traffic based on protocol, port range, and source/destination IPs.
- C. Security groups are associated with EC2 instances, not entire subnets like network ACLs.

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_SecurityGroups.html

Incorrect Options Explained:

Incorrect Options Explained:

- B. Security groups are stateful, meaning if you allow inbound traffic, return traffic is automatically allowed.
- D. By default, security groups allow all outbound traffic, but no inbound traffic.

Q83.

Which of the following are true about AWS Regions and Availability Zones? *(Choose 2 answers)*

- A. A Region consists of multiple isolated Availability Zones
- B. All Availability Zones within a Region share the same data center
- C. You can deploy AWS resources such as EC2 and EBS in specific Availability Zones
- D. AWS guarantees zero latency between Availability Zones in the same Region

Correct Answers:

- A. A Region consists of multiple isolated Availability Zones
- C. You can deploy AWS resources such as EC2 and EBS in specific Availability Zones

Explanation:

- A. Each AWS Region is a geographically distinct area and includes multiple AZs for fault tolerance and availability.
- C. You can choose specific AZs for deploying resources like EC2 instances and EBS volumes to improve performance or redundancy.

https://aws.amazon.com/about-aws/global-infrastructure/regions_az/

Incorrect Options Explained:

- B. AZs are physically separated data centers within a Region, not shared.
- D. While AZs are connected by low-latency, high-bandwidth networking, zero latency is not guaranteed.

Q84.

Which of the following are true about On-Demand Instances in AWS EC2? (*Choose 2 answers*)

- A.** On-Demand Instances require long-term contracts and upfront payments
- B.** They help eliminate the need to purchase and manage physical hardware
- C.** On-Demand Instances are ideal for applications with steady, predictable usage
- D.** They are suitable for unpredictable workloads or short-term, spiky traffic

Correct Answers:

- B.** They help eliminate the need to purchase and manage physical hardware
- D.** They are suitable for unpredictable workloads or short-term, spiky traffic

Explanation:

- B. On-Demand Instances allow you to run workloads without managing hardware, offering flexibility and speed.
- D. They are ideal for unpredictable or bursty workloads, as you pay by the hour or second without long-term commitments.

<https://aws.amazon.com/ec2/pricing/on-demand/>

Incorrect Options Explained:

- A. On-Demand Instances do not require long-term contracts or upfront payments.

- C. For steady, predictable workloads, Reserved Instances are typically more cost-effective than On-Demand.

Q85.

Which of the following best describes the AWS cloud benefit of trading capital expense for variable expense?

- A. You commit to a 3-year pricing model for reserved resources
- B. You only pay for the compute resources you use, eliminating upfront investments
- C. You purchase and manage servers in your on-premises data centre
- D. You pay the same monthly fee regardless of usage

Correct Answer:

- B.** You only pay for the compute resources you use, eliminating upfront investments

Explanation:

This benefit refers to replacing upfront capital investment in hardware with pay-as-you-go pricing, which is one of the foundational economic advantages of cloud computing. AWS lets you scale resources up or down and only charges for what you use.

<https://aws.amazon.com/economics/>

Incorrect Options Explained:

- A. Describes Reserved Instances, which involve commitment, not variable pricing
- C. It is the opposite of cloud-based resource provisioning
- D. A flat monthly fee doesn't reflect the usage-based billing model of AWS

Q87.

What is the purpose of an Amazon Machine Image (AMI) in AWS?

- A.** It stores encrypted S3 data for backup
- B.** It packages the operating system and custom software into a reusable template

- C. It monitors EC2 instance performance using CloudWatch
- D. It acts as a firewall controlling access to EC2 instances

Correct Answer:

- B. It packages the operating system and custom software into a reusable template

Explanation:

An Amazon Machine Image (AMI) provides a blueprint for launching EC2 instances. It includes the OS, application server, and custom configurations, allowing you to create multiple identical instances.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AMIs.html>

Incorrect Options Explained:

- A. Refers to Amazon S3 backups, not AMIs
- C. CloudWatch is used for monitoring, not creating instance templates
- D. Security groups or network ACLs manage access control, not AMIs

Q88.

Which of the following EC2 instance types are correctly matched with their optimization category? (*Choose 2 answers*)

- A. c-type – Memory optimized
- B. m-type – General purpose
- C. r-type – Storage optimized
- D. p-type – Accelerated computing

Correct Answers:

- B. m-type – General purpose
- D. p-type – Accelerated computing

Explanation:

- B. m-type and t-type are general purpose instances, offering a balance of compute, memory, and networking.

- D. p-type, g-type, and f-type are designed for accelerated computing using GPUs or FPGAs.

Incorrect Options Explained:

- A. c-type is for compute-optimized, not memory
- C. r-type is for memory-optimized, not storage; storage-optimized types include i-type, d-type, and h-type

Q89.

Which of the following best describes a containerized application?

- A.** An application that is deployed directly onto a physical server without any abstraction
- B.** An application that runs in a lightweight, portable environment containing code, dependencies, and runtime
- C.** An application hosted only in Amazon EC2 without scalability
- D.** An application that requires a dedicated database per deployment

Correct Answer:

B. An application that runs in a lightweight, portable environment containing code, dependencies, and runtime

Explanation:

Containers package everything an application needs to run: code, runtime, libraries, and dependencies. This allows them to run consistently across environments, making them ideal for microservices and modern cloud-native applications.

<https://aws.amazon.com/containers/>

Incorrect Options Explained:

- A. Describes bare metal deployments, not containers
- C. Containers can run on EC2, but are not limited to it and are inherently scalable
- D. Dedicated databases are not a requirement for containerized apps

Q90.

Which of the following statements are true about building and storing container images in AWS? (Choose 2 answers)

- A.** Images are built from a Docker file that defines all required components
- B.** Container images can only be stored locally and cannot be shared
- C.** Images can be stored in a registry for reuse across clusters
- D.** Docker files are binary files used to launch EC2 instances directly

Correct Answers:

- A.** Images are built from a Docker file that defines all required components
- C.** Images can be stored in a registry for reuse across clusters

Explanation:

- A. A Docker file is a text file that includes instructions to build a container image (e.g., base OS, dependencies, and commands).
- C. Built container images can be stored in a container registry like Amazon ECR (Elastic Container Registry) for deployment and reuse.

<https://docs.aws.amazon.com/AmazonECR/latest/userguide/what-is-ecr.html>

Incorrect Options Explained:

- B. Container images can be stored and shared via registries like ECR, Docker Hub, etc.
- D. Docker files are text-based and not used to launch EC2 instances directly — they build images for containers.

Q91.

What AWS service provides a private container image registry for storing and managing Docker images?

- A.** Amazon ECS
- B.** Amazon EKS
- C.** Amazon ECR
- D.** AWS Lambda

Correct Answer:

- C.** Amazon ECR

Explanation:

Amazon Elastic Container Registry (ECR) is a fully managed Docker container registry that allows AWS users to store, manage, and deploy container images. Each AWS account has its own private registry, and users can create repositories within it to store images.

<https://docs.aws.amazon.com/AmazonECR/latest/userguide/what-is-ecr.html>

Incorrect Options Explained:

- A. Amazon ECS is a container orchestration service, not a registry
- B. Amazon EKS is for Kubernetes orchestration, not image storage
- D. AWS Lambda runs functions, not container registries

Q92.

What must be done before an IAM user can create or manage resources in Amazon ECR?

- A. Enable ECR in the IAM dashboard
- B. Attach an appropriate IAM policy granting ECR permissions
- C. Use the root account to access ECR features
- D. Create an EC2 instance and configure ECR access from there

Correct Answer:

B. Attach an appropriate IAM policy granting ECR permissions

Explanation:

By default, IAM users do not have access to manage Amazon ECR resources. You must explicitly assign IAM policies (e.g., AmazonEC2ContainerRegistryFullAccess) to allow the user to perform tasks using the ECR API or console.

https://docs.aws.amazon.com/AmazonECR/latest/userguide/ecr_managed_policies.html

Incorrect Options Explained:

- A. ECR permissions are not "enabled" from the IAM dashboard — they are granted via policies
- C. Using the root account is not a recommended security practice
- D. EC2 instances can use ECR, but IAM users still need explicit permissions

Q93:

A Cloud Practitioner is building a new application and wants to integrate AWS service features directly into the application code. Which of the following is the most appropriate tool for this purpose?

- a) AWS Command Line Interface (CLI)
- b) AWS CodeDeploy
- c) AWS Software Development Kit (SDK)
- d) AWS CodePipeline

Correct Answer:

c) AWS Software Development Kit (SDK)

Explanation:

The AWS SDK allows developers to interact with AWS services directly from within their code using programming languages like Python, Java, and JavaScript. This makes it the most suitable tool for embedding AWS functionality into applications.

<https://aws.amazon.com/tools/>

Incorrect Options Explained:

- a) AWS CLI: Suitable for command-line operations, not for direct integration within application code.
- b) AWS CodeDeploy: Used for deploying applications to AWS environments, not for embedding service logic.
- d) AWS CodePipeline: Automates the software release process but doesn't embed AWS features into your code.

Q94:

A startup company has deployed their application and databases on AWS. To reduce operational overhead, they want AWS to handle as many routine infrastructure tasks as possible, so their team can focus on innovation and business logic.

Which of the following tasks can AWS manage on their behalf? (*Select TWO.*)

- a) Taking a backup of a database
- b) Patching database software
- c) Application source code auditing

- d) Application security testing
- e) Creating a database schema

Correct Answers:

- a) Taking a backup of a database
- b) Patching database software

Explanation:

- a) Services like Amazon RDS and DynamoDB automate database backups, allowing users to enable scheduled backups and point-in-time recovery.
- b) AWS manages software patching for managed services like RDS, relieving customers from manual maintenance tasks.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_WorkingWithAutomatedBackups.html

Incorrect Options Explained:

- c) AWS does not audit your source code — this is the responsibility of the customer or third-party tools.
- d) AWS provides tools to secure applications but does not perform security testing on customer applications.
- e) Designing and creating the database schema is an application-level decision handled by the customer.

Q95:

A company is evaluating different AWS compute and database services for hosting their application. They want to understand where they are responsible for tasks like patching the underlying operating system (OS).

In which of the following services is the customer responsible for patching the operating system?

- a) AWS Fargate
- b) Amazon DynamoDB
- c) Amazon EC2
- d) AWS Lambda

Correct Answer:

c) Amazon EC2

Explanation:

With Amazon EC2, customers manage the virtual machine, which includes patching the operating system and maintaining software updates — this is part of the shared responsibility model.

<https://aws.amazon.com/compliance/shared-responsibility-model/>

Incorrect Options Explained:

- a) AWS Fargate is a serverless compute engine — AWS handles all infrastructure and OS patching.
- b) Amazon DynamoDB is a fully managed NoSQL database — AWS manages all underlying infrastructure.
- d) AWS Lambda runs code without provisioning servers — OS management is handled entirely by AWS.

Q96:

A media company hosts all its video content in a single Amazon S3 bucket located in the US East (N. Virginia) region. As their global audience grows, users in distant regions are experiencing increased latency while streaming videos.

According to the performance pillar of the AWS Well-Architected Framework, which service should the company use to ensure low-latency access to the video content globally?

- a) Use AWS CloudFront to cache the video content closer to end users.
- b) Use Amazon ElastiCache to cache the video content closer to end users.
- c) Use AWS DynamoDB DAX to cache the video content closer to end users.
- d) Use Amazon Kinesis to cache the video content closer to end users.

Correct Answer:

- a) Use AWS CloudFront to cache the video content closer to end users.

Explanation:

AWS CloudFront is a global Content Delivery Network (CDN) that caches content at edge locations near end users, which significantly reduces latency and improves video streaming performance for users worldwide.

<https://aws.amazon.com/cloudfront/>

Incorrect Options Explained:

- b) Amazon ElastiCache is used to cache database query results, not large video content.
- c) DynamoDB DAX is a caching layer for DynamoDB, not for static content like videos.
- d) Amazon Kinesis is for real-time data streaming and analytics, not content delivery or caching.

Q97:

A database administrator is tasked with deploying and managing a highly available relational database for a production application. They are considering using Amazon RDS to reduce operational complexity and focus more on performance tuning and application logic.

Which of the following is a key advantage the administrator would gain by using Amazon RDS?

- a) RDS enables users to dynamically adjust CPU and RAM resources.
- b) RDS databases automatically scale based on load.
- c) RDS provides 99.9999999999% reliability and durability.
- d) RDS simplifies relational database administration tasks.

Correct Answer:

- d) RDS simplifies relational database administration tasks.

Explanation:

Amazon RDS is a fully managed service that automates tasks like backups, patching, monitoring, and failover, thereby reducing the administrative overhead for database admins.

Incorrect Options Explained:

- a) You must manually scale instance classes or use RDS Auto Scaling for read replicas; it's not dynamic like compute auto-scaling.
- b) RDS does not automatically scale compute resources based on load (except Aurora Serverless).

- c) 99.999999999999% refers to S3 durability, not RDS reliability or durability.

Q98:

A financial services company is planning to use Amazon RDS to host its production databases. As part of their compliance audit, the team wants to know who is responsible for applying security patches to the RDS instances under the AWS Shared Responsibility Model.

Who is responsible for patching Amazon RDS database instances?

- a) AWS
- b) Customer
- c) Database engine vendor
- d) Both the customer and AWS

Correct Answer: a) AWS

Explanation:

In managed services like Amazon RDS, AWS is responsible for infrastructure-level patching and maintenance, including the underlying OS and database engine software.

Q99:

An enterprise wants to migrate part of its on-premises file-based workloads to the cloud. The applications require standard NFS-based access and must be able to read/write objects to Amazon S3, with local caching to improve performance.

Which AWS service can best meet these requirements?

- a) EBS volumes
- b) Amazon Redshift
- c) AWS Storage Gateway — Amazon S3 File Gateway
- d) AWS EFS

Correct Answer: c) AWS Storage Gateway — Amazon S3 File Gateway

Explanation:

The Amazon S3 File Gateway allows on-premises applications to access Amazon S3 as if it were a file share using NFS or SMB protocols, with local caching capabilities.

Q100:

An eCommerce platform uses an Elastic Load Balancer (ELB) to route traffic to a fleet of EC2 instances. One of the instances starts failing health checks.

What will ELB do in response to this unhealthy instance?

- a) It only sends traffic to the remaining healthy instances.
- b) It restarts the unhealthy EC2 instance.
- c) It terminates the failed instance so that it is not part of the ELB target group.
- d) It continues to send traffic to the failed instance.

Correct Answer: a) It only sends traffic to the remaining healthy instances

Explanation:

ELB stops routing traffic to any EC2 instance that fails health checks, ensuring that only healthy instances receive traffic. It does not take actions like restarting or terminating the instance.

[Elastic Load Balancing Health Checks](#)

Q101:

A data analytics company is planning to archive a large volume of structured and unstructured data into Amazon S3. Their engineering lead wants to confirm the maximum storage limit per account.

How much data can be stored in Amazon S3?

- a) 100 PB
- b) 1 PB
- c) 100 TB
- d) Virtually unlimited

Correct Answer: d) Virtually unlimited

Explanation:

Amazon S3 is designed to provide virtually unlimited object storage. Each object can be up to 5 TB, and there's no limit to the total number of objects or amount of storage.

[Amazon S3 Limits](#)

Q102:

A cloud architect is designing a secure web hosting environment on AWS. The solution must follow security best practices to minimize risk from external threats.

Where should firewalling and traffic filtering be implemented in the design?

- a) At the perimeter
- b) At all design layers
- c) At the core
- d) For all access layer functions

Correct Answer: b) At all design layers

Explanation:

Defense in depth is a key principle in AWS. Firewalls (like security groups and network ACLs) should be applied at multiple layers, including perimeter, network, and host level, to ensure layered security.

[AWS Security Best Practices](#)

Q103:

A startup is building a new analytics platform on AWS. Their workloads vary dramatically—low usage during the day and high spikes at night and weekends. The company wants a cost-efficient solution without committing to large upfront infrastructure costs.

Which TWO reasons explain why AWS is more economical than traditional data centers for applications with varying compute workloads?

- a) Amazon EC2 costs are billed on a monthly basis.
- b) Users retain full administrative access to their Amazon EC2 instances.
- c) Amazon EC2 instances can be launched on demand when needed.
- d) AWS allows automatic scaling of compute resources based on demand.

Correct Answers:

- c) Amazon EC2 instances can be launched on demand when needed
- d) AWS allows automatic scaling of compute resources based on demand

Explanation for Correct Answers:

- c) Amazon EC2 instances can be launched on demand when needed:
This allows you to avoid paying for idle resources. You pay only when an instance is running, making it ideal for workloads with unpredictable spikes.
- d) AWS allows automatic scaling of compute resources based on demand:
Services like Auto Scaling and Elastic Load Balancing allow resources to grow and shrink dynamically, helping you avoid over-provisioning and reduce costs.

<https://aws.amazon.com/autoscaling/>

Incorrect Answers Explained:

- a) Amazon EC2 costs are billed on a monthly basis:
Not true — EC2 charges per second (Linux) or per hour (Windows), not monthly. Monthly billing is a characteristic of traditional hosting, not AWS.
- b) Users retain full administrative access to their Amazon EC2 instances:
While technically true, this is related to control and configuration, not cost efficiency.

Q104:

An organization recently migrated its applications to AWS. The security team wants to ensure that the infrastructure follows best practices, such as restricted security groups, unused access keys, and proper IAM permissions. They are looking for a tool that can provide real-time recommendations for improving infrastructure security and cost optimization.

Which AWS service should they use to get these recommendations?

- a) AWS Price List Application Programming Interface (API)
- b) Reserved Instances
- c) AWS Trusted Advisor
- d) Amazon Elastic Compute Cloud (Amazon EC2) Spot Fleet

Correct Answer:

- c) AWS Trusted Advisor

Explanation (Correct):

AWS Trusted Advisor is a web-based tool that provides real-time recommendations for improving security, fault tolerance, performance, service limits, and cost optimization. It checks things like open ports, multi-factor authentication, and IAM usage — making it ideal for infrastructure security optimization.

<https://docs.aws.amazon.com/awssupport/latest/user/trusted-advisor.html>

Incorrect Answers Explained:

- a) AWS Price List API:
Used for retrieving pricing information — not related to security recommendations.
- b) Reserved Instances:
Offer cost savings for predictable workloads but have no role in security optimization.
- d) Amazon EC2 Spot Fleet:
Helps you manage and request Spot Instances for cost efficiency — not for security assessments or advice.

Q105:

A system administrator is responsible for ensuring the health and performance of applications running on AWS. They need a solution that can collect and track operational metrics, such as CPU utilization, disk I/O, and custom application-level metrics, across various AWS services.

Which AWS service should the administrator use to meet this requirement?

- a) Amazon CloudFront
- b) Amazon CloudSearch
- c) Amazon CloudWatch
- d) Amazon Machine Learning (Amazon ML)

Correct Answer:

c) Amazon CloudWatch

Explanation (Correct):

Amazon CloudWatch is the AWS-native monitoring service that collects and tracks metrics, logs, and events from AWS resources and applications. It enables real-time monitoring and alarming to detect anomalies and operational issues.

<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/WhatIsCloudWatch.html>

Incorrect Answers Explained:

- a) Amazon CloudFront:
A Content Delivery Network (CDN) used to deliver content with low latency. It's not used for collecting metrics across AWS services.
- b) Amazon CloudSearch:
A search service for integrating search functionality into applications — not for performance monitoring or metrics collection.
- d) Amazon Machine Learning (Amazon ML):
Used for creating and training machine learning models — unrelated to infrastructure monitoring or metrics tracking.

Q105:

An operations engineer is tasked with monitoring AWS infrastructure for a rapidly growing SaaS platform. The engineer needs a solution to collect metrics, track resource utilization, set alarms, and automate responses when thresholds are breached (e.g., CPU usage over 80% for EC2 instances).

Which AWS service provides these monitoring and automation capabilities?

- a) Amazon CloudFront
- b) Amazon CloudSearch
- c) Amazon CloudWatch
- d) Amazon Machine Learning (Amazon ML)

Correct Answer:

c) Amazon CloudWatch

Explanation (Correct):

Amazon CloudWatch is the AWS service that collects and tracks metrics, logs, and events from AWS services and custom applications. It allows users to set alarms, visualize performance metrics, and automate actions based on thresholds (like scaling or notifications).

<https://docs.aws.amazon.com/cloudwatch/>

Incorrect Answers Explained:

- a) Amazon CloudFront:
A Content Delivery Network (CDN) used for low-latency delivery of content — it does not track metrics.
- b) Amazon CloudSearch:
A managed search service to index and search text — unrelated to system monitoring or metrics.
- d) Amazon Machine Learning (Amazon ML):
Used for building and deploying machine learning models — not for monitoring or metric tracking.

Q106:

An IT administrator is setting up a new web application for their company on AWS. As part of the setup, they need to register a custom domain name (e.g., example.com) to use for accessing the application. The administrator prefers to manage DNS settings and domain registration within AWS itself.

Which AWS service should the administrator use to register the domain name?

- a) Amazon Route 53
- b) Amazon CloudFront
- c) Elastic Load Balancing
- d) Amazon Virtual Private Cloud (Amazon VPC)

Correct Answer:

- a) Amazon Route 53

Explanation (Correct):

Amazon Route 53 is AWS's scalable Domain Name System (DNS) web service that also supports domain name registration. It allows administrators to register new domains, manage DNS records, and route end users to applications hosted on AWS or elsewhere.

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/domain-register.html>

Incorrect Answers Explained:

- b) Amazon CloudFront:
A Content Delivery Network (CDN) used for low-latency content delivery — not for registering domain names.

- c) Elastic Load Balancing:
Distributes traffic across multiple targets like EC2 instances — it has no role in domain registration.
- d) Amazon Virtual Private Cloud (Amazon VPC):
Let's you provision logically isolated cloud networks — unrelated to domain name registration or DNS.

Q107:

A DevOps engineer at a fast-growing startup is tasked with automating the deployment and scaling of cloud infrastructure. The team wants to avoid manual configuration through the AWS Management Console and instead integrate infrastructure provisioning directly into their CI/CD pipeline using scripts and tools like Terraform or AWS CLI.

What is the key benefit of having AWS services accessible through an Application Programming Interface (API)?

- a) Cloud resources can be managed programmatically
- b) AWS infrastructure use will always be cost-optimized
- c) All application testing is managed by AWS
- d) Customer-owned, on-premises infrastructure becomes programmable

Correct Answer:

- a) Cloud resources can be managed programmatically

Explanation (Correct):

Using AWS APIs allows developers and system administrators to automate tasks such as creating, managing, and terminating resources through scripts or third-party tools. This improves efficiency, repeatability, and supports infrastructure as code (IaC) practices.

<https://docs.aws.amazon.com/general/latest/gr/aws-apis.html>

Incorrect Answers Explained:

- b) AWS infrastructure use will always be cost-optimized:
Cost optimization is achieved through monitoring and right-sizing, not simply by using APIs.
- c) All application testing is managed by AWS:
Application testing remains the customer's responsibility. AWS provides tools but does not automatically manage testing.
- d) Customer-owned, on-premises infrastructure becomes programmable:
AWS APIs manage cloud resources, not on-premises infrastructure, unless additional integration (like AWS Outposts) is in place

Q108:

A cloud architect is tasked with designing a high-availability web application that must remain operational even if parts of the infrastructure experience failure. The architect is following the AWS Well-Architected Framework, particularly the principle of "Design for failure and nothing will fail."

Which of the following actions best aligns with this design principle?

- a) Adding an Elastic Load Balancer in front of a single Amazon EC2 instance
- b) Creating and deploying the most cost-effective solution
- c) Deploying the application in multiple Availability Zones
- d) Using Amazon CloudWatch alerts to monitor performance

Correct Answer:

- c) Deploying the application in multiple Availability Zones

Explanation (Correct):

Deploying across multiple Availability Zones (AZs) ensures that if one AZ fails, the application remains available in others. This is a core strategy for fault tolerance and aligns directly with the "design for failure" principle

<https://docs.aws.amazon.com/wellarchitected/latest/framework/general-design-principles.html>

Incorrect Answers Explained:

- a) Adding an Elastic Load Balancer in front of a single Amazon EC2 instance
This doesn't eliminate single points of failure. The EC2 instance is still a bottleneck.
- b) Creating and deploying the most cost-effective solution
While cost optimization is important, it is not related to fault tolerance or failure design.
- d) Using Amazon CloudWatch alerts to monitor performance
Monitoring helps with detection, not prevention of failure. It's reactive, not proactive design.

Q109:

A new system administrator at a company is tasked with setting up user access for different teams in their AWS account. The administrator needs to create individual users, assign specific permissions to them (e.g., read-only access to S3, full access to EC2), and ensure that only authorized individuals can access specific AWS services.

Which AWS service should the administrator use to manage these user permissions and roles?

- a) AWS Config
- b) AWS CloudTrail
- c) AWS Key Management Service (AWS KMS)
- d) AWS Identity and Access Management (IAM)

Correct Answer:

- d) AWS Identity and Access Management (IAM)

Explanation (Correct):

AWS IAM enables administrators to securely control access to AWS services and resources. It lets you create users, groups, and roles, and assign permissions via policies. IAM is the central service for managing authentication and authorization in AWS.

<https://docs.aws.amazon.com/IAM/latest/UserGuide/introduction.html>

Incorrect Answers Explained:

- a) AWS Config
Used to track configuration changes and compliance—not to assign permissions.
- b) AWS CloudTrail
Used to log and monitor API activity, not to manage user access.
- c) AWS KMS
Used to create and control encryption keys, not user or permission management.

Q110:

A data engineer is building a centralized data warehouse solution to handle large-scale analytics workloads for a retail company. The solution must be able to automatically provision infrastructure, perform administrative tasks such as backups and patching, and scale to accommodate complex analytical queries on structured data.

Which AWS service is the most appropriate for this use case?

- a) Amazon Redshift
- b) Amazon DynamoDB
- c) Amazon ElastiCache
- d) Amazon Aurora

Correct Answer:

- a) Amazon Redshift

Explanation (Correct):

Amazon Redshift is a fully managed, petabyte-scale data warehouse service designed specifically for analytical workloads. It automates infrastructure provisioning, patching, backups, and scaling, allowing teams to focus on data analysis instead of operations.

<https://aws.amazon.com/redshift/>

Incorrect Answers Explained:

- b) Amazon DynamoDB
A NoSQL key-value and document database service—not designed for complex analytical workloads.

- c) Amazon ElastiCache
Primarily used to cache frequently accessed data to improve application performance—not for data warehousing.
- d) Amazon Aurora
A relational database service suitable for transactional (OLTP) workloads—not optimized for large-scale analytical queries.

Q111:

A cloud engineer is reviewing their company's responsibilities under the AWS Shared Responsibility Model. They are specifically tasked with controlling user access to resources within their AWS environment and want to ensure that they are properly following best practices for securing the application stack.

According to the Shared Responsibility Model, which of the following is the customer responsible for?

- a) Managing AWS Identity and Access Management (IAM)
- b) Securing edge locations
- c) Monitoring physical device security
- d) Implementing Service Organization Control (SOC) standards

Correct Answer:

- a) Managing AWS Identity and Access Management (IAM)

Explanation (Correct):

In the AWS Shared Responsibility Model, customers are responsible for security *in* the cloud, which includes managing IAM roles and policies, configuring security groups, patching applications, and encrypting data. IAM helps control who can access what within your AWS environment.

<https://aws.amazon.com/compliance/shared-responsibility-model/>

Incorrect Answers Explained:

- b) Securing edge locations
AWS is responsible for securing its global infrastructure, including edge locations.
- c) Monitoring physical device security
Physical hardware and data center security fall under AWS's responsibility.
- d) Implementing Service Organization Control (SOC) standards
AWS obtains and manages SOC certifications for its infrastructure; customers may use them as part of their compliance reporting but are not responsible for implementation.

Q112:

A finance analyst at a company is investigating an unexpected spike in compute costs from three months ago. The analyst needs to review historical billing activity specifically related to Amazon EC2 usage in order to understand what caused the increase.

Which AWS service should the analyst use to access detailed historical billing and usage data?

- a) Amazon EC2 dashboard
- b) AWS Cost and Usage Reports
- c) AWS Trusted Advisor dashboard
- d) AWS CloudTrail logs stored in Amazon S3

Correct Answer:

- b) AWS Cost and Usage Reports

Explanation (Correct):

The AWS Cost and Usage Reports (CUR) provide detailed billing information, including usage by service, region, tags, and accounts. It's the most granular source for historical billing analysis and is ideal for reviewing EC2 activity from three months ago or earlier.

<https://docs.aws.amazon.com/cur/latest/userguide/what-is-cur.html>

Incorrect Answers Explained:

- a) Amazon EC2 dashboard
Shows current EC2 instance statuses and settings but does not include detailed billing history.
- c) AWS Trusted Advisor dashboard
Provides real-time recommendations for cost optimization, security, and performance, but not historical billing data.
- d) AWS CloudTrail logs stored in Amazon S3
Tracks API activity, not cost or billing data. It's useful for security audits, not cost analysis.

Q113:

A company is undergoing a cloud security audit and the auditor asks, "Who ultimately has ownership and control of the data stored in your AWS account, including decisions about data access, encryption, and deletion?"

The cloud administrator must identify which parties have direct control over the data stored and processed in their AWS environment.

Which TWO entities have control of the data in an AWS account?
(Choose TWO)

- a) AWS Support Team
- b) AWS Account Owner
- c) AWS Security Team
- d) AWS Technical Account Manager (TAM)
- e) IAM Users with delegated permissions

Correct Answers:

- b) AWS Account Owner
- e) IAM Users with delegated permissions

Explanation (Correct):

- b) AWS Account Owner: The account owner has full administrative control over all AWS resources and data. They define access policies, manage billing, and determine who can interact with data.

- e) IAM Users with delegated permissions: IAM users or roles granted access by the account owner can manage, modify, or delete data, depending on their permissions.

<https://aws.amazon.com/compliance/shared-responsibility-model/>

Incorrect Answers Explained:

- a) AWS Support Team
AWS Support can assist with infrastructure issues but does not access or control customer data unless explicitly granted permission for troubleshooting.
- c) AWS Security Team
AWS is responsible for securing the infrastructure, not managing customer data.
- d) AWS Technical Account Manager (TAM)
TAMs provide advisory support, but they do not have access or control over customer data.

Q114:

A solutions architect is tasked with improving the reliability and scalability of a monolithic application. To achieve this, they decide to break the application into independent components that can communicate through asynchronous messaging using services like Amazon SQS or SNS.

What are the primary benefits of decoupling an application into smaller, loosely connected components?

(Choose TWO)

- a) Create a tightly integrated application
- b) Reduce inter-dependencies so failures do not impact other components
- c) Enable data synchronization across the web application layer
- d) Have the ability to execute automated bootstrapping actions
- e) Increase fault tolerance and flexibility in scaling

Correct Answers:

- b) Reduce inter-dependencies so failures do not impact other components
- e) Increase fault tolerance and flexibility in scaling

Explanation (Correct Choices):

- b) Decoupling helps prevent cascading failures—if one component fails, others can continue functioning.
- e) Decoupled systems allow components to scale independently, enhancing fault tolerance and resource optimization.

<https://docs.aws.amazon.com/whitepapers/latest/microservices-on-aws/microservices-on-aws.pdf>

Incorrect Answers Explained:

- a) Tightly integrated applications are the opposite of decoupled; they increase failure impact and reduce flexibility.
- c) Data synchronization is not the goal of decoupling. The focus is on reducing direct dependencies.
- d) Automated bootstrapping (e.g., using EC2 user data scripts) is a deployment feature, not a benefit of decoupling the architecture.

Q115:

A company is designing a mission-critical application that must remain highly available even during failures in part of the infrastructure. The solutions architect recommends deploying the application across two Availability Zones (AZs) to meet this requirement.

What is the key benefit of deploying the application across multiple Availability Zones?

- a)** Performance is improved over running in a single Availability Zone
- b)** It is more secure than running in a single Availability Zone
- c)** It significantly reduces the total cost of ownership versus running in a single Availability Zone
- d)** It increases the availability of an application compared to running in a single Availability Zone

Correct Answer:

d) It increases the availability of an application compared to running in a single Availability Zone

Explanation (Correct):

Running your application across multiple Availability Zones provides redundancy and fault tolerance. If one AZ goes down, the other can continue serving traffic, which improves overall application availability.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-regions-availability-zones.html>

Incorrect Answers Explained:

- a) Performance gains are not guaranteed just by spanning AZs; latency may even slightly increase due to cross-AZ communication.
- b) Security posture is not automatically improved by spreading across AZs. Security depends on IAM, encryption, and security group settings.
- c) Costs may increase, not decrease, since you're running resources in more than one zone.

Q116:

A security engineer is reviewing their organization's responsibilities under the AWS Shared Responsibility Model. The engineer must ensure that security settings under the customer's control are properly configured while AWS manages the underlying infrastructure.

**Which TWO responsibilities fall under the control of the AWS customer?
(Choose TWO)**

- a) Password policies
- b) User permissions
- c) Physical security of data centers
- d) Disk disposal
- e) Hardware patching

Correct Answers:

- a) Password policies
- b) User permissions

Explanation:

- a) Password policies: Customers define and enforce IAM password policies such as complexity rules and rotation periods.

- b) User permissions: Customers are responsible for managing IAM users, groups, and the policies that govern their access to AWS resources.

<https://aws.amazon.com/compliance/shared-responsibility-model/>

Incorrect Answers Explained:

- c) Physical security: This is AWS's responsibility—they manage physical data center access and security.
- d) Disk disposal: AWS handles the secure decommissioning and physical destruction of storage media.
- e) Hardware patching: AWS is responsible for maintaining and patching the underlying hardware infrastructure.

Q117:

A startup is launching a new e-commerce platform on AWS and expects fluctuating traffic, especially during seasonal sales. The cloud architect recommends applying the cloud architecture principle of elasticity to ensure optimal performance and cost-efficiency during these traffic spikes.

How does designing for elasticity benefit the application in this scenario?

- a) Minimize storage requirements by reducing logging and auditing activities
- b) Create systems that scale to the required capacity based on changes in demand
- c) Enable AWS to automatically select the most cost-effective services
- d) Accelerate the design process because recovery from failure is automated, reducing the need for testing

Correct Answer:

- b) Create systems that scale to the required capacity based on changes in demand

Explanation (Correct):

Elastic systems can automatically scale up or down in response to demand, which is ideal for unpredictable or bursty workloads like online retail. This optimizes both performance and cost.

<https://docs.aws.amazon.com/wellarchitected/latest/framework/wellarchitected-framework.pdf>

Incorrect Answers Explained:

- a) Logging and auditing are unrelated to elasticity; reducing them may compromise compliance or troubleshooting.
- c) AWS offers cost tools, but elasticity is about scaling, not selecting services.
- d) Elasticity and automated recovery improve reliability, but don't eliminate the need for testing or significantly accelerate the design process.

Q118:

A data analytics company is looking to reduce its compute costs on AWS. They have several batch processing and data transformation jobs that run multiple times a day but do not require guaranteed uptime. A cloud engineer recommends using Amazon EC2 Spot Instances to handle these workloads.

Which TWO characteristics make Spot Instances appropriate for this scenario?
(Choose TWO)

- a) Workloads that are only run in the morning and stopped at night
- b) Workloads where the availability of the Amazon EC2 instances can be flexible
- c) Workloads that need to run for long periods of time without interruption
- d) Workloads that are critical and need Amazon EC2 instances with termination protection
- e) Workloads that can tolerate interruptions and be restarted from checkpoints

Correct Answers:

- b) Workloads where the availability of the Amazon EC2 instances can be flexible
- e) Workloads that can tolerate interruptions and be restarted from checkpoints

Explanation:

- b) Spot instances are ideal when you can be flexible about when your workloads run, as they may be interrupted when capacity is no longer available.
- e) They are most effective for interruptible workloads that can resume from a saved state or checkpoint (like data analysis or rendering).

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-spot-instances.html>

Incorrect Answers Explained:

- a) This describes a predictable schedule, better suited for Scheduled Reserved Instances or Savings Plans, not Spot Instances.
- c) Long-running, non-interruptible workloads are not ideal for Spot, as they could be terminated mid-process.
- d) Spot instances can be terminated at any time by AWS and do not support termination protection, making them unsuitable for critical applications.

Q119:

A new DevOps engineer joins your team and needs to interact with AWS services for provisioning resources, managing configurations, and monitoring infrastructure. The engineer wants to use a web-based interface for visibility and control but also wants to automate some tasks via code.

Which TWO AWS features will allow the engineer to manage AWS services in these ways? (Choose TWO)

- a) AWS Management Console
- b) AWS Application Programming Interface (API)
- c) AWS Software Development Kit (SDK)
- d) Amazon CloudWatch

Correct Answers:

- a) AWS Management Console
- b) AWS Application Programming Interface (API)

Explanation (Correct Answers):

- a) AWS Management Console is a web-based user interface that enables users to visually interact with AWS services, launch resources, view logs, configure security settings, and more.
- b) AWS APIs provide programmatic access to AWS services and are commonly used in automation tools and script.

<https://aws.amazon.com/console/>

Incorrect Answers Explained:

- c) AWS SDKs are programming libraries for various languages (like Python, Java, etc.) and are built on top of AWS APIs. SDKs help simplify API calls, but they are not a web-based UI.
- d) Amazon CloudWatch is a monitoring and observability service, not a management interface. It's used for logging, metrics, and alerting, not for provisioning or managing AWS services.

Q120:

A cloud financial analyst at a company is tasked with reviewing the distribution of AWS costs across departments and services. They need a tool that provides visualizations, filtering by account or service, and cost trends over time to help with budget forecasting and chargebacks.

Which TWO AWS tools can assist in analyzing and displaying AWS spending distribution? (Choose TWO)

- a) AWS Organizations
- b) Amazon DevPay
- c) AWS Trusted Advisor
- d) AWS Cost Explorer

Correct Answers:

- a) AWS Organizations
- d) AWS Cost Explorer

Explanation:

- a) AWS Organizations – Allows consolidated billing and cost allocation across multiple linked accounts. It lets you group and track spend by organizational unit or account.
- d) AWS Cost Explorer – Provides interactive dashboards to view and analyze costs by service, linked account, usage type, and more. Ideal for visualizing spending trends and forecasting future costs.

Incorrect Answers Explained:

- b) Amazon DevPay – An outdated tool used for billing third-party developers. It's not used to analyze your AWS spending.

- c) AWS Trusted Advisor – Focuses on best practices related to security, performance, fault tolerance, and cost optimization recommendations, not cost distribution visualization.

Q121:

A cloud administrator is tasked with strengthening the security of the AWS Management Console for all users in an organization. The goal is to prevent unauthorized access and enhance account protection, especially for accounts with elevated privileges.

Which TWO measures should the administrator implement to secure access to the AWS Management Console?

(Choose TWO)

- a) Apply Multi-Factor Authentication (MFA)
- b) Set up a secondary password
- c) Request root access privileges
- d) Disable AWS console access for users who don't need it

Correct Answers:

- a) Apply Multi-Factor Authentication (MFA)
- d) Disable AWS console access for users who don't need it

Explanation (Correct Answers):

- a) MFA adds an extra layer of security by requiring a second factor (like an OTP from an authenticator app), making unauthorized access significantly harder.
- d) Disabling AWS console access for users who don't need it follows the principle of least privilege, reducing the attack surface by limiting access only to those who require it.

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_mfa.html

Incorrect Answers Explained:

- b) AWS does not support a “secondary password” feature. MFA is the recommended secure practice instead.
- c) Root access should be used sparingly and never requested unless necessary. Encouraging more root access increases security risks.

Q122:

A security engineer is tasked with improving the account protection measures across an organization's AWS environment. As part of the security enhancement, the engineer must enforce Multi-Factor Authentication (MFA) for all AWS users to ensure that access to the environment is properly secured.

Which TWO AWS services can the engineer use to enable and manage MFA?
(Choose TWO)

- a) AWS Identity and Access Management (IAM)
- b) Amazon Elastic Compute Cloud (Amazon EC2)
- c) AWS Config
- d) AWS Management Console

Correct Answers:

- a) AWS Identity and Access Management (IAM)
- d) AWS Management Console

Explanation (Correct Answers):

- a) AWS IAM is the core service used to configure and enforce MFA. IAM lets you enable MFA for users and roles, manage devices, and require MFA for privileged actions.
- d) AWS Management Console provides the user interface where users can register MFA devices and configure MFA settings under their IAM security credentials.

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_mfa.html

Incorrect Answers Explained:

- b) Amazon EC2 is used for virtual servers and has no role in configuring or enforcing MFA for user accounts.
- c) AWS Config tracks resource configuration changes and compliance but does not configure security credentials or MFA

Q123:

A company is building a disaster recovery (DR) plan to ensure business continuity in case of a major infrastructure failure. The cloud architect is asked to design the strategy using AWS services so that critical workloads can be recovered and run independently from the affected environment in the event of a disaster.

Which TWO AWS infrastructure components should the company consider using to launch disaster recovery resources in a separate location?

(Choose TWO)

- a) Subnet
- b) AWS Region
- c) AWS edge location
- d) Amazon Virtual Private Cloud (Amazon VPC)

Correct Answers:

- b) AWS Region
- d) Amazon Virtual Private Cloud (Amazon VPC)

Explanation (Correct Answers):

- b) AWS Region: A DR strategy typically requires launching backup infrastructure in a different Region to protect against regional failures (e.g., natural disasters).
- d) Amazon VPC: For isolation and control, disaster recovery environments are usually created in a separate VPC within the DR Region, allowing for network-level configuration and security.

<https://docs.aws.amazon.com/whitepapers/latest/disaster-recovery-workloads-on-aws/disaster-recovery-workloads-on-aws.html>

Incorrect Answers Explained:

- a) Subnet: Subnets are subdivisions within a VPC and do not provide geographic separation. They cannot serve as independent DR environments.
- c) AWS edge location: These are used for content delivery via CloudFront and not intended for hosting full infrastructure or DR environments.

Q124:

A company is preparing to migrate their infrastructure to the AWS Cloud. The cloud architect has been asked to estimate the Total Cost of Ownership (TCO) to evaluate the financial benefits of the move. As part of this process, the architect must consider all relevant factors that will impact cost savings and ROI.

Which TWO factors should the architect consider when calculating the Total Cost of Ownership (TCO) for the AWS Cloud?

(Choose TWO)

- a) The number of servers migrated to AWS
- b) The number of users migrated to AWS
- c) The number of passwords migrated to AWS
- d) The number of keys migrated to AWS

Correct Answers:

- a) The number of servers migrated to AWS
- b) The number of users migrated to AWS

Explanation (Correct Answers):

- a) Number of servers migrated to AWS: The count of physical or virtual servers directly affects compute, storage, and licensing costs—a critical TCO component.
- b) Number of users migrated to AWS: This impacts the type and volume of services needed, including IAM permissions, support plans, and user-level configurations (e.g., Workspaces, SSO).

<https://aws.amazon.com/tco-calculator/>

Incorrect Answers Explained:

- c) Number of passwords migrated to AWS: Passwords themselves do not contribute to cost. They are related to security practices, not TCO.
- d) Number of keys migrated to AWS: While important for security, migrating cryptographic keys has minimal to no impact on cost estimation in TCO models.

Q125:

A media company is expanding globally and needs to deliver high-quality video content to users in multiple countries with low latency and high transfer speeds. To optimize performance and reduce load on the origin servers, the company wants to implement a global content delivery strategy using AWS services.

Which TWO AWS services can help achieve this goal by acting as or supporting a Content Delivery Network (CDN)?

(Choose TWO)

- a) Amazon SES
- b) Amazon CloudTrail
- c) Amazon CloudFront
- d) Amazon S3

Correct Answers:

- c) Amazon CloudFront
- d) Amazon S3

Explanation (Correct Answers):

- c) Amazon CloudFront is AWS's global CDN service. It caches content at edge locations to deliver it with low latency and high transfer speeds to users worldwide.
- d) Amazon S3 stores the original static content (like images, videos, HTML) that CloudFront can cache and serve through its edge locations globally.

<https://aws.amazon.com/cloudfront/>

Incorrect Answers Explained:

- a) Amazon SES (Simple Email Service) is used for email delivery, not for content distribution or CDN functionality.
- b) Amazon CloudTrail records API calls for auditing and compliance. It does not serve or distribute content to end users.

Q126:

A company performs annual compliance audits and wants to store large volumes of rarely accessed historical data, such as logs and reports. They are looking for a cost-effective storage solution that provides long-term durability but does not require frequent access.

Which TWO AWS services should the company consider for this use case?

(Choose TWO)

- a) Amazon Storage Gateway
- b) Amazon S3 Glacier (now part of Amazon S3 Archive classes)
- c) Amazon EBS
- d) Amazon S3 (Standard or Infrequent Access classes)

Correct Answers:

- b) Amazon S3 Glacier
- d) Amazon S3

Explanation (Correct Answers):

- b) Amazon S3 Glacier is ideal for long-term, infrequent access and archival storage. It offers low-cost storage with retrieval times ranging from minutes to hours.
- d) Amazon S3 offers different storage classes, including S3 Standard-Infrequent Access and S3 Intelligent-Tiering, which are suitable for cost-efficient storage of data that is not accessed often

<https://aws.amazon.com/s3/storage-classes/glacier/>

Incorrect Answers Explained:

- a) Amazon Storage Gateway is a hybrid cloud storage service for integrating on-premises environments with AWS. It is not the most cost-effective choice for archive-only data.
- c) Amazon EBS (Elastic Block Store) is designed for block storage with high IOPS and low latency. It is more expensive and suited for active, frequently accessed data, not archival.

Q127:

A startup is building a scalable e-commerce platform that requires a fully managed NoSQL database capable of handling millions of read/write requests per second with low latency and high availability. The development team wants to avoid managing infrastructure, scaling, or software patching.

Which TWO database services should they consider using on AWS?

(Choose TWO)

- a) AWS RDS
- b) AWS DynamoDB
- c) AWS Redshift
- d) AWS MongoDB (self-hosted on EC2)

Correct Answers:

- b) AWS DynamoDB
- d) AWS MongoDB (self-hosted)

Explanation (Correct Answers):

- b) AWS DynamoDB is a fully managed NoSQL database provided by AWS. It supports key-value and document-based models, offers single-digit millisecond

latency, automatic scaling, and serverless operation.

- d) AWS MongoDB is not natively provided by AWS as a managed service, but you can run MongoDB on EC2 or use Amazon DocumentDB, which is MongoDB-compatible and fully managed. So technically, MongoDB on AWS (self-managed or via DocumentDB) can be considered, although it's not a native AWS-managed service like DynamoDB.

<https://aws.amazon.com/dynamodb/>

Incorrect Answers Explained:

- a) AWS RDS is a managed relational database service, not a NoSQL solution.
- c) AWS Redshift is a data warehousing service used for OLAP (Online Analytical Processing), not for NoSQL use cases or high-throughput transactional workloads.

Q128:

A media company needs to store archived video footage and compliance-related documents that are rarely accessed, but must be kept for regulatory purposes. They want a cost-efficient solution for long-term storage while still ensuring durability.

Which AWS service offers the most suitable and cost-effective storage option for this use case?

- A.** Amazon Storage Gateway
- B.** Amazon S3 Glacier
- C.** Amazon EBS
- D.** Amazon S3 (Standard)

Correct Answer:

B. Amazon S3 Glacier

Explanation (Correct Answer):

Amazon S3 Glacier is specifically designed for long-term archival storage of infrequently accessed data. It provides very low-cost storage with options for retrieving data within minutes to hours, making it ideal for backup and compliance scenarios.

<https://aws.amazon.com/s3/storage-classes/glacier/>

Incorrect Answers Explained:

- **A. Amazon Storage Gateway:**
This is used for hybrid cloud storage to integrate on-premises environments with AWS, not optimal for long-term archival storage alone.
- **C. Amazon EBS:**
Amazon EBS is for block-level storage attached to EC2 instances, typically used for high-performance tasks, not for archival and rarely accessed data.
- **D. Amazon S3 (Standard):**
While durable and scalable, S3 Standard is more expensive than S3 Glacier and is meant for frequently accessed data, not long-term storage.

Q129:

You are managing an AWS-hosted infrastructure where most of your EC2 instances are running continuously, and their CPU utilization consistently remains between 90–100%. To optimize long-term cost efficiency without sacrificing availability, you want to choose the most cost-effective pricing model.

Which EC2 instance pricing model is best suited for this scenario?

- A. Reserved Instances**
- B. On-Demand Instances**
- C. Spot Instances**
- D. Regular Instances**

Correct Answer:

A. Reserved Instances

Explanation:

- Reserved Instances are ideal when you have predictable and steady workloads (like instances running near 100% all the time).
- By committing to a 1-year or 3-year term, you can save up to 75% compared to On-Demand pricing.

<https://aws.amazon.com/ec2/pricing/reserved-instances/>

Incorrect Answers Explained:

- **B. On-Demand Instances:**
These are more flexible but more expensive when used continuously. Ideal for short-term or unpredictable workloads, not for always-on servers.
- **C. Spot Instances:**
While extremely cost-effective, they can be terminated by AWS with short notice, making them unsuitable for critical workloads that must run 24/7 at high utilization.
- **D. Regular Instances:**
This is a vague term, not an official AWS pricing model. It lacks the specificity of Reserved, Spot, or On-Demand pricing structures.

Q130:

A data analytics company needs to store large amounts of both structured and unstructured data such as logs, images, and application files. They want to use AWS services that allow them to store these files persistently and retrieve them as needed for processing and analysis.

Which TWO AWS services are most suitable for storing files in this use case?

- A. Amazon CloudWatch**
- B. Amazon Simple Storage Service (Amazon S3)**
- C. Amazon Elastic Block Store (Amazon EBS)**
- D. AWS Config**
- E. Amazon Athena**

Correct Answers:

- B. Amazon S3
- C. Amazon EBS

Explanation (Correct Answers):

- **Amazon S3** – A scalable object storage service used to store files such as images, logs, backups, and static website content. It is highly durable and ideal for unstructured data.
- **Amazon EBS** – Block-level storage that acts like a virtual hard drive. It can store files persistently when attached to an EC2 instance. Ideal for structured data or file systems.

Explanation (Incorrect Answers):

- A. Amazon CloudWatch:
Used for monitoring resources and performance metrics—not for storing files.
- D. AWS Config:
Tracks configuration changes of AWS resources; does not offer file storage.
- E. Amazon Athena:
A query service for analyzing data stored in S3; it does not store files itself.

Q131

You work for a company that is migrating to AWS and plans to use Amazon EC2 to host its applications. Your operations team currently maintains golden images of operating systems, which are pre-configured with all required patches, software, and system settings. They want to know the equivalent concept in AWS that would allow them to replicate this approach and easily launch standardized EC2 instances from these templates.

Which AWS service or resource should they use to meet this requirement?

- A. EBS Volumes
- B. EBS Snapshots
- C. Amazon Machine Image (AMI)
- D. EC2 Copies

Correct Answer:

C. Amazon Machine Image (AMI)

Explanation (Correct Answer):

Amazon Machine Image (AMI) is the AWS equivalent of a golden image. It includes an operating system, application server, and applications pre-installed, and can be used to launch new EC2 instances with the same configuration.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AMIs.html>

Explanation (Incorrect Answers):

- A. EBS Volumes:
EBS volumes are block storage devices attached to EC2 instances, but they do not serve as templates for launching new instances.

- **B. EBS Snapshots:**
Snapshots are backups of EBS volumes. While you can create AMIs from snapshots, snapshots alone do not include full bootable configurations like AMIs do.
- **D. EC2 Copies:**
"EC2 Copies" is not a defined AWS concept or resource. It is not used to represent golden images.

Q132

You are designing a payment processing application that will run on AWS. As part of your responsibilities, you must ensure the application complies with Payment Card Industry Data Security Standards (PCI DSS) before deployment.

Which of the following actions will help you meet the PCI compliance requirements for this application? (*Choose TWO*)

- A. Choose AWS services which are PCI compliant**
- B. Ensure the right steps are taken during application development for PCI compliance**
- C. Ensure the AWS Services are made PCI Compliant**
- D. Do an audit after the deployment of the application for PCI compliance**

Correct Answers:

- A. Choose AWS services which are PCI compliant**
- B. Ensure the right steps are taken during application development for PCI compliance**

Explanation for Correct Answers:

- **A. Choose AWS services which are PCI compliant:**
AWS maintains a list of services that are pre-certified for PCI DSS. You must use only those compliant services when handling cardholder data.

Explanation for Incorrect Answers:

- **C. Ensure the AWS Services are made PCI Compliant:**
You cannot modify or certify AWS services yourself. AWS is responsible for service-level compliance, which is already handled and published in their compliance documentation.
- **D. Do an audit after the deployment of the application for PCI Compliance:**
While audits are important, waiting until after deployment can expose you to

compliance risks. PCI requirements should be addressed during design and development phases, not just afterward.

Question 133 (Scenario-Based MCQ – Choose One Answer)

Scenario:

Your team is building a document-sharing platform where users need to upload files, and those files must be downloadable via publicly shareable URLs. The files are expected to be accessed over the web using HTTP or HTTPS.

Which AWS storage service is the most suitable for this requirement?

- A.** Amazon S3
- B.** Amazon Glacier
- C.** Amazon Storage Gateway
- D.** Amazon EBS

Correct Answer: A. Amazon S3

Explanation:

- Amazon S3 (Simple Storage Service) is designed for storing objects (like files, documents, images) and allows access to these objects via public or pre-signed URLs.
- It supports HTTP/HTTPS-based downloads, making it ideal for building web-accessible storage platforms.
- S3 is highly durable, scalable, and integrates with many AWS services.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html>

Explanation for Incorrect Answers:

B. Amazon Glacier:

Designed for long-term archival of infrequently accessed data. Downloading data via URL is not instant and may take hours to retrieve.

C. Amazon Storage Gateway:

Used for hybrid cloud storage—integrates on-premises environments with cloud storage. It's not designed for direct web-based file access via URLs.

D. Amazon EBS:

Block storage that attaches to EC2 instances. It's not accessible over HTTP/HTTPS and doesn't support URL-based file access natively.

Q134.

Your team needs a development and testing environment for an experimental project that will last only 3 months. The usage will be irregular, depending on developer availability and feature testing schedules.

Which EC2 pricing model is the most cost-effective and flexible for this case?

- A. Spot Instances
- B. On-Demand Instances
- C. Reserved Instances with No Upfront Costs
- D. Reserved Instances with Partial Upfront Costs

Correct Answer: B. On-Demand Instances

Explanation:

On-Demand Instances are ideal for short-term and unpredictable workloads, such as development and test environments. They require no long-term commitment and offer full flexibility.

Incorrect Options:

- A: Spot Instances may be interrupted anytime—not ideal for development stability.
- C/D: Reserved Instances are optimized for long-term, predictable workloads and not cost-efficient for 3 months.

Q135.

Your company plans to run automated batch jobs overnight for the next few weeks to test performance on large datasets. You are looking to minimize compute costs without concern for interruptions.

What EC2 pricing model best fits this use case?

- A. Reserved Instances
- B. On-Demand Instances
- C. Spot Instances
- D. Dedicated Hosts

Correct Answer: C. Spot Instances

Explanation:

Spot Instances are highly cost-effective and are great for flexible, fault-tolerant workloads like batch processing. You can save up to 90% over On-Demand pricing.

Incorrect Options:

- A/D: Not suitable for short-term, interruptible jobs.
- B: More expensive than Spot Instances for this use case.

Q136.

You are running a staging environment to simulate production conditions for 6 months. Your usage is steady and predictable, and you're looking for a cost-effective solution.

Which EC2 pricing model is most suitable?

- A. Spot Instances
- B. On-Demand Instances
- C. Reserved Instances (No Upfront)
- D. Reserved Instances (Partial Upfront)

Correct Answer: C. Reserved Instances (No Upfront)

Explanation:

No Upfront Reserved Instances offer a commitment-based discount without initial capital investment—perfect for medium-term, steady use.

Incorrect Options:

- A: Not reliable due to possible interruptions.
- B: Costlier over time.
- D: Partial upfront offers more savings but requires upfront payment.

Q137.

You're working on a short-term software prototype. The app is in early development, and you expect intermittent use over the next couple of months.

Which EC2 instance pricing model offers the best cost-control and flexibility?

- A. Spot Instances
- B. Reserved Instances (1-year term)
- C. On-Demand Instances
- D. Reserved Instances (3-year term)

Correct Answer: C. On-Demand Instances

Explanation:

On-Demand Instances let you pay only when you use them, offering flexibility during frequent stop/start cycles of early development.

Incorrect Options:

- A: Risk of interruption
- B/D: Not cost-effective for short-term, unpredictable workloads

Q138.

You are setting up a new highly available web application on AWS using an Application Load Balancer (ALB). As part of the deployment process, you're required to configure public subnets for the load balancer.

Which of the following is a correct requirement when configuring subnets for the ALB to ensure high availability?

- A. You must specify two private subnets in one Availability Zone.
- B. You must specify one public subnet in each of at least two different Availability Zones.
- C. You can specify multiple public subnets within the same Availability Zone.
- D. You can deploy the ALB in a single public subnet without any Availability Zone restriction.

Correct Answer: B. You must specify one public subnet in each of at least two different Availability Zones.

Explanation:

To make your ALB highly available, AWS requires that it be deployed across at least two Availability Zones, with one public subnet in each. This ensures redundancy and fault tolerance.

Incorrect Answers:

- A: Private subnets do not allow public access, and the ALB requires public subnets to route traffic from the internet.
- C: You can't have multiple public subnets in the same Availability Zone for an ALB.
- D: A single public subnet provides no redundancy or high availability.

Q139.

You are deploying a web application with an Application Load Balancer (ALB) for fault tolerance. What configuration ensures high availability?

- A. You must specify two private subnets in one Availability Zone
- B. You must specify one public subnet in each of at least two different Availability Zones
- C. You can specify multiple public subnets within the same Availability Zone
- D. You can deploy the ALB in a single public subnet without any Availability Zone restriction

Correct Answer: B

Correct Explanation: ALBs require at least two public subnets in different Availability Zones for fault tolerance.

Incorrect Explanations:

A – Private subnets are not suitable for internet-facing ALBs.

C – ALB requires different AZs for HA, not multiple subnets in the same AZ.

D – Using only one AZ defeats high availability.

Q140.

Your application needs to route HTTP and HTTPS traffic. Which feature of the ALB allows this?

- A. Path-based routing
- B. Network Load Balancer
- C. Sticky sessions
- D. Target Groups

Correct Answer: A

Correct Explanation: Path-based routing lets the ALB route traffic to different targets based on URL paths.

Incorrect Explanations:

- B – NLB works at the transport layer, not for HTTP-specific features.
- C – Sticky sessions deal with session persistence, not routing.
- D – Target groups define backend resources, not routing logic.

Q141. ELB

What does an ALB use to determine which target to send the request to?

- A. IP address
- B. Listener rules
- C. Security group
- D. Route table

Correct Answer: B

Correct Explanation: Listener rules define how the ALB forwards traffic to target groups.

Incorrect Explanations:

- A – IP address is not used for routing decisions in ALB.
- C – Security groups control access, not routing.
- D – Route tables are used in VPCs, not by ALBs.

Q142.

Your ALB reports a target as "unhealthy." What does this mean?

- A. The EC2 instance is terminated
- B. The target failed a health check
- C. Security group blocks traffic
- D. The load balancer is misconfigured

Correct Answer: B

Correct Explanation: Targets are marked unhealthy if they fail health checks.

Incorrect Explanations:

- A – Load balancer doesn't terminate instances.
- C – It could cause failure, but health check determines status.
- D – Health checks determine status, not configuration errors.

Q143.

A developer is packaging dependencies separately from their Lambda function code. What AWS feature allows this?

- A. Layers
- B. AMIs
- C. Snapshots
- D. Volumes

Correct Answer: A

Correct Explanation: Lambda layers allow reuse of libraries and dependencies.

Incorrect Explanations:

- B – AMIs are for EC2 instances.
- C – Snapshots are for storage.
- D – Volumes don't apply to Lambda.

Q144.

How does Lambda handle scaling under load?

- A. Launches EC2 instances
- B. Queues requests until capacity frees
- C. Automatically runs multiple instances of the function
- D. Requires manual scaling

 **Correct Answer:** C

Correct Explanation: Lambda automatically scales by running multiple instances.

Incorrect Explanations:

- A – EC2 is unrelated to Lambda autoscaling.
- B – Lambda does not queue unless integrated with services like SQS.
- D – Scaling is automatic in Lambda.

Q145.

Your function needs environment-specific variables. How should you configure this in Lambda?

- A. Use EC2 user data
- B. Use Lambda Layers
- C. Use environment variables
- D. Use VPC subnets

Correct Answer: C

Correct Explanation: Lambda allows setting environment variables for functions.

Incorrect Explanations:

A – User data is for EC2.

B – Layers are for dependencies, not config.

D – VPC affects networking, not environment setup.

Q146.

How does Lambda handle function timeouts?

A. Sends error to CloudWatch

B. Restarts the function

C. Charges for full hour

D. Automatically retries without logging

Correct Answer: A

Correct Explanation: Timeouts result in error logs in CloudWatch.

Incorrect Explanations:

B – It does not restart automatically.

C – Lambda charges per 1ms, not per hour.

D – Logs are always captured.

Q147.

You are using Amazon RDS for your application. Which of the following benefits does RDS provide?

A. Access to the underlying OS

B. Automatic patching and backups

C. Ability to install custom monitoring agents

D. Requires manual scaling

Correct Answer: B

Correct Explanation: RDS handles patching and backups automatically.

Incorrect Explanations:

A – RDS does not allow OS-level access.

C – Custom agents are not permitted.

D – RDS supports both auto and manual scaling.

Q148.

What happens during an RDS multi-AZ failover?

- A. Database becomes read-only
- B. Traffic is switched to a standby instance
- C. Application must be restarted
- D. DNS is updated manually

Correct Answer: B

Correct Explanation: RDS automatically switches to standby in another AZ.

Incorrect Explanations:

- A – The standby becomes fully writable.
- C – No restart is required.
- D – DNS is updated automatically.

Q149.

How can you improve read performance in RDS?

- A. Create a multi-AZ setup
- B. Use Read Replicas
- C. Resize the storage
- D. Add a NAT gateway

Correct Answer: B

Correct Explanation: Read replicas help distribute read load.

Incorrect Explanations:

- A – Multi-AZ is for failover, not performance.
- C – Storage affects capacity, not read throughput.
- D – NAT gateway is unrelated.

Q150.

Which database engines are supported by Amazon Aurora?

- A. SQL Server and MongoDB
- B. MySQL and PostgreSQL
- C. Oracle and Cassandra
- D. Redis and DynamoDB

Correct Answer: B

Correct Explanation: Aurora is compatible with MySQL and PostgreSQL.

Incorrect Explanations:

A/C/D – These engines are not supported by Aurora.

Q151.

How does DynamoDB handle scaling?

- A. Manual sharding
- B. Auto scaling of read/write capacity
- C. Replication to EBS
- D. Using RDS as backend

Correct Answer: B

Correct Explanation: DynamoDB auto-scales based on workload.

Incorrect Explanations:

- A – Sharding is automatic.
- C – EBS is not used by DynamoDB.
- D – RDS is unrelated.

Q152.

You need millisecond read/write access. Which AWS database should you use?

- A. Aurora
- B. DynamoDB
- C. Redshift
- D. Neptune

Correct Answer: B

Correct Explanation: DynamoDB is optimized for low-latency reads/writes.

Incorrect Explanations:

- A – Aurora is not as fast at scale.
- C – Redshift is for analytics.
- D – Neptune is for graphs.

Q153.

What feature of DynamoDB helps with backup and restore?

- A. Dynamo Streams

- B. Point-in-time recovery
- C. Kinesis
- D. EBS snapshots

Correct Answer: B

Correct Explanation: DynamoDB supports PITR for restoring data.

Incorrect Explanations:

- A – Streams are for real-time changes.
- C – Kinesis is unrelated.
- D – EBS is for EC2.

Q154.

You want to detect changes to a DynamoDB table in real time. What should you use?

- A. S3 Events
- B. CloudWatch
- C. DynamoDB Streams
- D. SNS

Correct Answer: C

Correct Explanation: DynamoDB Streams capture changes for processing.

Incorrect Explanations:

- A – S3 Events are for object changes.
- B – CloudWatch logs metrics, not real-time changes.
- D – SNS sends notifications, not change logs.

Q155.

Which of the following is true about Elastic IP addresses?

- A. They are automatically released when not in use
- B. They are charged when not attached
- C. They are dynamic
- D. They change after a reboot

Correct Answer: B

Correct Explanation: You are charged for unused Elastic IPs.

Incorrect Explanations:

- A – You must manually release them.

- C – They are static, not dynamic.
- D – EIPs persist through reboots.

Q156.

Your team needs temporary compute power for testing. What is the best instance type?

- A. Reserved
- B. On-Demand
- C. Spot
- D. Dedicated Host

Correct Answer: B

Correct Explanation: On-Demand is best for short-term, flexible workloads.

Incorrect Explanations:

- A – Reserved is for long-term.
- C – Spot can be interrupted.
- D – Dedicated Hosts are costly.

Q157.

Which workload suits Spot Instances best?

- A. Real-time transaction processing
- B. Mission-critical databases
- C. Batch data processing
- D. Web servers needing high uptime

Correct Answer: C

Correct Explanation: Spot Instances are cost-effective for interruption-tolerant workloads.

Incorrect Explanations:

A/B/D – These require consistent availability.

Q158.

A company runs a steady workload 24/7. What is the most cost-effective option?

- A. Spot Instances
- B. On-Demand
- C. Reserved Instances
- D. Lambda

Correct Answer: C

Correct Explanation: Reserved Instances offer discounted pricing for predictable workloads.

Incorrect Explanations:

A – Spot is not suitable for critical apps.

B – On-demand is costlier over time.

D – Lambda is for event-driven, not consistent compute.

Q159.

An organization runs several EC2 instances inside a VPC using three subnets: one for Development, one for Test, and one for Production. The security team wants to restrict communication across the EC2 instances using Security Groups.

Which of the following options is true for Security Groups in this scenario?

A. You can change a Security Group associated with an instance if the instance is in the running state.

B. You can change a Security Group associated with an instance if the instance is in the hibernate state.

C. You can change a Security Group only if there are no instances associated with it.

D. The only Security Group you can change is the Default Security Group.

Correct Answer: A

Explanation:

You can change the Security Group of a **running** EC2 instance at any time without needing to stop or restart it.

- **A is correct:** Security Groups can be modified on running instances.
- **B is incorrect:** While changes can be made during hibernation, it's not required to be in hibernation to change security groups.
- **C is incorrect:** You can change security group settings even if it is associated with instances.
- **D is incorrect:** All Security Groups (not just the default) can be modified and reassigned.

Q160.

You are deploying a web application that must be highly available. The application is

fronted by an Application Load Balancer (ALB). During deployment, you're required to configure the public subnets for the load balancer.

Which of the following is a correct requirement when configuring subnets for the ALB to ensure high availability?

- A. You must specify two private subnets in one Availability Zone.
- B. You must specify one public subnet in each of at least two different Availability Zones.
- C. You can specify multiple public subnets within the same Availability Zone.
- D. You can deploy the ALB in a single public subnet without any Availability Zone restriction.

Correct Answer: B

Explanation:

An ALB must be deployed in **at least two Availability Zones** to ensure high availability and fault tolerance.

- **A is incorrect:** Private subnets won't allow public access to the ALB.
- **B is correct:** ALB must span at least two public subnets across different AZs.
- **C is incorrect:** You can only specify one subnet per AZ for an ALB.
- **D is incorrect:** Using a single AZ defeats the purpose of high availability.

Q161.

Your company's EC2 instances are grouped by function and reside in different subnets. You want to restrict access between different tiers of the application—for example, only allowing the app tier to talk to the database tier.

What is the most secure and scalable way to manage this using Security Groups?

- A. Use different NACLs for each tier.
- B. Allow IP-based access from the application tier to the database tier.
- C. Reference the application tier's Security Group in the inbound rule of the database tier's Security Group.
- D. Attach the same Security Group to all instances to allow inter-tier communication.

Correct Answer: C

Explanation:

Referencing a Security Group within another Security Group's inbound rules is a **secure and scalable** way to manage inter-instance communication.

- **A is incorrect:** NACLs are not ideal for this fine-grained instance-level control.
- **B is incorrect:** IP-based rules are static and not scalable.
- **C is correct:** This allows only instances in the app tier Security Group to communicate with the DB tier.
- **D is incorrect:** Using the same SG everywhere doesn't enforce isolation.

Q162.

You have a web server hosted on an EC2 instance in a public subnet. The instance's Security Group allows inbound HTTP traffic on port 80, but users cannot access the website.

Which of the following is the MOST likely cause?

- A. The Security Group does not allow outbound traffic.
- B. The instance is using the default VPC.
- C. The subnet's route table is not pointing to the Internet Gateway.
- D. You forgot to assign a key pair to the instance.

Correct Answer: C

Explanation:

To allow external HTTP traffic, the subnet must route to an Internet Gateway.

- **A is incorrect:** Outbound rules typically allow all traffic by default, and inbound is more relevant here.
- **B is incorrect:** Being in a default VPC doesn't automatically block access.
- **C is correct:** Without a route to an Internet Gateway, no public traffic can reach the instance.
- **D is incorrect:** A key pair is for SSH access, not HTTP availability.

Q163.

You have an EC2 instance in a running state. The instance has been accidentally associated with the wrong Security Group. What is the best course of action?

- A. Terminate and relaunch the instance with the correct Security Group.
- B. Stop the instance, change the Security Group, and restart it.

- C. Change the Security Group while the instance is running.
- D. You must detach the ENI before you can modify the Security Group.

Correct Answer: C

Explanation:

You can change the Security Group of a running EC2 instance without stopping it.

- A is incorrect: Termination is unnecessary and disruptive.
- B is incorrect: Stopping is not required to change Security Groups.
- C is correct: Security Groups are dynamic and can be modified during the runtime.
- D is incorrect: Detaching the ENI is not required for Security Group changes.

Q164.

Your company wants to move an existing Oracle database to the AWS Cloud. Which of the following services can help facilitate this move?

- A. AWS Database Migration Service
- B. AWS VM Migration Service
- C. AWS Inspector
- D. AWS Trusted Advisor

Correct Answer: A. AWS Database Migration Service

Explanation:

- **A is correct:** AWS DMS (Database Migration Service) is specifically designed to migrate relational databases like Oracle to AWS with minimal downtime.
- **B is incorrect:** AWS VM Migration Service helps migrate virtual machines, not databases.
- **C is incorrect:** AWS Inspector is a security assessment tool, unrelated to database migration.
- **D is incorrect:** AWS Trusted Advisor gives best practice recommendations but does not perform migrations.

Q165.

A team is planning to migrate their on-premises VMware-based applications to AWS without rearchitecting them. Which AWS service is best suited for this purpose?

- A. AWS DMS
- B. AWS Migration Hub
- C. AWS Application Discovery Service
- D. AWS VM Import/Export

Correct Answer: D. AWS VM Import/Export

Explanation:

- **D is correct:** VM Import/Export allows you to import virtual machines from your on-premises environment into Amazon EC2.
- **A is incorrect:** DMS is for database migration, not full virtual machines.
- **B is incorrect:** Migration Hub provides a central location to track migration, but it doesn't handle the actual VM import.
- **C is incorrect:** Application Discovery Service helps gather information, not perform the actual migration.

Q166.

Your company is preparing to migrate critical workloads to AWS. They want to assess the security posture of these workloads before and after the migration. Which AWS service should they use?

- A. AWS Trusted Advisor
- B. AWS Inspector
- C. AWS Shield
- D. AWS WAF

Correct Answer: B. AWS Inspector

Explanation:

- **B is correct:** AWS Inspector automatically assesses applications for vulnerabilities or deviations from best practices.
- **A is incorrect:** Trusted Advisor provides general recommendations, not detailed security assessments.
- **C is incorrect:** AWS Shield protects against DDoS attacks, but doesn't scan for vulnerabilities.
- **D is incorrect:** AWS WAF protects against web attacks but does not perform assessments.

Q167.

You are moving a mission-critical application to AWS and want to monitor cost, performance, and fault tolerance post-migration. Which service provides checks and recommendations in these areas?

- A. AWS Config
- B. AWS Inspector
- C. AWS Trusted Advisor
- D. AWS CloudTrail

Correct Answer: C. AWS Trusted Advisor

Explanation:

- **C is correct:** AWS Trusted Advisor offers real-time best practices checks for cost optimization, performance, security, and fault tolerance.
- **A is incorrect:** AWS Config tracks configuration changes but doesn't give optimization advice.
- **B is incorrect:** AWS Inspector is only for security assessments.
- **D is incorrect:** AWS CloudTrail logs API activity but does not provide best practice recommendations.

Q168.

You want to continuously monitor configuration changes across your AWS resources and evaluate them for compliance with internal policies. Which AWS service is best suited for this?

- A. AWS Config
- B. AWS CloudTrail
- C. Amazon Inspector
- D. AWS DMS

Correct Answer: A. AWS Config

Explanation:

- **A is correct:** AWS Config continuously records configuration changes and evaluates them against compliance rules.

- **B is incorrect:** AWS CloudTrail tracks API activity but not resource configuration states.
- **C is incorrect:** Amazon Inspector checks for security vulnerabilities, not configuration compliance.
- **D is incorrect:** AWS DMS is for migrating databases, not tracking configurations.

169.

You are planning to serve a web application on the AWS Platform by using EC2 Instances. Which of the following principles would you adopt to ensure that even if some of the EC2 instances crash, you still have a working application?

- A. Using a scalable system
- B. Using an elastic system
- C. Using a regional system
- D. Using a fault tolerant system

Correct Answer: D. Using a fault tolerant system

Explanation:

- **D is correct** because a fault-tolerant system is designed to continue functioning even when some components fail. This is essential for high availability.
- **A is incorrect** because scalability refers to handling increased load, not fault tolerance.
- **B is incorrect** because elasticity refers to automatically adjusting resources, which doesn't guarantee fault tolerance.
- **C is incorrect** because regional systems refer to location coverage, not necessarily resilience.

170.

A company wants to migrate an existing Oracle database to AWS with minimal downtime. Which AWS service should they use?

- A. AWS Database Migration Service
- B. AWS VM Migration Service

- C. AWS Inspector
- D. AWS Trusted Advisor

Correct Answer: A. AWS Database Migration Service

Explanation:

- **A is correct** because AWS DMS is purpose-built for migrating databases with minimal downtime.
- **B is incorrect** as it deals with VM migration, not databases.
- **C is incorrect** because AWS Inspector is used for security assessments.
- **D is incorrect** because Trusted Advisor provides recommendations, not migration services.

171.

Your team is tasked with launching a web application that must remain available even during infrastructure failures. What is the best AWS architectural principle to adopt?

- A. Use Amazon S3 to store logs
- B. Use a single AZ deployment
- C. Deploy in multiple Availability Zones
- D. Use AWS IAM to secure access

Correct Answer: C. Deploy in multiple Availability Zones

Explanation:

- **C is correct** because distributing workloads across multiple AZs ensures high availability and fault tolerance.
- **A is incorrect** as S3 is for storage, not availability.
- **B is incorrect** because a single AZ poses a risk of downtime if it fails.
- **D is incorrect** because IAM is about access control, not availability.

172. You need to enable developers to programmatically interact with AWS services using familiar programming languages. Which AWS feature should you recommend?

- A. AWS Management Console
- B. AWS SDK
- C. AWS CLI
- D. AWS CloudShell

Correct Answer: B. AWS SDK

Explanation:

- **B is correct** because the AWS Software Development Kit (SDK) allows developers to interact with AWS services in various programming languages.
- **A is incorrect** because the console is graphical, not programmatic.
- **C is incorrect** because the CLI is for command-line interaction, not integration into applications.
- **D is incorrect** as CloudShell is a browser-based CLI terminal.

173. A company wants to monitor the CPU utilization of its EC2 instances. Which AWS service should be used?

- A. AWS CloudTrail
- B. Amazon CloudWatch
- C. AWS Config
- D. AWS Shield

Correct Answer: B. Amazon CloudWatch

Explanation:

- **B is correct** because CloudWatch provides metrics such as CPU utilization, disk I/O, and memory usage.
- **A is incorrect** because CloudTrail tracks API activity.
- **C is incorrect** because AWS Config tracks configuration changes.
- **D is incorrect** because Shield is a DDoS protection service.

174.

You are running several EC2 instances in a VPC across multiple subnets for development,

testing, and production. The security team has requested that you restrict communication between these instances using Security Groups.

Which of the following statements about Security Groups is correct?

- A) You can change a Security Group associated with an instance if the instance is in the running state.
- B) You can change a Security Group associated with an instance if the instance is in the hibernate state.
- C) You can change a Security Group only if there are no instances associated to it.
- D) The only Security Group you can change is the Default Security Group.

Correct Answer: A

Explanation:

You can change the security groups associated with a running EC2 instance at any time. This allows you to update or restrict access dynamically without stopping the instance.

Incorrect Answers:

- B) Hibernate state is not a requirement for changing security groups.
- C) Security groups can be modified even if they are attached to instances.
- D) All security groups can be changed, not just the default one.

175.

Your company wants to migrate its Oracle database to AWS. Which service should you use to simplify the migration?

- A) AWS Database Migration Service
- B) AWS VM Migration Service
- C) AWS Inspector
- D) AWS Trusted Advisor

 **Correct Answer: A**

Explanation:

AWS Database Migration Service (DMS) helps you migrate databases to AWS easily and securely, supporting homogenous (Oracle to Oracle) and heterogeneous migrations.

Incorrect Answers:

- B) AWS VM Migration Service is for virtual machines, not databases.
- C) AWS Inspector is for security assessments, not migration.
- D) AWS Trusted Advisor provides best practice checks, not migration.

176.

You are deploying a highly available web application using EC2 instances. To ensure that your application remains accessible even if some instances fail, which principle should you apply?

- A) Using a scalable system
- B) Using an elastic system
- C) Using a regional system
- D) Using a fault tolerant system

Correct Answer: D

Explanation:

A fault-tolerant system is designed to continue operating properly even if some components fail. This ensures high availability.

Incorrect Answers:

- A) Scalability refers to the ability to handle increased load, not failure.
- B) Elasticity means dynamically adjusting resources, not fault tolerance.
- C) “Regional system” is not a defined design principle in AWS.

177.

Your security team requires that changes to AWS resources be tracked for compliance and auditing. Which service should you use?

- A) Amazon CloudWatch
- B) AWS Config
- C) Amazon Inspector
- D) AWS Trusted Advisor

 **Correct Answer: B**

Explanation:

AWS Config enables you to assess, audit, and evaluate the configurations of your AWS resources and tracks changes over time.

Incorrect Answers:

- A) CloudWatch is for monitoring metrics and logs.
- C) Inspector focuses on security vulnerabilities.
- D) Trusted Advisor offers best practice checks but doesn't track resource changes.

178.

You are designing a backup strategy for your EC2 instances using EBS volumes. Which service provides a fully managed solution to automate and centralize this backup?

- A) AWS Backup
- B) Amazon S3
- C) AWS CloudTrail
- D) AWS Config

Correct Answer: A

Explanation:

AWS Backup provides centralized backup management for EBS, RDS, DynamoDB, and other services, enabling policy-based backup scheduling and retention.

Incorrect Answers:

- B) Amazon S3 is an object storage service, not a backup management tool.
- C) CloudTrail records API activity, not backups.
- D) AWS Config monitors configurations but doesn't back up data.

179.

Your company is hosting multiple web applications on a single EC2 instance. You want to ensure that requests are routed to different applications based on the URL path (e.g., /app1, /app2).

Which feature of Application Load Balancer should you use?

- A) IP-based routing
- B) Path-based routing
- C) Port-based routing
- D) Protocol-based routing

Correct Answer: B

Explanation:

ALB supports path-based routing, which allows routing of requests to different applications on the same instance depending on the URL path.

Incorrect Answers:

- A) IP-based routing is used for routing to different backend IPs.

- C) Port-based routing is not a native ALB feature.
- D) Protocol-based routing is used in Network Load Balancers.

180.

You need to route requests to different applications hosted on the same EC2 instance, based on the hostname in the request (e.g., api.example.com and web.example.com).

Which feature will help achieve this?

- A) Host-based routing
- B) Port forwarding
- C) Lambda target group
- D) Proxy protocol support

Correct Answer: A

Explanation:

Host-based routing allows you to route traffic based on the Host header, enabling multiple applications to be served from the same instance under different domain names.

Incorrect Answers:

- B) Port forwarding is not managed by ALB.
- C) Lambda targets are not relevant to hostname-based routing.
- D) Proxy protocol support is for preserving client IP, not routing.

181.

Your company is running workloads on-premises, and you want to use ALB to route traffic to servers outside of your VPC.

Which ALB feature should be used to register these external targets?

- A) Host-based routing
- B) IP address-based target registration
- C) Lambda function integration
- D) EC2 instance auto-discovery

Correct Answer: B

Explanation:

ALB allows you to register IP addresses as targets, including addresses outside the VPC, making it suitable for hybrid or on-premises use.

Incorrect Answers:

- A) Host-based routing affects how requests are routed, not where.
- C) Lambda targets are for serverless compute, not external IPs.
- D) EC2 instance auto-discovery is not an ALB feature.

182.

You are deploying a containerized application using Amazon ECS and need to ensure that each container can be targeted individually by the load balancer.

Which ALB feature allows this?

- A) Classic Load Balancer
- B) Registering containers using ECS with service discovery
- C) IP-based routing
- D) Container-based target registration via dynamic ports

Correct Answer: D

Explanation:

ALB supports container-based target registration, where it can route traffic to individual containers using dynamic ports exposed by ECS.

Incorrect Answers:

- A) Classic Load Balancer lacks container-awareness.
- B) Service discovery is related but not specific to ALB routing.
- C) IP-based routing does not account for multiple containers on the same host.

183.

You want to redirect HTTP traffic to HTTPS to improve your application's security posture. Which ALB feature can help enforce this redirection?

- A) Path-based routing
- B) Host-based routing
- C) ALB redirect rules
- D) Target group stickiness

Correct Answer: C

Explanation:

ALB supports redirect rules, which allow you to redirect requests (e.g., from HTTP to HTTPS) based on rules defined at the listener level.

Incorrect Answers:

- A) Path-based routing is used for URL-based routing, not redirection.
- B) Host-based routing determines routing, not redirection.
- D) Stickiness ensures session persistence, unrelated to redirection.

184.

You are deploying an internal application and want to authenticate users before requests are routed to the backend. You want to use either Amazon Cognito or a third-party OIDC provider.

Which ALB feature enables this?

- A) Web Application Firewall (WAF) integration
- B) Listener rules
- C) OIDC authentication support in ALB
- D) Lambda authorizers

Correct Answer: C

Explanation:

ALB supports OIDC and Amazon Cognito user authentication directly, allowing user authentication at the load balancer level before reaching the backend.

Incorrect Answers:

- A) WAF protects against web exploits, not used for authentication.
- B) Listener rules route traffic but don't handle authentication.
- D) Lambda authorizers are used with API Gateway, not ALB.

185.

You need to host a public-facing web application in AWS. The web server must be accessible via the internet. Which subnet and gateway configuration is required?

- A) Private subnet with Internet Gateway
- B) Public subnet with Internet Gateway
- C) VPN-only subnet with NAT Gateway
- D) Private subnet with NAT Gateway

Correct Answer: B

Explanation:

A public subnet with an Internet Gateway and proper route table allows inbound internet traffic.

- A) Private subnets cannot directly receive internet traffic.
- C) VPN-only subnets are for VPN connections, not public access.
- D) Private subnet with NAT only allows outbound internet, not inbound.

186.

Your private subnet's EC2 instances need outbound internet access to download updates. What solution ensures this while keeping instances unreachable from the internet?

- A) Attach an Elastic IP to each instance
- B) Create a NAT Gateway in a public subnet
- C) Associate the subnet with a public route table
- D) Add Internet Gateway to the private subnet

Correct Answer: B

Explanation:

A NAT Gateway in a public subnet enables outbound internet access for private subnet instances.

- A) Elastic IP provides inbound access.
- C) Public route table allows inbound.
- D) Internet Gateway on private subnet breaks isolation.

187.

You are connecting two VPCs owned by different AWS accounts. You want to keep routing private between them. Which AWS feature should you use?

- A) Internet Gateway
- B) VPC Peering
- C) Virtual Private Gateway
- D) Transit Gateway

Correct Answer: B

Explanation:

VPC Peering provides private connectivity between VPCs in the same or different accounts.

- A) Internet Gateway is for internet access.
- C) Virtual Private Gateway is for on-prem VPN.
- D) Transit Gateway supports hub-and-spoke across multiple VPCs and accounts (but more complex than needed).

188.

Which CIDR range is valid for creating a subnet within 10.0.0.0/16?

- A) 10.1.0.0/16
- B) 10.0.0.0/8
- C) 10.0.100.0/24
- D) 10.0.0.0/15

Correct Answer: C

Explanation:

10.0.100.0/24 fits within the parent /16 block.

- A) /16 equals the parent and cannot be nested.
- B) /8 is broader than the parent.
- D) /15 overlaps but is larger than the parent.

189.

A subnet in your VPC has no associated route table. Which route table will it use?

- A) Internet route table
- B) No route table – traffic blocked
- C) Main route table
- D) Randomly assigned route table

Correct Answer: C

Explanation:

Unassociated subnets default to the main route table.

- A) Internet route requires explicit association.

- B) Default traffic path exists via main route table.
- D) AWS does not randomly assign route tables.

190.

You want to allow instances in a private subnet to receive IPv6 internet access. What do you use?

- A) NAT Gateway
- B) Egress-Only Internet Gateway
- C) Internet Gateway
- D) Virtual Private Gateway

Correct Answer: B

Explanation:

Egress-Only Internet Gateway enables outbound IPv6 without inbound access.

- A) NAT only handles IPv4.
- C) Internet Gateway would allow both inbound and outbound.
- D) Virtual Private Gateway is for VPN.

191.

You plan to connect your on-prem network to AWS over IPsec VPN. Which component must you attach to your VPC?

- A) Internet Gateway
- B) NAT Gateway
- C) Virtual Private Gateway
- D) Egress-Only Internet Gateway

Correct Answer: C

Explanation:

Virtual Private Gateway is required for AWS VPN connections.

- A) Internet Gateway is for internet, not VPN.
- B) NAT Gateway is for outbound IPv4.
- D) IPv6 only, not needed for VPN.

192.

Which VPC component controls traffic at the instance level?

- A) Route Table
- B) Network ACL
- C) Security Group
- D) Internet Gateway

Correct Answer: C

Explanation:

Security Groups act as virtual firewalls for instances.

- A) Route tables control subnet traffic.
- B) NACLs control subnet-level traffic.
- D) Internet Gateway enables internet access.

193.

Which VPC component provides stateless inbound and outbound filtering at the subnet boundary?

- A) Security Group
- B) Network ACL
- C) Route Table
- D) Internet Gateway

Correct Answer: B

Explanation:

NACLs are stateless and apply at the subnet level.

- A) Security Groups are stateful.
- C) Route tables manage routing, not filtering.
- D) IGW is a gateway, not a firewall.

194.

You created a VPC with a /16 CIDR. How many non-overlapping /24 subnets can you carve?

- A) 16
- B) 64
- C) 256
- D) 512

Correct Answer: C

Explanation:

A /16 divides into 256 /24 subnets.

- A & B) Too few.
- D) Larger than available addresses.

195.

You need to assign static IPv4 addresses to EC2 instances. Which service should you use?

- A) Private IP assignments
- B) Elastic IP addresses
- C) DHCP options set
- D) Amazon Route 53

Correct Answer: B

Explanation:

Elastic IP addresses provide static, public IPv4 addresses.

- A) Private IPs are local to the VPC.
- C) DHCP options are for domain info.
- D) Route 53 provides DNS, not IP assignment.

196.

Your VPC needs to support IPv6 for both clients and servers. What must you configure?

- A) IPv6 CIDR block and dual-stack subnets
- B) NAT Gateway
- C) Egress-only IGW only
- D) Transit Gateway

Correct Answer: A

Explanation:

Assign IPv6 CIDR at VPC level and enable dual-stack in subnets.

- B) NAT is for IPv4.
- C) Only outbound, lacks IPv4.
- D) Transit Gateway is for network connectivity.

197.

Two VPCs are peered. Can instances in VPC A access public internet through VPC B's Internet Gateway via the peering connection?

A) Yes

B) No

Correct Answer: B

Explanation:

Peering does not support gateway transit. Instances cannot route internet via peer's IGW.

198.

Which VPC feature helps you monitor IP traffic to and from network interfaces?

A) CloudWatch Logs

B) Flow Logs

C) CloudTrail

D) Config

Correct Answer: B

Explanation:

VPC Flow Logs capture network interface traffic metadata.

- A) CloudWatch is for metrics/logs.
- C) CloudTrail logs API calls.
- D) Config tracks configuration changes.

199.

Your Network ACL allows inbound SSH (22) and outbound HTTPS (443). Client connects via SSH. Will return traffic be allowed?

- A) Yes
- B) No

Correct Answer: B

Explanation:

NACLs are stateless; return traffic must be allowed by the inbound and outbound rules. Without outbound port 1024+ allowed, return traffic is blocked.

200.

Which option allows you to extend your VPC IP address range later?

- A) Add a secondary IPv4 CIDR block
- B) Increase the existing CIDR mask
- C) Change the CIDR entirely
- D) Delete and recreate the VPC

Correct Answer: A

Explanation:

You can add secondary CIDR blocks to expand IP range.

- B) Masks cannot be changed.
- C/D) Not possible without recreation.

201.

What is the AWS-recommended maximum size for a single VPC CIDR block?

- A) /16
- B) /24
- C) /12
- D) /8

Correct Answer: A

Explanation:

AWS allows a maximum /16 CIDR per VPC. Larger ranges are not supported.

202.

You need out-of-service targeting for unhealthy hosts in a load-balanced VPC. Where should health checks be configured?

- A) Security Groups
- B) Route Tables
- C) Application Load Balancer
- D) Network ACL

Correct Answer: C

Explanation:

ALB health checks determine target availability.

- Others do not perform health checks.

203.

You want to enforce that all subnets have tags and follow naming standards. Which VPC feature helps enforce this?

- A) CloudFormation
- B) AWS Config rules
- C) IAM policies
- D) Route Table associations

Correct Answer: B

Explanation:

AWS Config can enforce tagging compliance via rules.

- IAM governs permissions, not tags.
- CFN is for deployment templates.
- Route tables do not enforce tags.

204.

Your VPC spans multiple AZs. You want to enable cross-zone communication but restrict cross-subnet traffic. Which can achieve this?

- A) Use NACLs to block cross-AZ traffic
- B) Use Security Groups referencing specific subnets

- C) Use subnet route tables to isolate traffic
- D) It's not possible to restrict cross-subnet traffic within the same VPC

Correct Answer: B

Explanation:

Security Groups can allow traffic only from instances in certain subnets (via SG referencing).

- A) NACLs are subnet-wide, not AZ-based.
- C) Route tables are for routing, not filtering.
- D) It is possible via SG rules.

205.

Your application needs a DNS service that includes health checks and automatic failover between multiple endpoints. Which Route 53 feature should you use?

- A) Simple routing
- B) Geolocation routing
- C) Failover routing with health checks
- D) Latency-based routing

Correct Answer: C

Explanation:

Failover routing with health checks lets Route 53 detect unhealthy endpoints and route to healthy alternatives.

- A) Simple routing offers no health checks or failover.
- B) Geolocation routing sends traffic based on user location, not endpoint health.
- D) Latency-based routing directs to lowest latency, not health.

206.

Your CloudFront distribution should serve only over HTTPS and redirect all HTTP traffic to HTTPS. Which setting should you configure?

- A) Viewer Protocol Policy = Redirect HTTP to HTTPS
- B) Origin Protocol Policy = HTTPS Only

- C) SSL Certificate = Default CloudFront certificate
- D) Security Policy = TLSv1

Correct Answer: A

Explanation:

“Redirect HTTP to HTTPS” ensures viewer traffic is redirected to secure connections.

- B) Origin protocol is between CF and origin, irrelevant to viewer.
- C) Certificate selection does not enforce redirect.
- D) Security policy defines TLS versions, not redirect behavior.

207.

You need to deploy infrastructure via IaC, with parameter validation and resource dependency. Which CloudFormation feature helps define conditions and dependencies?

- A) Mappings
- B) Conditions and DependsOn
- C) Outputs
- D) Transform

Correct Answer: B

Explanation:

Conditions allow conditional resource creation; DependsOn ensures order.

- A) Mappings provide static values.
- C) Outputs export values, not manage creation logic.
- D) Transform integrates macros or nested stacks.

208.

Your CloudTrail logs must be centralized and encrypted with SSE-KMS for compliance. Which settings are required?

- A) Set CloudTrail to use S3 encryption
- B) Configure KMS CMK and enable log file encryption
- C) Use default S3 bucket
- D) Enable CloudWatch Logs only

Correct Answer: B

Explanation:

Creating a KMS CMK and enabling CloudTrail encryption ensures compliant log storage.

- A) CloudTrail doesn't have a simple "S3 encryption" option.
- C) Default bucket is not encrypted by default.
- D) CloudWatch Logs doesn't encrypt S3 storage.

209.

You have an application issue and need to monitor EC2 memory and disk usage. Which best approach?

- A) CloudWatch default metrics
- B) Install CloudWatch agent
- C) Use CloudTrail logs
- D) CloudFormation drift detection

Correct Answer: B

Explanation:

Installing the CloudWatch agent provides memory and disk metrics.

- A) CloudWatch default lacks memory/disk.
- C) CloudTrail logs API calls, not performance.
- D) Drift detection is for config management.

210.

A security audit requires visibility into every IAM user's API activity. Which service should you enable?

- A) CloudWatch Events
- B) CloudTrail multi-region, all-events
- C) CloudFront logs
- D) Route 53 query logs

Correct Answer: B

Explanation:

CloudTrail records all API activity across regions for IAM users.

- A) Events capture changes, not full API log.
- C/D) CloudFront and Route 53 logs don't cover IAM API calls.

211.

Your CloudFormation template needs to include reusable sub-components that can be maintained independently. Which feature helps achieve that?

- A) AWS::Include transform
- B) Nested stacks
- C) Mappings
- D) Custom resources

Correct Answer: B

Explanation:

Nested stacks allow modular and reusable cloud templates.

- A) Include is for code snippets.
- C) Mappings are for static values.
- D) Custom resources invoke Lambda for custom logic, not reuse.

212.

You want to send CloudWatch Alarms to a Slack channel via SNS. Which IAM permission is required?

- A) cloudwatch:PutMetricAlarm
- B) sns:Publish
- C) logs:CreateLogStream
- D) cloudtrail:StartLogging

Correct Answer: B

Explanation:

SNS needs publish permissions to route alarm notifications.

- A) Creates alarms but not sending alerts.
- C) Logs relate to CloudWatch Logs, not SNS.
- D) CloudTrail logging is unrelated.

213.

Your CloudFront distribution isn't caching dynamic content sufficiently. Which feature should you adjust?

- A) Increase TTL in Cache Behavior
- B) Change SSL protocol
- C) Enable IPv6 support
- D) Use origin path rewrite

Correct Answer: A

Explanation:

Increasing TTL ensures CloudFront caches content longer.

- B) SSL doesn't affect caching.
- C) IPv6 is for connectivity only.
- D) Path rewrites don't impact caching.

214.

Which Route 53 record type supports aliasing to AWS resources like CloudFront or S3?

- A) CNAME only
- B) A (alias) or AAAA (alias)
- C) MX record
- D) PTR record

Correct Answer: B

Explanation:

A/AAAA alias records resolve directly to AWS resources without extra charges.

- A) Alias is required for root domain to AWS resources.
- C/D) MX and PTR aren't used for aliasing.

215.

Your CloudTrail logs are delivered to S3 but not visible in CloudWatch Logs. What must you do?

- A) Set up a CloudWatch Logs log group and subscription in the trail
- B) Re-enable CloudTrail
- C) Create a CloudWatch Alarm
- D) Add SNS notification

Correct Answer: A

Explanation:

You must explicitly configure CloudTrail to deliver logs to CloudWatch Logs.

- B) Trail is working.
- C) Alarms react, not transport logs.
- D) SNS alerts, not log transport.

216.

You want to deploy identical environments in multiple regions using CloudFormation. What should you do?

- A) Manually copy templates
- B) Use Change Sets
- C) Use StackSets
- D) Enable Drift Detection

Correct Answer: C

Explanation:

StackSets allow deploying a template across multiple accounts/regions.

- A) Error-prone.
- B) ChangeSets preview changes, not cross-region deployment.
- D) Drift detection checks drift only.

217.

Your CloudWatch dashboard shows zeros for API call metrics. Which log source should you verify?

- A) CloudTrail is logging management events
- B) CloudFront access logs

- C) Route 53 logs
- D) CloudFormation logs

Correct Answer: A

Explanation:

CloudTrail provides API call tracking data for CloudWatch metric filters.

- B/C/D) Other logs unrelated to AWS API activity.

218.

Your CloudFront distribution uses geo restriction to block a country. Which policy type should you use?

- A) Whitelist
- B) Blacklist
- C) Origin Access Identity
- D) Edge function

Correct Answer: B

Explanation:

Blacklist blocks selected countries from accessing the distribution.

- A) Whitelist allows only specified countries.
- C) OAI restricts S3 access.
- D) Edge function can customize logic but isn't needed here.

219.

You need to deploy EC2 instances with custom IAM roles using CloudFormation. Which section in the template defines those roles?

- A) Mappings
- B) Resources
- C) Parameters
- D) Outputs

Correct Answer: B

Explanation:

IAM roles are AWS resources and go in the Resources section.

- A) Only static value mapping.
- C) Used for external input.
- D) Used to output results.

220.

Your application must track S3 object changes through Lambda invocation. Which CloudWatch feature helps?

- A) Metric filters
- B) EventBridge (CloudWatch Events)
- C) Alarms
- D) Logs Insights

Correct Answer: B

Explanation:

EventBridge can detect S3 events and trigger Lambda functions.

- A) Filters logs to metrics.
- C) Alarms trigger based on thresholds.
- D) Logs Insights analyze log data.

221.

Route 53 queries are not being logged for your VPC Resolver. What must be enabled?

- A) Route 53 health checks
- B) Query logging in Resolver
- C) Server-side encryption
- D) CloudTrail on Route 53

Correct Answer: B

Explanation:

Query logging must be explicitly enabled for Route 53 Resolver.

- A) Health checks are not logs.
- C) Encryption doesn't control logs.
- D) CloudTrail logs configuration, not queries.

222.

Your CloudFormation stack fails on a resource you no longer need. What is the best way to remove it without impacting the rest?

- A) Delete the entire stack
- B) Use Change Set to remove the resource
- C) Modify template and deploy with update
- D) Roll back entire change set

Correct Answer: B

Explanation:

Change Sets let you preview and apply selective deletions.

- A) Overkill.
- C) Would remove resource but no preview.
- D) Rollback discards all modifications.

223.

Your CloudFront origin returns 500 errors for large files. You need to retry automatically. Where do you configure this behavior?

- A) on the origin server
- B) CloudFront origin settings
- C) Route 53 TTL
- D) CloudWatch retry policy

Correct Answer: B

Explanation:

CloudFront has built-in origin retry settings.

- A) Server might retry, but CF settings needed.
- C) TTL not related.
- D) CloudWatch does not retry.

224.

You need to monitor user login failures across AWS accounts. Which service helps analyze failures over time?

- A) CloudWatch Logs Insights
- B) CloudTrail Insights
- C) Route 53 query logs
- D) CloudFront usage stats

Correct Answer: A

Explanation:

Logs Insights can query login failure events in CloudTrail logs.

- B) CloudTrail Insights warns on anomalies but doesn't analyze history.
- C/D unrelated.

225.

Domain owners want to route traffic based on user IP regions, not instance health. Which Route 53 policy should you choose?

- A) Failover routing
- B) Geoproximity routing
- C) Latency-based routing
- D) Weighted routing

Correct Answer: B

Explanation:

Geoproximity routing directs traffic based on user's geographic location.

- A) Uses health, not location.
- C) Uses latency.
- D) Distributes by weights.

226.

Your CloudTrail audits show no events in some regions. How do you ensure global logging?

- A) Enable multi-region trail
- B) Use CloudTrail Insights

- C) Enable S3 replication
- D) Increase CloudTrail retention

Correct Answer: A

Explanation:

Multi-region trails automatically log events across all AWS regions.

- B) Insights detect anomalies only.
- C/D) Not relevant to log region capture.

227.

You need CloudWatch alarms triggered only during business hours. What feature supports this?

- A) Alarm scheduling by default
- B) Use CloudWatch composite alarms and math expressions
- C) Use CloudWatch anomaly detection
- D) Configure alarm actions only via AWS Config

Correct Answer: B

Explanation:

Composite alarms with math expressions allow conditional triggering based on time dimension.

- A) Doesn't exist.
- C) Detects anomalies, not time based.
- D) Config not for alarm scheduling.

228.

Your CloudFront distribution must block access to URLs with specific query strings. Which feature will help?

- A) Query string caching whitelist
- B) Geo restriction
- C) Custom error pages
- D) WAF integration

Correct Answer: D

Explanation:

AWS WAF lets you block or allow traffic based on query string patterns.

- A) Caching only, not blocking.
- B) Location-based.
- C) Only custom responses on errors.

229.

You are using CloudFormation to deploy a VPC and need to ensure subnet resource deletion order depends on route table association. What should you do?

- A) Use DependsOn attribute for subnet
- B) Use Mappings to order
- C) Use Outputs to reference route tables
- D) Define subnets before route tables

Correct Answer: A

Explanation:

DependsOn ensures resources are created or deleted in the required order.

- B) Mappings contain static data, not ordering.
- C) Outputs share values, not affect order.
- D) Definition order doesn't guarantee resource operations order.

230.

You are monitoring an EC2 instance's memory and disk usage using Amazon CloudWatch. However, you notice that only CPU and network-related metrics are available. What is the most likely reason for this?

- A) CloudWatch only supports CPU metrics for EC2
- B) Detailed monitoring must be enabled to view disk and memory metrics
- C) The CloudWatch Agent has not been installed on the instance
- D) Metrics older than two weeks are deleted by CloudWatch

Correct Answer: C

Explanation:

CloudWatch does not collect memory and disk space usage by default. You must install the CloudWatch Agent to collect these additional metrics.

- A) Incorrect. CloudWatch can support more than CPU metrics with the agent.
- B) Incorrect. Detailed monitoring provides 1-minute granularity but still doesn't include memory/disk.
- D) Incorrect. Metrics expire after 15 months, not two weeks.

231.

You created a custom CloudWatch metric to monitor your application, but it stopped receiving new data two months ago. What will happen to this metric?

- A) It will be deleted immediately
- B) It will be retained indefinitely
- C) It will automatically expire after 15 months if no data is published
- D) It will be archived into S3 for cost efficiency

Correct Answer: C

Explanation:

CloudWatch metrics expire automatically after 15 months of inactivity.

- A) Incorrect. Metrics are not deleted immediately.
- B) Incorrect. CloudWatch doesn't retain inactive metrics forever.
- D) Incorrect. CloudWatch does not automatically archive expired metrics.

232.

You're sending custom metric data points to CloudWatch with timestamps. However, some points are not appearing in the dashboard. What could be a reason?

- A) You didn't enable detailed monitoring
- B) You used timestamps more than two weeks in the past
- C) You didn't assign a region to the metric
- D) CloudWatch only accepts metrics from EC2 instances

Correct Answer: B

Explanation:

CloudWatch accepts timestamps up to two weeks in the past and two hours in the future. Data outside this range will be discarded.

- A) Incorrect. Detailed monitoring is unrelated to custom metrics.
- C) Incorrect. Metrics are region-specific but that's not the cause of the issue.
- D) Incorrect. CloudWatch supports custom metrics beyond EC2.

233.

You want to calculate the average CPU utilization across 5 EC2 instances using CloudWatch. Which feature should you use?

- A) Detailed Monitoring
- B) Metric Math
- C) Composite Alarms
- D) Log Insights

Correct Answer: B

Explanation:

Metric Math allows you to use mathematical expressions on multiple metrics, such as averaging CPU utilization.

- A) Incorrect. Detailed monitoring improves frequency, not computation.
- C) Incorrect. Composite alarms evaluate multiple alarms, not metrics directly.
- D) Incorrect. Log Insights analyzes log data, not metric data.

234.

You are publishing a custom metric to CloudWatch every minute. What will happen if you stop publishing this metric without deleting it?

- A) The metric will remain forever
- B) CloudWatch will alert you immediately
- C) The metric will expire 15 months after the last data point was published
- D) The metric will be automatically converted to a standard metric

Correct Answer: C

Explanation:

CloudWatch metrics automatically expire 15 months after the last data point is received.

- A) Incorrect. Inactive metrics are not stored indefinitely.
- B) Incorrect. CloudWatch does not alert by default unless alarms are configured.
- D) Incorrect. Metrics types don't convert automatically.

235.

Your organization is designing a disaster recovery strategy for a mission-critical web application. You need to ensure that the application remains available even if an entire data center fails. What should you do?

- A) Deploy all application components into a single Availability Zone in a region
- B) Deploy application resources across multiple AWS Regions for fault isolation
- C) Use only AWS edge locations for deploying the application
- D) Enable CloudTrail logs in the AWS Region where the application is hosted

Correct Answer: B

Explanation:

Deploying across multiple AWS Regions provides maximum isolation and fault tolerance in the event of an entire regional failure, supporting disaster recovery strategies.

- A) Incorrect. A single Availability Zone does not protect against data center failure.
- C) Incorrect. Edge locations are used for caching content, not hosting applications.
- D) Incorrect. CloudTrail logging is for auditing, not availability or disaster recovery.

236.

A company is launching a new service that must comply with data sovereignty laws requiring user data to be stored within a specific country. What AWS design choice satisfies this requirement?

- A) Deploy resources in a Region located within that specific country
- B) Use multiple Availability Zones in the nearest AWS Region
- C) Deploy to a Local Zone in a different country
- D) Use Amazon CloudFront to cache user data closer to the country

Correct Answer: A

Explanation:

To meet data sovereignty requirements, AWS recommends using a Region physically located within the country so that data never leaves the jurisdiction.

- B) Incorrect. Availability Zones within another country still violate data sovereignty rules.
- C) Incorrect. Local Zones are extensions of AWS Regions and may be outside the country.
- D) Incorrect. CloudFront is a CDN and caches content temporarily, not for primary data storage.

237.

You are configuring an Application Load Balancer (ALB) for a multi-service application running on multiple EC2 instances. You want the ALB to inspect incoming client requests and forward them to different target groups based on the requested path (e.g., /api, /web). However, before implementing routing rules, you must ensure the listener is properly configured.

What must you do when creating a listener for this configuration to work?

- A) Define an auto-scaling policy that matches each target group
- B) Specify the instance type for the target group in the listener
- C) Define a default rule for the listener with a target group, condition, and priority
- D) Create a Lambda function to map URLs to the correct target group

Correct answer: C) Define a default rule for the listener with a target group, condition, and priority

Explanation (Correct Answer):

A listener is a process that checks for connection requests using the configured protocol and port. When creating a listener on an ALB, you must define a default rule. This rule tells the load balancer where to send traffic when no other rule matches. You can also define additional rules with specific conditions and priorities.

<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-listeners.html>

Explanations (Incorrect Answers):

A) Define an auto-scaling policy that matches each target group

- Auto-scaling helps scale EC2 instances based on load, but it is not required to create or configure a listener.

B) Specify the instance type for the target group in the listener

- Target groups reference EC2 instances or Lambda functions by instance ID or IP address, not by instance type.

D) Create a Lambda function to map URLs to the correct target group

- This is unnecessary. Path-based routing can be configured directly in the ALB using listener rules without custom Lambda logic.

239.

You are deploying an app across multiple Availability Zones using an ALB. For high availability, which requirement must be met when configuring subnets?

A) Deploy ALB in a single public subnet.

B) Use only private subnets in each AZ.

C) Specify one public subnet in at least two AZs.

D) Enable VPC peering for load balancer communication.

Correct Answer: C) Specify one public subnet in at least two AZs

Explanation:

ALB must be in at least two public subnets (one per AZ) for high availability.

- A: Single subnet = no AZ-level redundancy.
- B: Private subnets won't expose ALB to internet traffic.
- D: VPC peering is not required for this setup.

240.

You are running a container-based app in ECS. You want to route traffic to different containers on the same EC2 instance using a single load balancer.

Which feature of ALB helps achieve this?

- A) Path-based routing
- B) Host-based routing
- C) IP-based target registration
- D) Dynamic port mapping

Correct Answer: D) Dynamic port mapping

Explanation:

Dynamic port mapping lets ALB route traffic to multiple containers on the same instance using ephemeral ports.

- A & B: Useful but do not address multiple containers on the same host.
- C: Allows IP registration but doesn't handle port differentiation.

241.

Which ELB feature allows your application to maintain session state across multiple requests from the same user?

- A) Health checks
- B) Sticky sessions
- C) Route 53 integration
- D) SSL offloading

Correct Answer: B) Sticky sessions

Explanation:

Sticky sessions (session affinity) use cookies to ensure a client's traffic is routed to the same backend target.

- A: Used to check target health.
- C: Used for DNS-based load balancing.
- D: SSL offloading handles encryption, not session persistence.

242.

You've noticed that your NLB is not detecting failures in EC2 targets fast enough. What should you do to improve the speed of failure detection?

- A) Increase the health check interval
- B) Decrease the health check timeout and threshold

- C) Enable sticky sessions
- D) Change the listener protocol from TCP to HTTP

Correct Answer: B) Decrease the health check timeout and threshold

Explanation:

Lower timeout and threshold means NLB will mark a target as unhealthy faster.

- A: Increases the delay in detecting failures.
- C: Irrelevant to health checks.
- D: NLB operates at Layer 4; does not support HTTP listeners

243.

Which of the following programming languages is supported by AWS Elastic Beanstalk?

- A) Swift
- B) Python
- C) Perl
- D) COBOL

Correct Answer: B) Python

Explanation: Elastic Beanstalk supports Python among other modern languages. Swift, Perl, and COBOL are not officially supported by Elastic Beanstalk.

244.

A developer wants to deploy a web application written in Go. Which AWS service allows for easy deployment and management of applications developed in Go?

- A) AWS CodePipeline
- B) AWS CloudFormation
- C) AWS Elastic Beanstalk
- D) AWS Snowball

Correct Answer: C) AWS Elastic Beanstalk

Explanation: Elastic Beanstalk supports Go and handles deployment, scaling, and monitoring. The other options are not designed specifically for application deployment.

245. Under the shared responsibility model, which of the following is the customer responsible for?

- A. Ensuring that disk drives are wiped after use
- B. Ensuring that firmware is updated on hardware devices
- C. Ensuring that data is encrypted at rest
- D. Ensuring that network cables are category six or higher

☒ Correct Answer: C. Ensuring that data is encrypted at rest

Explanation:

Customers are responsible for securing their data, including encryption at rest or in transit.

Incorrect Answers:

- A. Wiping disk drives is AWS's responsibility.
- B. Firmware updates are handled by AWS.
- D. Network hardware is managed by AWS.

246. Which services are parts of the AWS serverless platform?

- A. Amazon EC2, Amazon S3, Amazon Athena
- B. Amazon Kinesis, Amazon SQS, Amazon EMR
- C. AWS Step Functions, Amazon DynamoDB, Amazon SNS
- D. Amazon Athena, Amazon Cognito, Amazon EC2

Correct Answer: C. AWS Step Functions, Amazon DynamoDB, Amazon SNS

Explanation:

These services require no server provisioning or management, making them serverless.

Incorrect Answers:

- A. EC2 is not serverless.
- B. EMR is not serverless.
- D. EC2 is not serverless.

247. Which of the following services is in the category of AWS serverless platform?

- A. Amazon EMR
- B. Elastic Load Balancing
- C. AWS Lambda
- D. AWS Mobile Hub

Correct Answer: C. AWS Lambda

Explanation:

AWS Lambda lets you run code without provisioning or managing servers.

Incorrect Answers:

- A. EMR involves server setup.
- B. ELB requires managing resources.
- D. Mobile Hub is a development tool, not a compute platform.

248. An administrator needs to rapidly deploy a popular IT solution and start using it immediately. Where can the administrator find assistance?

- A. AWS Well-Architected Framework documentation
- B. Amazon CloudFront
- C. AWS CodeCommit
- D. AWS Quick Start reference deployments

Correct Answer: D. AWS Quick Start reference deployments

Explanation:

Quick Starts are automated reference deployments to help launch popular workloads fast.

Incorrect Answers:

- A. The Well-Architected Framework is for design guidance, not deployment.
- B. CloudFront is a CDN.
- C. CodeCommit is for source control.

249. One benefit of On-Demand Amazon Elastic Compute Cloud (Amazon EC2) pricing is:

- A. the ability to bid for a lower hourly cost
- B. paying a daily rate regardless of time used
- C. paying only for time used
- D. pre-paying for instances and paying a lower hourly rate

Correct Answer: C. paying only for time used

Explanation:

On-Demand pricing allows you to pay per second or hour, depending on the instance type.

Incorrect Answers:

- A. That describes Spot Instances.

- B. No daily rate model exists.
- D. That applies to Reserved Instances.

250. Which of the following tasks is the responsibility of AWS?

- A. Encrypting client-side data
- B. Configuring AWS Identity and Access Management (IAM) roles
- C. Securing the Amazon EC2 hypervisor
- D. Setting user password policies

Correct Answer: C. Securing the Amazon EC2 hypervisor

Explanation:

AWS manages infrastructure, including physical security and virtualization layers.

Incorrect Answers:

- A. Client-side encryption is customer responsibility.
- B. IAM setup is customer-managed.
- D. Password policies are defined by customers.

251. Which is the MINIMUM AWS Support plan that provides technical support through phone calls?

- A. Enterprise
- B. Business
- C. Developer
- D. Basic

Correct Answer: B. Business

Explanation:

The Business plan is the minimum support tier offering 24x7 phone, chat, and email access.

Incorrect Answers:

- A. Enterprise provides more services but isn't the minimum.
- C. Developer plan only offers email support.
- D. Basic has no technical support.

251. Which of the following tasks is the responsibility of AWS?

- A. Encrypting client-side data
- B. Configuring AWS Identity and Access Management (IAM) roles
- C. Securing the Amazon EC2 hypervisor
- D. Setting user password policies

Correct Answer: C. Securing the Amazon EC2 hypervisor

Explanation:

Under the AWS Shared Responsibility Model, AWS is responsible for the security *of* the cloud infrastructure, which includes the physical facilities, network infrastructure, and hypervisors. The customer is responsible for configuring IAM roles, client-side encryption, and password policies.

Incorrect Answers:

- A: Client-side encryption is the customer's responsibility.
- B: IAM configuration is managed by the customer.
- D: Password policies fall under customer responsibility.

252. Which is the MINIMUM AWS Support plan that provides technical support through phone calls?

- A. Enterprise
- B. Business
- C. Developer
- D. Basic

Correct Answer: B. Business

Explanation:

The Business support plan includes 24x7 access to Cloud Support Engineers via email, chat, and phone. Developer plans only include email support during business hours, and Basic has no technical support.

Incorrect Answers:

- A: Enterprise includes phone support but is not the minimum plan for it.
- C: Developer does not include phone support.
- D: Basic support is limited to documentation and forums.

253. How should a customer forecast the future costs for running a new web application?

- A. Amazon Aurora Backtrack
- B. Amazon CloudWatch Billing Alarms
- C. AWS Simple Monthly Calculator
- D. AWS Cost and Usage report

Correct Answer: C. AWS Simple Monthly Calculator

Explanation:

The AWS Pricing Calculator (formerly Simple Monthly Calculator) helps customers estimate the cost of AWS services before launching resources.

Incorrect Answers:

- A: Aurora Backtrack is a feature for reverting databases to a previous time.
- B: Billing alarms monitor thresholds, not forecasts.
- D: Cost and Usage reports show past usage, not future projections.

254. A company will be moving from an on-premises data center to the AWS Cloud. What would be one financial difference after the move?

- A. Moving from variable operational expense (opex) to upfront capital expense (capex).
- B. Moving from upfront capital expense (capex) to variable capital expense (capex).
- C. Moving from upfront capital expense (capex) to variable operational expense (opex).
- D. Elimination of upfront capital expense (capex) and elimination of variable operational expense(opex).

Correct Answer: C. Moving from upfront capital expense (capex) to variable operational expense (opex).

Explanation:

Cloud computing uses a pay-as-you-go model, eliminating the need for large upfront investments in hardware (capex) and instead introduces operating costs (opex) based on consumption.

Incorrect Answers:

- A, B: These do not reflect the shift from CapEx to OpEx.
- D: Ongoing OpEx remains in the cloud; only CapEx is reduced.

255. A solution that is able to support growth in users, traffic, or data size with no drop in performance aligns with which cloud architecture principle?

- A. Think parallel
- B. Implement elasticity
- C. Decouple your components
- D. Design for failure

Correct Answer: B. Implement elasticity

Explanation:

Elasticity in cloud architecture allows you to automatically scale resources up or down based on demand, ensuring consistent performance without overprovisioning.

Incorrect Answers:

- A: Think parallel refers to parallel processing, not scalability.
- C: Decoupling improves resilience and flexibility, not necessarily scaling.
- D: Design for failure ensures high availability, not elasticity.

256. Which of the following can limit Amazon Simple Storage Service (Amazon S3) bucket access to specific users?

- A. A public and private key-pair
- B. Amazon Inspector
- C. AWS Identity and Access Management (IAM) policies
- D. Security Groups

Correct Answer: C. AWS Identity and Access Management (IAM) policies

Explanation:

IAM policies control access to AWS resources by defining permissions. You can use IAM to restrict which users or roles can access a specific S3 bucket.

Incorrect Answers:

- A: Key-pairs are used for SSH access, not S3 permissions.
- B: Amazon Inspector is for security assessment, not access control.
- D: Security groups apply to EC2 instances, not S3.

257. Which of the following tasks is the responsibility of AWS?

- A. Encrypting client-side data
- B. Configuring AWS Identity and Access Management (IAM) roles

- C. Securing the Amazon EC2 hypervisor
- D. Setting user password policies

Correct Answer: C. Securing the Amazon EC2 hypervisor

Explanation:

Under the AWS Shared Responsibility Model, AWS is responsible for the security *of* the cloud infrastructure, which includes the physical facilities, network infrastructure, and hypervisors. The customer is responsible for configuring IAM roles, client-side encryption, and password policies.

Incorrect Answers:

- A: Client-side encryption is the customer's responsibility.
- B: IAM configuration is managed by the customer.
- D: Password policies fall under customer responsibility.

258. Which is the MINIMUM AWS Support plan that provides technical support through phone calls?

- A. Enterprise
- B. Business
- C. Developer
- D. Basic

Correct Answer: B. Business

Explanation:

The Business support plan includes 24x7 access to Cloud Support Engineers via email, chat, and phone. Developer plans only include email support during business hours, and Basic has no technical support.

Incorrect Answers:

- A: Enterprise includes phone support but is not the minimum plan for it.
- C: Developer does not include phone support.
- D: Basic support is limited to documentation and forums.

259. How should a customer forecast the future costs for running a new web application?

- A. Amazon Aurora Backtrack
- B. Amazon CloudWatch Billing Alarms

- C. AWS Simple Monthly Calculator
- D. AWS Cost and Usage report

Correct Answer: C. AWS Simple Monthly Calculator

Explanation:

The AWS Pricing Calculator (formerly Simple Monthly Calculator) helps customers estimate the cost of AWS services before launching resources.

Incorrect Answers:

- A: Aurora Backtrack is a feature for reverting databases to a previous time.
- B: Billing alarms monitor thresholds, not forecasts.
- D: Cost and Usage reports show past usage, not future projections.

260. A company will be moving from an on-premises data center to the AWS Cloud. What would be one financial difference after the move?

- A. Moving from variable operational expense (opex) to upfront capital expense (capex).
- B. Moving from upfront capital expense (capex) to variable capital expense (capex).
- C. Moving from upfront capital expense (capex) to variable operational expense (opex).
- D. Elimination of upfront capital expense (capex) and elimination of variable operational expense(opex).

Correct Answer: C. Moving from upfront capital expense (capex) to variable operational expense (opex).

Explanation:

Cloud computing uses a pay-as-you-go model, eliminating the need for large upfront investments in hardware (capex) and instead introduces operating costs (opex) based on consumption.

Incorrect Answers:

- A, B: These do not reflect the shift from CapEx to OpEx.
- D: Ongoing OpEx remains in the cloud; only CapEx is reduced.

261. A solution that is able to support growth in users, traffic, or data size with no drop in performance aligns with which cloud architecture principle?

- A. Think parallel
- B. Implement elasticity

- C. Decouple your components
- D. Design for failure

Correct Answer: B. Implement elasticity

Explanation:

Elasticity in cloud architecture allows you to automatically scale resources up or down based on demand, ensuring consistent performance without overprovisioning.

Incorrect Answers:

- A: Think parallel refers to parallel processing, not scalability.
- C: Decoupling improves resilience and flexibility, not necessarily scaling.
- D: Design for failure ensures high availability, not elasticity.

262. Which of the following can limit Amazon Simple Storage Service (Amazon S3) bucket access to specific users?

- A. A public and private key-pair
- B. Amazon Inspector
- C. AWS Identity and Access Management (IAM) policies
- D. Security Groups

Correct Answer: C. AWS Identity and Access Management (IAM) policies

Explanation:

IAM policies control access to AWS resources by defining permissions. You can use IAM to restrict which users or roles can access a specific S3 bucket.

Incorrect Answers:

- A: Key-pairs are used for SSH access, not S3 permissions.
- B: Amazon Inspector is for security assessment, not access control.
- D: Security groups apply to EC2 instances, not S3.

263. What can AWS edge locations be used for? (Choose two.)

- A. Hosting applications
- B. Delivering content closer to users
- C. Running NoSQL database caching services
- D. Reducing traffic on the server by caching responses
- E. Sending notification messages to end users

Correct Answers: B. Delivering content closer to users

D. Reducing traffic on the server by caching responses

Explanation:

Edge locations are used by services like Amazon CloudFront to cache content close to users, improving performance and reducing the load on the origin server.

Incorrect Answers:

A: Applications are not hosted at edge locations; they're hosted in AWS Regions.

C: NoSQL databases (e.g., DynamoDB) are not run from edge locations.

E: Notifications are handled by services like Amazon SNS, not edge locations.

264. How does AWS shorten the time to provision IT resources?

A. It supplies an online IT ticketing platform for resource requests.

B. It supports automatic code validation services.

C. It provides the ability to programmatically provision existing resources.

D. It automates the resource request process from a company's IT vendor list.

Correct Answer: C. It provides the ability to programmatically provision existing resources.

Explanation:

AWS enables rapid provisioning through APIs, SDKs, and tools like AWS CloudFormation, which automates resource deployment.

Incorrect Answers:

A: AWS does not use a ticketing platform for provisioning.

B: Code validation isn't directly tied to resource provisioning.

D: AWS does not rely on a company's vendor list for automation.

265. Which AWS service can serve a static website?

A. Amazon S3

B. Amazon Route 53

C. Amazon QuickSight

D. AWS X-Ray

Correct Answer: A. Amazon S3

Explanation:

Amazon S3 supports static website hosting, allowing you to store HTML, CSS, and JavaScript files and serve them via a public endpoint.

Incorrect Answers:

B: Route 53 is a DNS service, not a web server.

C: QuickSight is for business intelligence and visualization.

D: X-Ray is used for application tracing and debugging.

266. Which is the minimum AWS Support plan that includes Infrastructure Event Management without additional costs?

A. Enterprise

B. Business

C. Developer

D. Basic

Correct Answer: A. Enterprise

Explanation:

Infrastructure Event Management (IEM) is included at no additional cost with the Enterprise support plan and is not available in lower-tier plans.

Incorrect Answers:

B: Business support doesn't include IEM unless you pay extra.

C: Developer support is limited to business hours and doesn't include IEM.

D: Basic support offers no technical support features.

267. Which AWS feature should a customer leverage to achieve high availability of an application?

A. AWS Direct Connect

B. Availability Zones

C. Data centers

D. Amazon Virtual Private Cloud (Amazon VPC)

Correct Answer: B. Availability Zones

Explanation:

Deploying across multiple Availability Zones ensures that your application remains available even if one zone fails, improving fault tolerance and availability.

Incorrect Answers:

A: Direct Connect is for dedicated network connections, not HA.

C: AWS abstracts data center management from the customer.

D: Amazon VPC provides network isolation but not high availability by itself.

268. In which scenario should Amazon EC2 Spot Instances be used?

A. A company wants to move its main website to AWS from an on-premises web server.

B. A company has a number of application services whose Service Level Agreement (SLA) requires 99.999% uptime.

C. A company's heavily used legacy database is currently running on-premises.

D. A company has a number of infrequent, interruptible jobs that are currently using On-Demand Instances

Correct Answer: D. A company has a number of infrequent, interruptible jobs that are currently using On-Demand Instances

Explanation:

Spot Instances are best suited for workloads that are flexible and fault-tolerant, such as batch processing, big data, and testing, where interruptions are acceptable.

Incorrect Answers:

A: Critical workloads like websites should avoid Spot due to interruption risk.

B: Spot Instances do not guarantee SLAs.

C: Legacy databases should use persistent, stable instances, not Spot.

269. Which of the following common IT tasks can AWS cover to free up company IT resources? (Choose two.)

A. Patching database software

B. Testing application releases

C. Backing up databases

D. Creating database schema

E. Running penetration tests

Correct Answers: A. Patching database software
C. Backing up databases

Explanation:

AWS services like RDS and DynamoDB automate common administrative tasks such as patching and backups, reducing operational burden.

Incorrect Answers:

B: Application release testing is the customer's responsibility.

D: Designing schema is part of application development, which AWS doesn't manage.

E: Penetration testing is controlled and must be authorized by AWS.

270. Which AWS services can be used to gather information about AWS account activity?
(Choose two.)

- A. Amazon CloudFront
- B. AWS Cloud9
- C. AWS CloudTrail
- D. AWS CloudHSM
- E. Amazon CloudWatch

Correct Answers: C. AWS CloudTrail
E. Amazon CloudWatch

Explanation:

CloudTrail logs API calls and activity across AWS, while CloudWatch monitors metrics and logs. Together, they offer comprehensive visibility into AWS account usage.

Incorrect Answers:

A: CloudFront is a CDN, not a logging or monitoring service.

B: Cloud9 is a development environment.

D: CloudHSM provides encryption and key storage, not activity tracking.

271. How do customers benefit from Amazon's massive economies of scale?

- A. Periodic price reductions as the result of Amazon's operational efficiencies
- B. New Amazon EC2 instance types providing the latest hardware
- C. The ability to scale up and down when needed
- D. Increased reliability in the underlying hardware of Amazon EC2 instances

Correct Answer: A. Periodic price reductions as the result of Amazon's operational efficiencies

Explanation:

AWS's scale allows it to reduce costs and pass those savings to customers. This is one of the key financial advantages of cloud computing.

Incorrect Answers:

B: New hardware is an innovation benefit, not an economy of scale.

C: Auto scaling is a feature, not directly a result of economies of scale.

D: Reliability is important but unrelated to pricing efficiency.

272. If each department within a company has its own AWS account, what is one way to enable consolidated billing?

A. Use AWS Budgets on each account to pay only to budget.

B. Contact AWS Support for a monthly bill.

C. Create an AWS Organization from the payer account and invite the other accounts to join.

D. Put all invoices into one Amazon Simple Storage Service (Amazon S3) bucket, load data into Amazon Redshift, and then run a billing report.

Correct Answer: C. Create an AWS Organization from the payer account and invite the other accounts to join.

Explanation:

AWS Organizations allows consolidated billing for multiple accounts under a single payer account, streamlining billing and management.

Incorrect Answers:

A: Budgets help with forecasting but not consolidation.

B: AWS does not manually consolidate bills.

D: This is an overly complex and unnecessary process for billing.

273. Which of the following features can be configured through the Amazon Virtual Private Cloud (Amazon VPC) Dashboard? (Choose two.)

A. Amazon CloudFront distributions

B. Amazon Route 53

C. Security Groups

- D. Subnets
- E. Elastic Load Balancing

Correct Answers: C. Security Groups
D. Subnets

Explanation:

Security Groups and Subnets are core VPC components and can be directly managed through the VPC dashboard.

Incorrect Answers:

- A: CloudFront is managed through its own service dashboard.
- B: Route 53 is DNS-based and not part of VPC.
- E: Elastic Load Balancers are configured through the EC2 or ELB dashboard.

274. Which options does AWS make available for customers who want to learn about security in the cloud in an instructor-led setting? (Choose two.)

- A. AWS Trusted Advisor
- B. AWS Online Tech Talks
- C. AWS Blog
- D. AWS Forums
- E. AWS Classroom Training

Correct Answers: B. AWS Online Tech Talks
E. AWS Classroom Training

Explanation:

AWS offers formal classroom training and online webinars (Tech Talks) that cover security topics in depth.

Incorrect Answers:

- A: Trusted Advisor offers best-practice checks, not training.
- C: The blog is self-paced reading material.
- D: Forums are community discussions, not formal instruction.

275. Which of the following is a component of the shared responsibility model managed entirely by AWS?

- A. Patching operating system software
- B. Encrypting data
- C. Enforcing multi-factor authentication
- D. Auditing physical data center assets

Correct Answer: D. Auditing physical data center assets

Explanation:

AWS is solely responsible for physical infrastructure, including securing and auditing their data centers.

Incorrect Answers:

A: Customers must patch OS on EC2 instances.

B: Data encryption responsibilities are shared depending on the service.

C: MFA is configured by the customer.

276. Which service is best for storing common database query results, which helps to alleviate database access load?

- A. Amazon Machine Learning
- B. Amazon SQS
- C. Amazon ElastiCache
- D. Amazon EC2 Instance Store

Correct Answer: C. Amazon ElastiCache

Explanation:

ElastiCache is designed for caching frequent queries, improving performance and reducing database load.

Incorrect Answers:

A: Amazon ML is used for machine learning models, not caching.

B: SQS is a message queue, not a cache.

D: EC2 Instance Store is temporary storage tied to instance lifecycle.

277. Amazon Relational Database Service (Amazon RDS) offers which of the following benefits over traditional database management?

- A. AWS manages the data stored in Amazon RDS tables.
- B. AWS manages the maintenance of the operating system.

- C. AWS automatically scales up instance types on demand.
- D. AWS manages the database type.

Correct Answer: B. AWS manages the maintenance of the operating system.

Explanation:

RDS simplifies database management by handling OS-level tasks, backups, patching, and maintenance.

Incorrect Answers:

- A: The customer is still responsible for managing their own data.
- C: Scaling must be configured manually or with Auto Scaling; it is not automatic by default.
- D: Customers choose the DB engine (MySQL, PostgreSQL, etc.); AWS doesn't decide it.

278. Your team is deploying a multi-tier web application in a VPC. You want to control which resources can initiate connections to your application servers (hosted in private subnets), and you also want to log and restrict IP traffic entering and leaving the subnets. Which combination of AWS features should you use to achieve this?

- A. Only use Security Groups for all traffic control and logging
- B. Use Security Groups for traffic control and enable VPC Flow Logs for traffic logging
- C. Use Network ACLs for granular access control and AWS WAF for traffic filtering
- D. Use Security Groups for outbound filtering and Route Tables for inbound access

Correct Answer: B. Use Security Groups for traffic control and enable VPC Flow Logs for traffic logging

Explanation:

- Security Groups act at the instance level and control inbound/outbound traffic. They are *stateful* and ideal for managing access to individual EC2 instances.
- VPC Flow Logs allow you to capture IP traffic information flowing to and from network interfaces in your VPC, which helps in monitoring and troubleshooting.
- Together, they help secure and monitor access at both the instance and subnet levels.

Incorrect Answers:

- A: Security Groups do not provide logging capabilities. You need VPC Flow Logs for traffic visibility.
- C: While Network ACLs provide *stateless* subnet-level filtering, they don't log traffic on their own. AWS WAF is used for filtering at the application layer, not for subnet-level traffic control.
- D: Route Tables are used for routing network traffic, not for controlling or filtering traffic like Security Groups or ACLs.

279. Which of the following services falls under the responsibility of the customer to maintain operating system configuration, security patching, and networking?

- A. Amazon RDS
- B. Amazon EC2
- C. Amazon ElastiCache
- D. AWS Fargate

Correct Answer: B. Amazon EC2

Explanation:

With Amazon EC2, customers are responsible for managing the guest OS, patching, and network configuration. EC2 offers the most control, which also means more responsibility.

Incorrect Answers:

- A. Amazon RDS – AWS manages OS and patching for RDS instances.
- C. Amazon ElastiCache – AWS handles patching and maintenance.
- D. AWS Fargate – This is a serverless container service where AWS manages the infrastructure.

280. Which service provides a hybrid storage service that enables on-premises applications to seamlessly use cloud storage?

- A. Amazon Glacier
- B. AWS Snowball
- C. AWS Storage Gateway
- D. Amazon Elastic Block Storage (Amazon EBS)

☒ Correct Answer: C. AWS Storage Gateway

Explanation:

AWS Storage Gateway connects on-premises environments with cloud storage, enabling hybrid use cases like backup, caching, and file sharing.

Incorrect Answers:

- A. Amazon Glacier – Cold storage for archival purposes.
- B. AWS Snowball – A data transport device, not for hybrid storage.
- D. Amazon EBS – Block storage for use within AWS only.

281. Which of the following security measures protect access to an AWS account?
(Choose two.)

- A. Enable AWS CloudTrail.
- B. Grant least privilege access to IAM users.
- C. Create one IAM user and share with many developers and users.
- D. Enable Amazon CloudFront.
- E. Activate multi-factor authentication (MFA) for privileged users.

Correct Answers: B, E

Explanation:

- B: IAM policies should follow the principle of least privilege.
- E: MFA adds a layer of security to the account login process.

Incorrect Answers:

- A: CloudTrail tracks API activity but doesn't prevent unauthorized access.
- C: Sharing IAM users is a bad security practice.
- D: CloudFront is a CDN and doesn't affect account-level access.

282. Which of the following is an AWS Cloud architecture design principle?

- A. Implement single points of failure.
- B. Implement loose coupling.
- C. Implement monolithic design.
- D. Implement vertical scaling

Correct Answer: B. Implement loose coupling

Explanation:

Loose coupling allows components to operate independently, making the system more resilient and scalable.

Incorrect Answers:

- A: Single points of failure reduce fault tolerance.
- C: Monolithic design is less scalable.
- D: Vertical scaling is limited and not ideal for cloud-native architectures.

283. Which of the following can an AWS customer use to launch a new Amazon Relational Database Service (Amazon RDS) cluster? (Choose two.)

- A. AWS Concierge
- B. AWS CloudFormation
- C. Amazon Simple Storage Service (Amazon S3)
- D. Amazon EC2 Auto Scaling
- E. AWS Management Console

Correct Answers: B, E

Explanation:

- B: AWS CloudFormation enables infrastructure as code to provision resources like RDS.
- E: The AWS Management Console offers a GUI to launch RDS.

Incorrect Answers:

- A: AWS Concierge is for Enterprise Support, not resource provisioning.
- C: Amazon S3 is for object storage, not launching databases.
- D: EC2 Auto Scaling is for EC2, not RDS.

284. Which AWS Cost Management tool allows you to view the most granular data about your AWS bill?

- A. AWS Cost Explorer
- B. AWS Budgets

- C. AWS Cost and Usage report
- D. AWS Billing dashboard

Correct Answer: C. AWS Cost and Usage report

Explanation:

It provides the most detailed usage and billing data in CSV format, allowing in-depth analysis.

Incorrect Answers:

- A: Cost Explorer is for visual summaries, not raw data.
- B: Budgets help track costs but don't give granular usage details.
- D: Billing dashboard gives high-level views, not line-item breakdowns.

285. The financial benefits of using AWS are: (Choose two.)

- A. Reduced Total Cost of Ownership (TCO)
- B. Increased capital expenditure (capex)
- C. Reduced operational expenditure (opex)
- D. Deferred payment plans for startups
- E. Business credit lines for startups

Correct Answers: A, C

Explanation:

- A: AWS reduces TCO by eliminating upfront infrastructure costs.
- C: With pay-as-you-go pricing, operational costs are optimized.

Incorrect Answers:

- B: AWS lowers capex, not increases it.
- D & E: These are not standard AWS financial benefits.

286. A company is migrating an application that is running non-interruptible workloads for a three-year time frame. Which pricing construct would provide the MOST cost-effective solution?

- A. Amazon EC2 Spot Instances
- B. Amazon EC2 Dedicated Instances

C. Amazon EC2 On-Demand Instances

D. Amazon EC2 Reserved Instances

Correct Answer: D. Amazon EC2 Reserved Instances

Explanation:

Reserved Instances provide significant savings for long-term, steady workloads.

Incorrect Answers:

- A: Spot is for interruptible jobs.
- B: Dedicated is more expensive and used for compliance.
- C: On-Demand is the most flexible but also costly over time.

287. Which AWS service can be used to manually launch instances based on resource requirements?

A. Amazon EBS

B. Amazon S3

C. Amazon EC2

D. Amazon ECS

Correct Answer: C. Amazon EC2

Explanation:

EC2 is the core compute service that allows launching instances manually or automatically.

Incorrect Answers:

- A: EBS is for block storage.
- B: S3 is for object storage.
- D: ECS is for running containers, not launching EC2 directly.

288. Under the shared responsibility model, which of the following tasks are the responsibility of the AWS customer? (Choose two.)

A. Ensuring that application data is encrypted at rest

B. Ensuring that AWS NTP servers are set to the correct time

- C. Ensuring that users have received security training in the use of AWS services
- D. Ensuring that access to data centers is restricted

Correct Answers: A, C

Explanation:

- A: Customers must manage their data security.
- C: User education and security training are customer responsibilities.

Incorrect Answers:

- B: AWS manages its own time services.
- D: Physical data center security is AWS's responsibility.

289. Which AWS service would you use to obtain compliance reports and certificates?

- A. AWS Artifact
- B. AWS Lambda
- C. Amazon Inspector
- D. AWS Certificate Manager

Correct Answer: A. AWS Artifact

Explanation:

AWS Artifact provides access to AWS compliance documentation and AWS agreements.

Incorrect Answers:

- B. AWS Lambda is for serverless computing, not compliance.
- C. Amazon Inspector is used for security assessments.
- D. AWS Certificate Manager manages SSL/TLS certificates, not compliance reports.

290. Which AWS services are defined as global instead of regional? (Choose two.)

- A. Amazon Route 53
- B. Amazon EC2
- C. Amazon S3
- D. Amazon CloudFront
- E. Amazon DynamoDB

Correct Answers: A, D

Explanation:

Amazon Route 53 and CloudFront are global services and not restricted to a single AWS Region.

Incorrect Answers:

- B. EC2, C. S3, and E. DynamoDB are regional services.

291. What technology enables compute capacity to adjust as loads change?

- A. Load balancing
- B. Automatic failover
- C. Round robin
- D. Auto Scaling

☒ Correct Answer: D. Auto Scaling

Explanation:

Auto Scaling automatically adjusts compute resources based on traffic/load.

Incorrect Answers:

- A. Load balancing distributes traffic but doesn't adjust capacity.
- B. Automatic failover is for availability, not scaling.
- C. Round robin is a simple load balancing method, not for scaling.

292. How would an AWS customer easily apply common access controls to a large set of users?

- A. Apply an IAM policy to an IAM group.
- B. Apply an IAM policy to an IAM role.
- C. Apply the same IAM policy to all IAM users with access to the same workload.
- D. Apply an IAM policy to an Amazon Cognito user pool.

Correct Answer: A. Apply an IAM policy to an IAM group

Explanation:

IAM groups make it easy to assign permissions to multiple users by applying a policy to the group.

Incorrect Answers:

- B. IAM roles are used for temporary access, not managing user groups.
- **C. Assigning individually increases complexity.
- **D. Cognito is for app authentication, not general IAM access control.

293. Which of the following AWS features enables a user to launch a pre-configured Amazon EC2 instance?

- A. Amazon Elastic Block Store (Amazon EBS)
- B. Amazon Machine Image
- C. Amazon EC2 Systems Manager
- D. Amazon AppStream 2.0

Correct Answer: B. Amazon Machine Image

Explanation:

An AMI contains a pre-configured OS, software, and settings used to launch EC2 instances.

Incorrect Answers:

- A. EBS is used for storage, not launching instances.
- C. EC2 Systems Manager helps with managing resources.
- D. AppStream 2.0 is for streaming applications.

294. Which of the following steps should be taken by a customer when conducting penetration testing on AWS?

- A. Conduct penetration testing using Amazon Inspector, and then notify AWS support.
- B. Request and wait for approval from the customer's internal security team, and then conduct testing.
- C. Notify AWS support, and then conduct testing immediately.
- D. Request and wait for approval from AWS support, and then conduct testing.

Correct Answer: D. Request and wait for approval from AWS support, and then conduct testing.

Explanation:

Penetration testing must be approved by AWS in advance as per AWS policy.

295. Which of the following is an advantage of consolidated billing on AWS?

- A. Volume pricing qualification
- B. Shared access permissions
- C. Multiple bills per account
- D. Eliminates the need for tagging

Correct Answer: A. Volume pricing qualification

Explanation:

Consolidated billing allows organizations to combine usage across accounts for volume discounts.

296. Which AWS service provides a customized view of the health of specific AWS services that power a customer's workloads running on AWS?

- A. AWS Service Health Dashboard
- B. AWS X-Ray
- C. AWS Personal Health Dashboard
- D. Amazon CloudWatch

Correct Answer: C. AWS Personal Health Dashboard

Explanation:

It provides account-specific insights into service health and planned maintenance.

297. Where can AWS compliance and certification reports be downloaded?

- A. AWS Artifact
- B. AWS Concierge
- C. AWS Certificate Manager
- D. AWS Trusted Advisor

Correct Answer: A. AWS Artifact

Explanation:

AWS Artifact provides access to compliance documents such as SOC and ISO reports.

298. Which is the MINIMUM AWS Support plan that allows for one-hour target response time for support cases?

- A. Enterprise
- B. Business
- C. Developer
- D. Basic

Correct Answer: B. Business

Explanation:

The Business support plan includes 24/7 access and a 1-hour response time for urgent cases.

299. Which design principles for cloud architecture are recommended when re-architecting a large monolithic application? (Choose two.)

- A. Use manual monitoring.
- B. Use fixed servers.
- C. Implement loose coupling.
- D. Rely on individual components.
- E. Design for scalability.

Correct Answers: C, E

Explanation:

Loose coupling and scalability are key cloud-native design principles.

300. Which Amazon EC2 pricing model adjusts based on supply and demand of EC2 instances?

- A. On-Demand Instances
- B. Reserved Instances
- C. Spot Instances
- D. Convertible Reserved Instances

Correct Answer: C. Spot Instances

Explanation:

Spot pricing varies with supply and demand, offering up to 90% cost savings.

301. Which of the following services could be used to deploy an application to servers running on-premises? (Choose two.)

- A. AWS Elastic Beanstalk
- B. AWS OpsWorks
- C. AWS CodeDeploy
- D. AWS Batch
- E. AWS X-Ray

Correct Answers: B, C

Explanation:

- B. AWS OpsWorks can manage on-premises infrastructure using Chef/Puppet.
- C. AWS CodeDeploy supports hybrid deployments including on-premises servers.

302. Which service allows a company with multiple AWS accounts to combine its usage to obtain volume discounts?

- A. AWS Server Migration Service
- B. AWS Organizations
- C. AWS Budgets
- D. AWS Trusted Advisor

Correct Answer: B. AWS Organizations

Explanation:

AWS Organizations lets companies consolidate billing and apply service control policies.

303. What is Amazon CloudWatch?

- A. A code repository with customizable build and team commit features.
- B. A metrics repository with customizable notification thresholds and channels.
- C. A security configuration repository with threat analytics.
- D. A rule repository of a web application firewall with automated vulnerability prevention features.

Correct Answer: B. A metrics repository with customizable notification thresholds and channels.

304. If a customer needs to audit the change management of AWS resources, which of the following AWS services should the customer use?

- A. AWS Config
- B. AWS Trusted Advisor
- C. Amazon CloudWatch
- D. Amazon Inspector

Correct Answer: A. AWS Config

Explanation:

AWS Config tracks configuration changes of AWS resources for auditing and compliance.

305. Which AWS service provides the ability to manage infrastructure as code?

- A. AWS CodePipeline
- B. AWS CodeDeploy
- C. AWS Direct Connect
- D. AWS CloudFormation

Correct Answer: D. AWS CloudFormation

Explanation:

CloudFormation uses templates to provision and manage resources in a repeatable manner.

306. When performing a cost analysis that supports physical isolation of a customer workload, which compute hosting model should be accounted for in the Total Cost of Ownership (TCO)?

- A. Dedicated Hosts
- B. Reserved Instances
- C. On-Demand Instances
- D. No Upfront Reserved Instances

Correct Answer: A. Dedicated Hosts

Explanation:

Dedicated Hosts provide physical isolation, which affects cost and licensing.

307. Which of the following is a benefit of using the AWS Cloud?

- A. Permissive security removes the administrative burden.
- B. Ability to focus on revenue-generating activities.
- C. Control over cloud network hardware.
- D. Choice of specific cloud hardware vendors.

Correct Answer: B. Ability to focus on revenue-generating activities

Explanation:

AWS removes infrastructure overhead, letting customers focus on their core business.